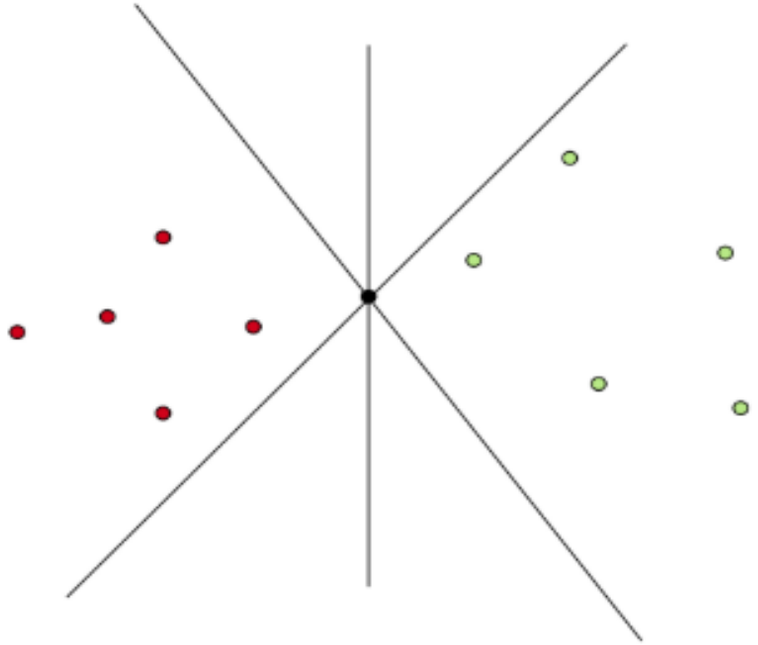
MLT: Week-10

Support Vector Machine
(theory)

Pulkit Mangal

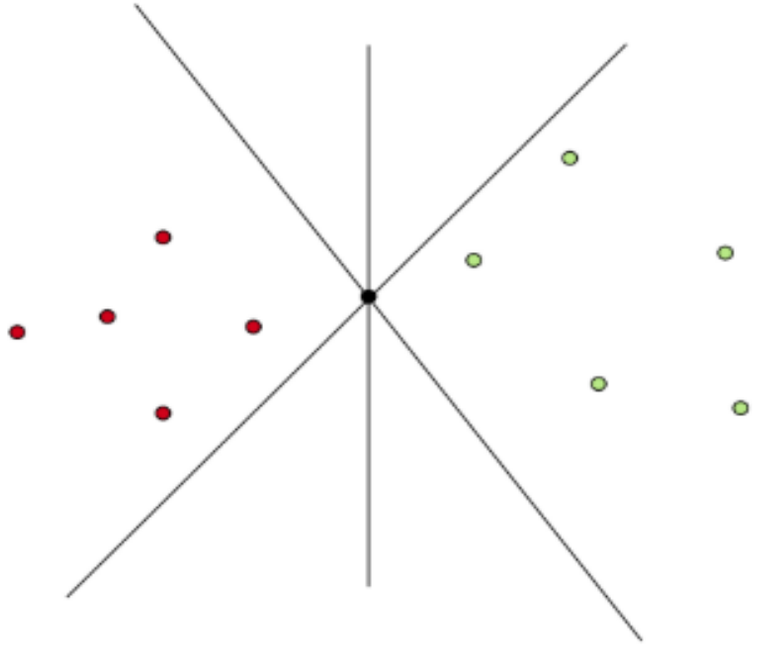
From Perceptrons to SVM

From Perceptrons to SVM



For linearly separable data with γ margin:

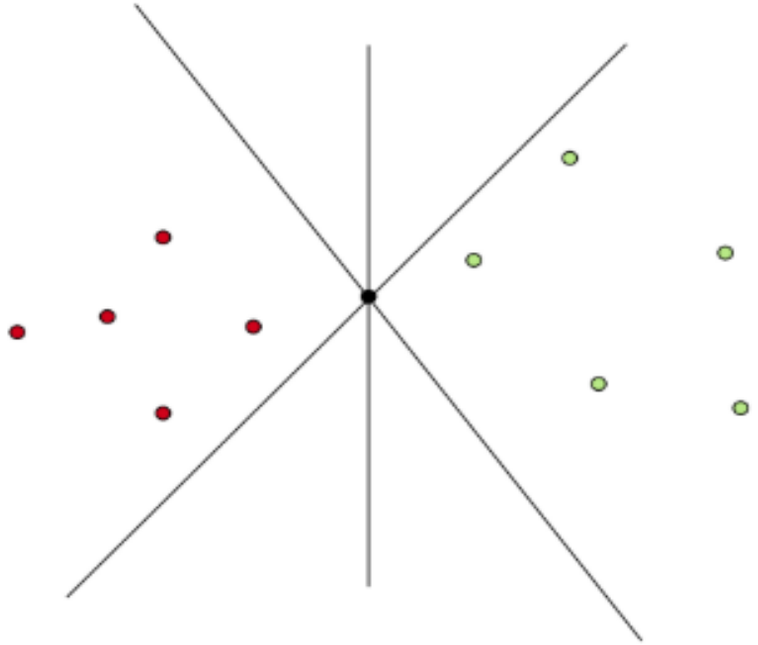
From Perceptrons to SVM



For linearly separable data with γ margin:

(1) Infinite number of valid linear classifiers.

From Perceptrons to SVM

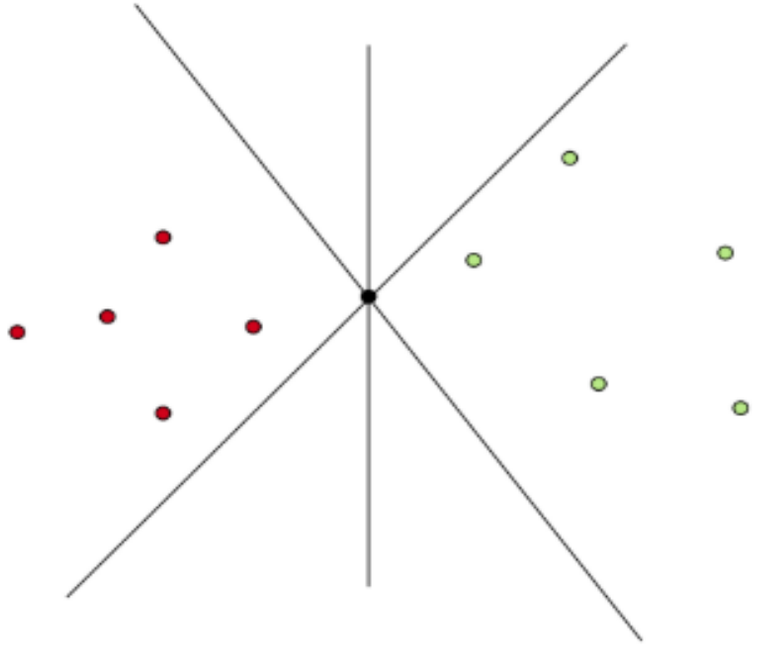


For linearly separable data with γ margin:

(1) Infinite number of valid linear classifiers.

(2) Perceptron returns a valid linear classifier.

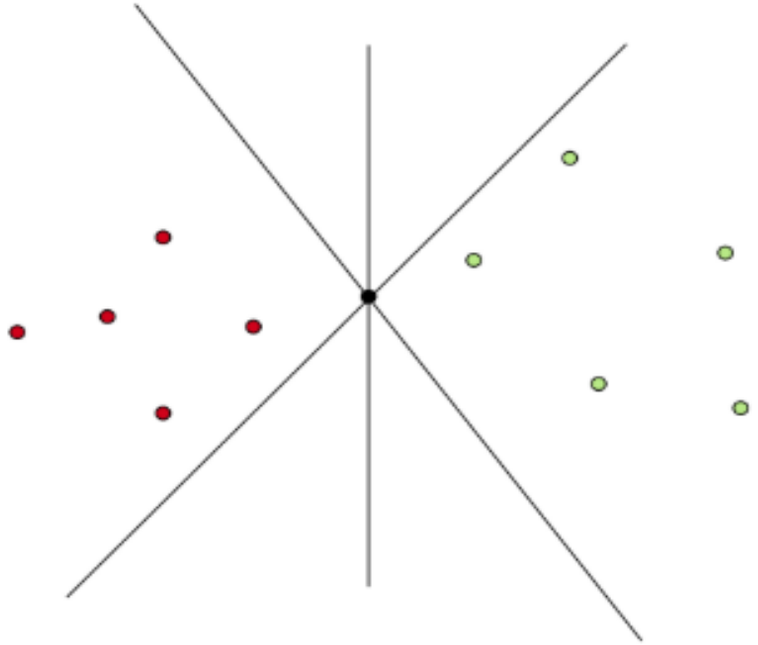
From Perceptrons to SVM



For linearly separable data with γ margin:

(1) Infinite number of valid linear classifiers.
(2) Perceptron returns a valid linear classifier.
(3) Is it the "best"?

From Perceptrons to SVM



For linearly separable data with γ margin:

(1) Infinite number of valid linear classifiers.
(2) Perceptron returns a valid linear classifier.
(3) Is it the "best"?
(4) What is a good notion of "best"?

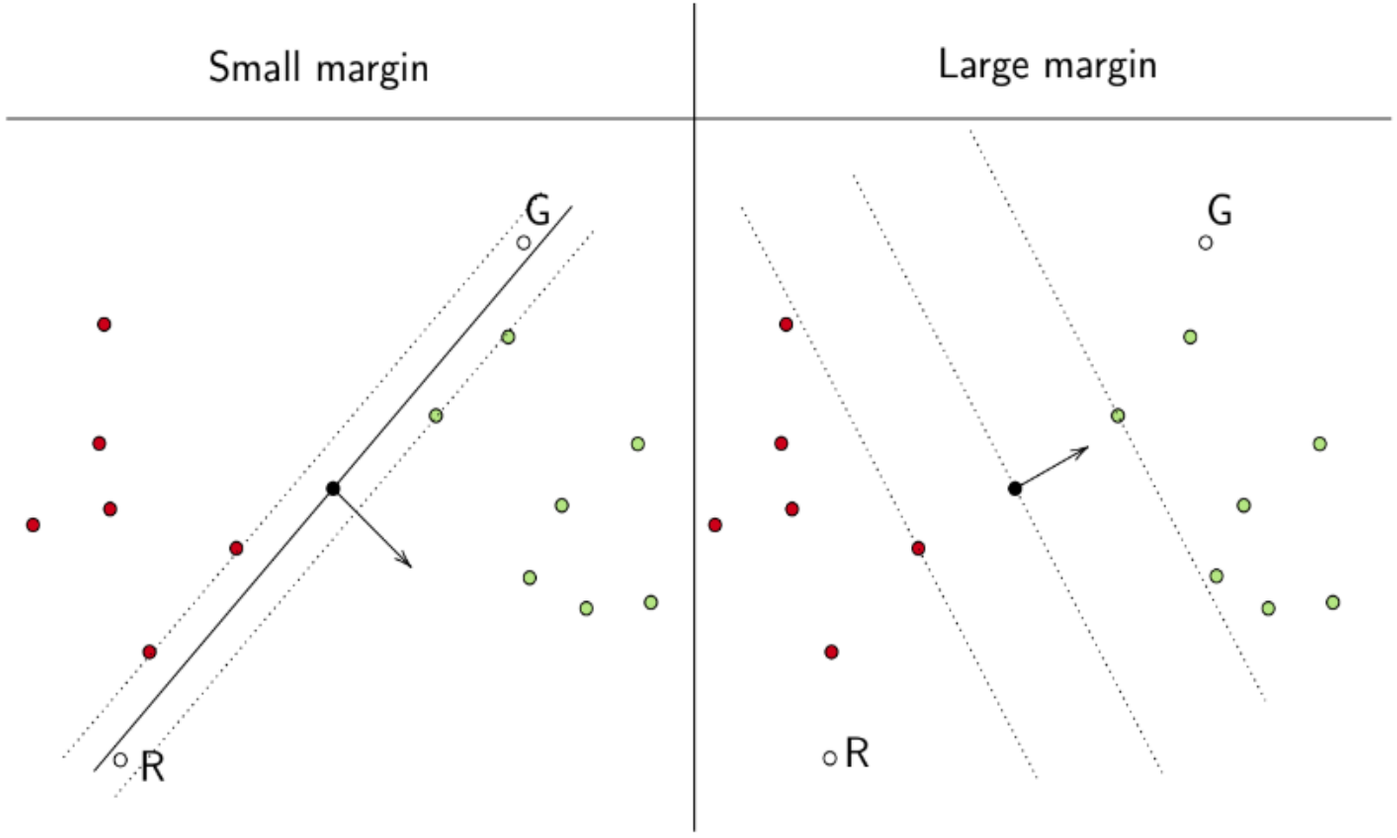
Margin

Margin

Classifiers with a larger margin generalize better. That is, they perform better on test data-points compared to classifiers that have a small margin. Here is a visual intuition for to justify this:

Margin

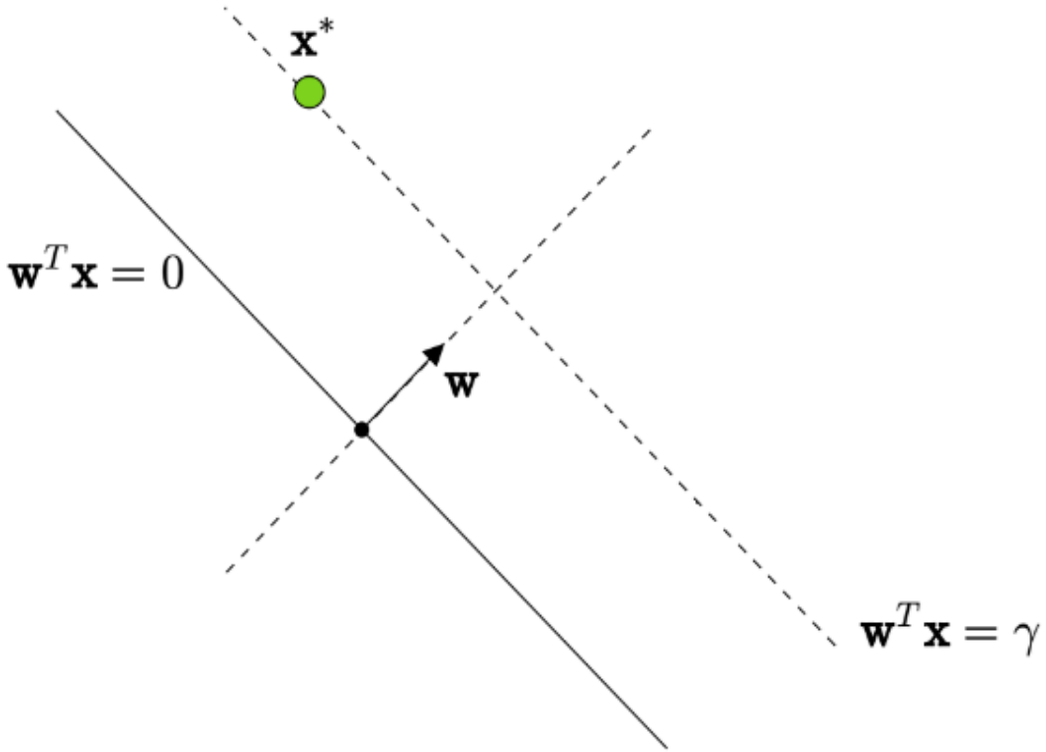
Classifiers with a larger margin generalize better. That is, they perform better on test data-points compared to classifiers that have a small margin. Here is a visual intuition for to justify this:



Margin

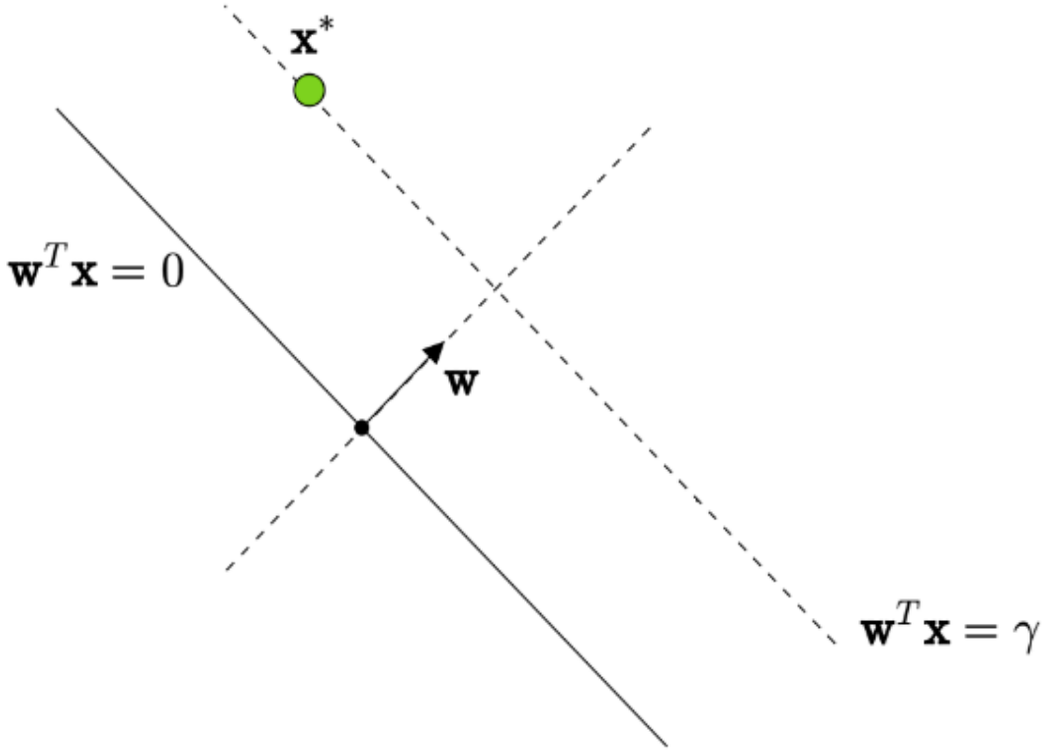
Margin

Consider a linearly separable dataset with a positive margin. Let $\mathbf{w}$ be any valid classifier that perfectly separates the dataset.



Margin

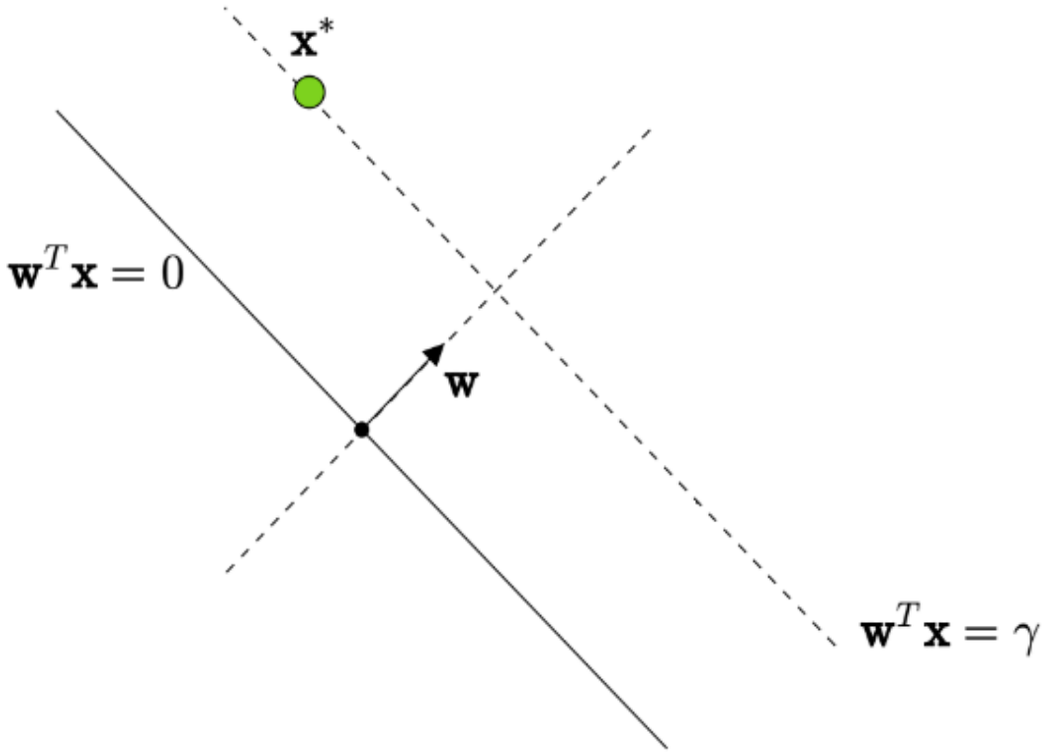
Consider a linearly separable dataset with a positive margin. Let $\mathbf{w}$ be any valid classifier that perfectly separates the dataset.



The distance of between the lines $\mathbf{w}^T\mathbf{x} = 0$ and is called the $\mathbf{w}^T\mathbf{x} = \gamma$ geometric margin and is given by:

Margin

Consider a linearly separable dataset with a positive margin. Let $\mathbf{w}$ be any valid classifier that perfectly separates the dataset.

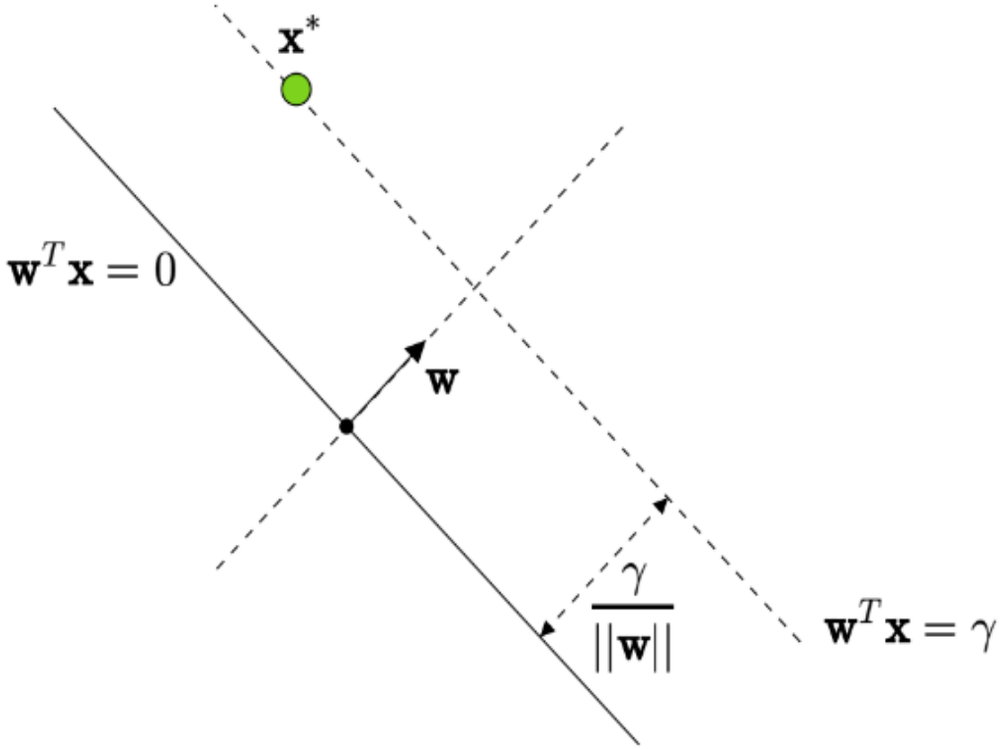


The distance of between the lines $\mathbf{w}^T\mathbf{x} = 0$ and is called the $\mathbf{w}^T\mathbf{x} = \gamma$ geometric margin and is given by:

$$\frac{\mathbf{w}^T\mathbf{x}^*}{\|\mathbf{w}\|} = \frac{\gamma}{\|\mathbf{w}\|}$$

Margin

Consider a linearly separable dataset with a positive margin. Let $\mathbf{w}$ be any valid classifier that perfectly separates the dataset.

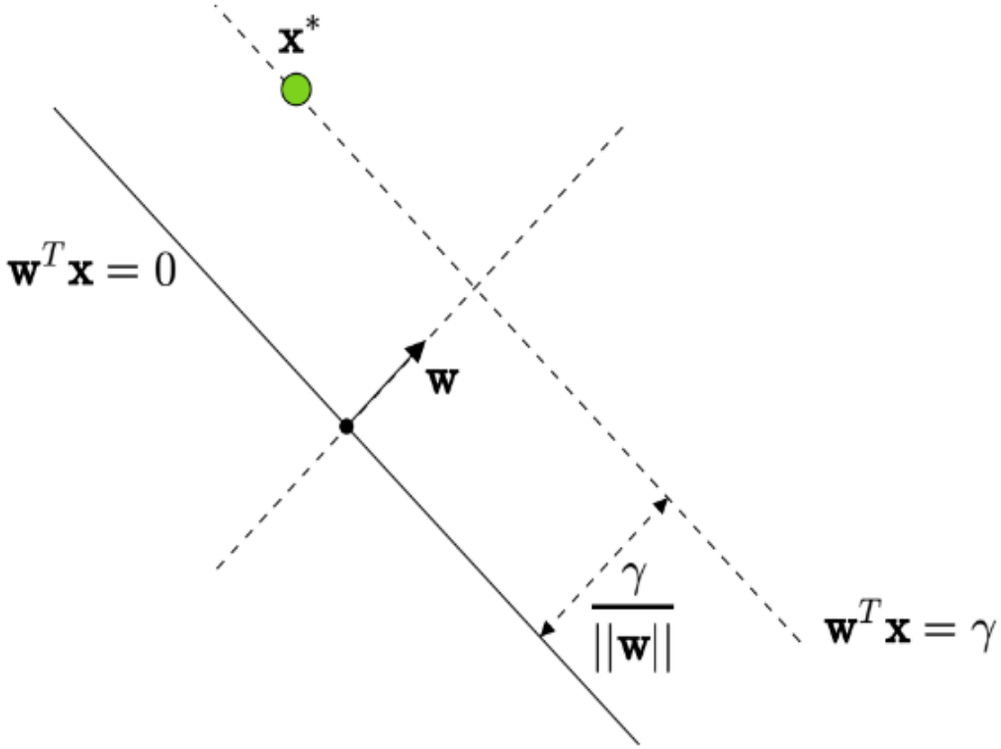


The distance of between the lines $\mathbf{w}^T\mathbf{x} = 0$ and is called the $\mathbf{w}^T\mathbf{x} = \gamma$ geometric margin and is given by:

$$\frac{\mathbf{w}^T\mathbf{x}^*}{\|\mathbf{w}\|} = \frac{\gamma}{\|\mathbf{w}\|}$$

Margin

Consider a linearly separable dataset with a positive margin. Let $\mathbf{w}$ be any valid classifier that perfectly separates the dataset.



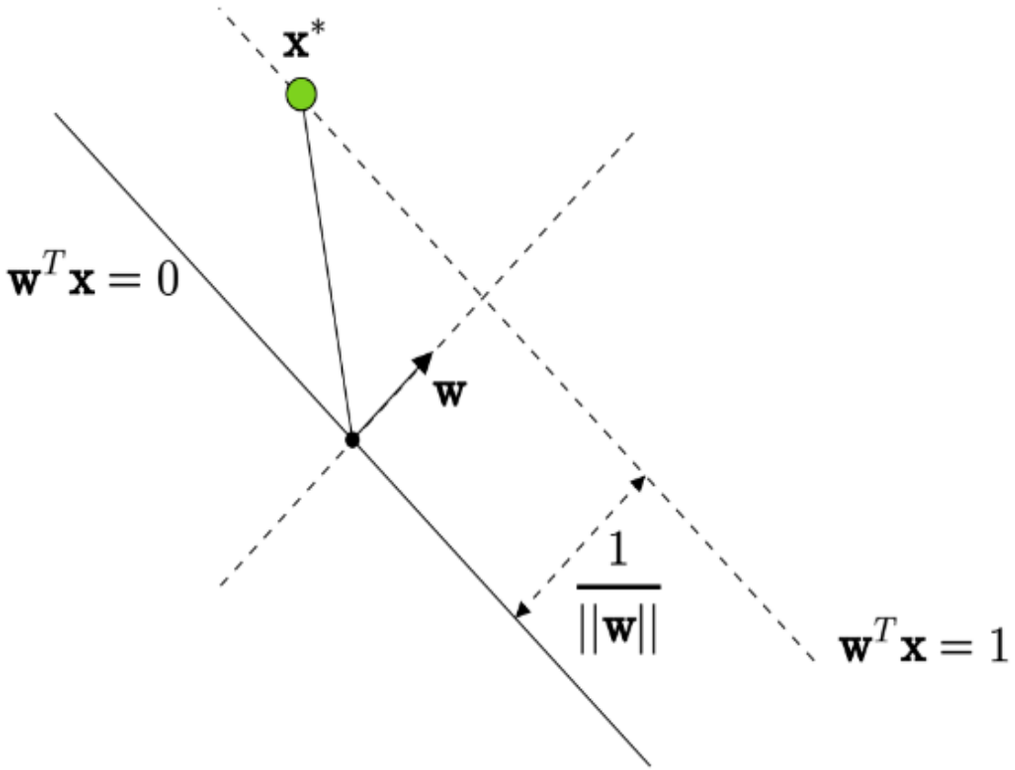
The distance of between the lines $\mathbf{w}^T\mathbf{x} = 0$ and is called the $\mathbf{w}^T\mathbf{x} = \gamma$ geometric margin and is given by:

$$\frac{\mathbf{w}^T\mathbf{x}^*}{\|\mathbf{w}\|} = \frac{\gamma}{\|\mathbf{w}\|}$$

Since both $\mathbf{w}$ and γ can be scaled simultaneously without affecting the setup, we will set $\gamma = 1$.

Margin

Consider a linearly separable dataset with a positive margin. Let $\mathbf{w}$ be any valid classifier that perfectly separates the dataset.



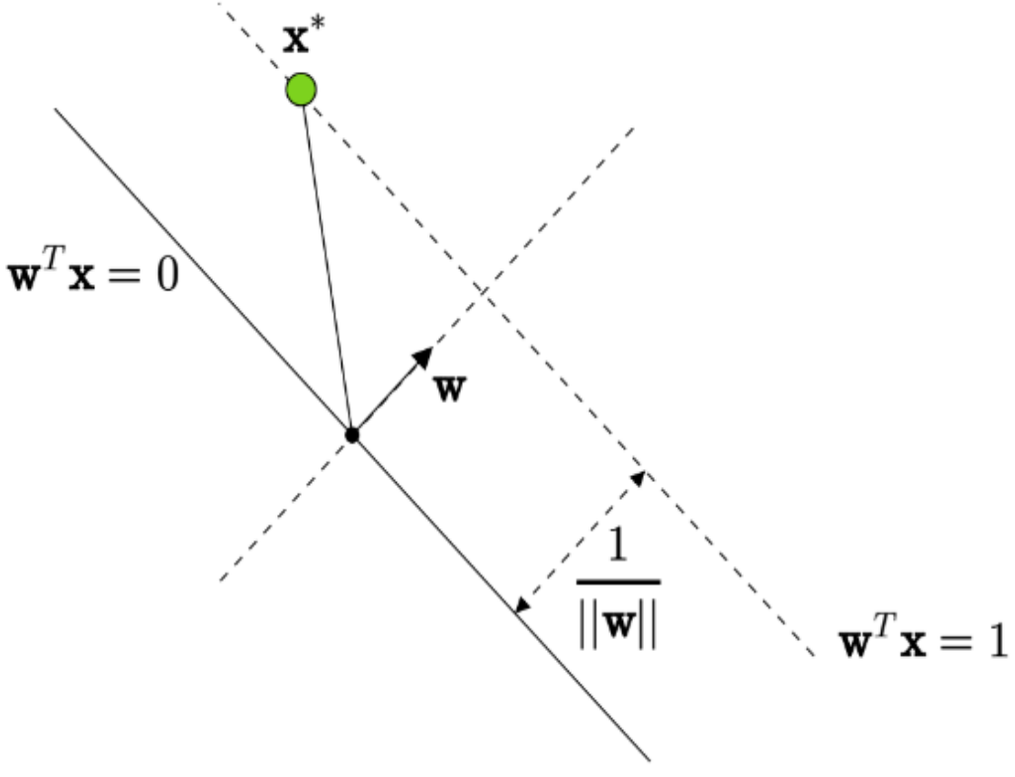
The distance of between the lines $\mathbf{w}^T\mathbf{x} = 0$ and is called the $\mathbf{w}^T\mathbf{x} = \gamma$ geometric margin and is given by:

$$\frac{\mathbf{w}^T\mathbf{x}^*}{\|\mathbf{w}\|} = \frac{\gamma}{\|\mathbf{w}\|}$$

Since both $\mathbf{w}$ and γ can be scaled simultaneously without affecting the setup, we will set $\gamma = 1$.

Margin

Therefore, for any linearly separable dataset with a positive margin, and a valid classifier $\mathbf{w}$, we can set the functional margin $\gamma = 1$ and the geometric margin becomes:



$$\text{(geometric) margin} = \frac{1}{\|\mathbf{w}\|}$$

Hard-Margin Support Vector Machines

Hard-Margin Support Vector Machines

We would like to find a classifier that has the maximum margin: $\max_{\mathbf{w}} \frac{1}{\|\mathbf{w}\|}$

Hard-Margin Support Vector Machines

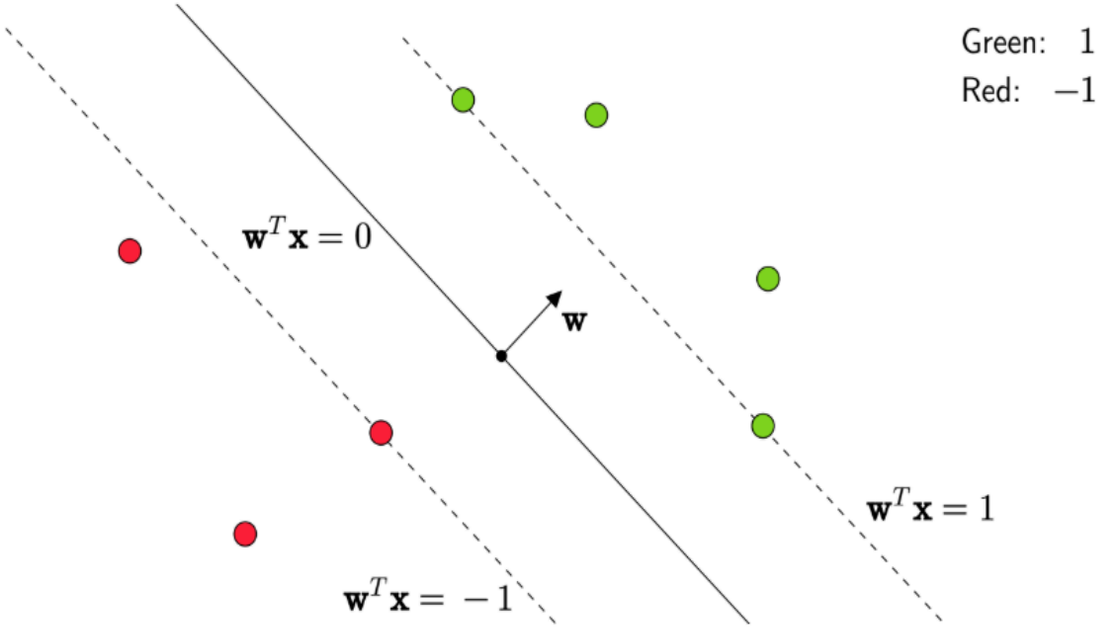
We would like to find a classifier that has the maximum margin: $\max_{\mathbf{w}} \frac{1}{\|\mathbf{w}\|}$

But we also have constraints:

Hard-Margin Support Vector Machines

We would like to find a classifier that has the maximum margin: $\max_{\mathbf{w}} \frac{1}{\|\mathbf{w}\|}$

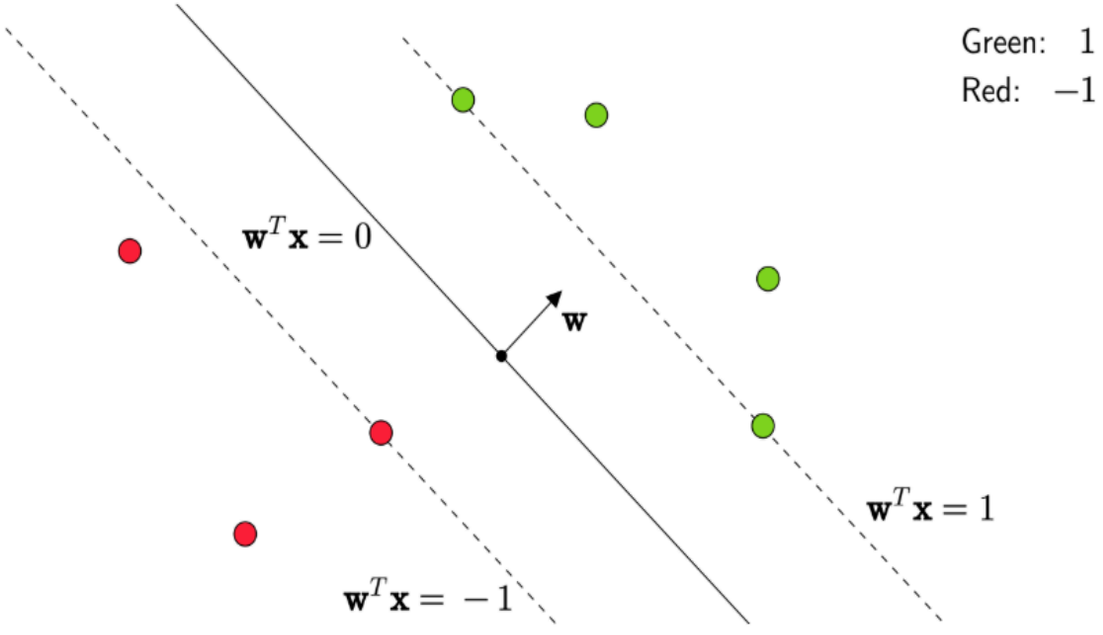
But we also have constraints:



Hard-Margin Support Vector Machines

We would like to find a classifier that has the maximum margin: $\max_{\mathbf{w}} \frac{1}{\|\mathbf{w}\|}$

But we also have constraints:

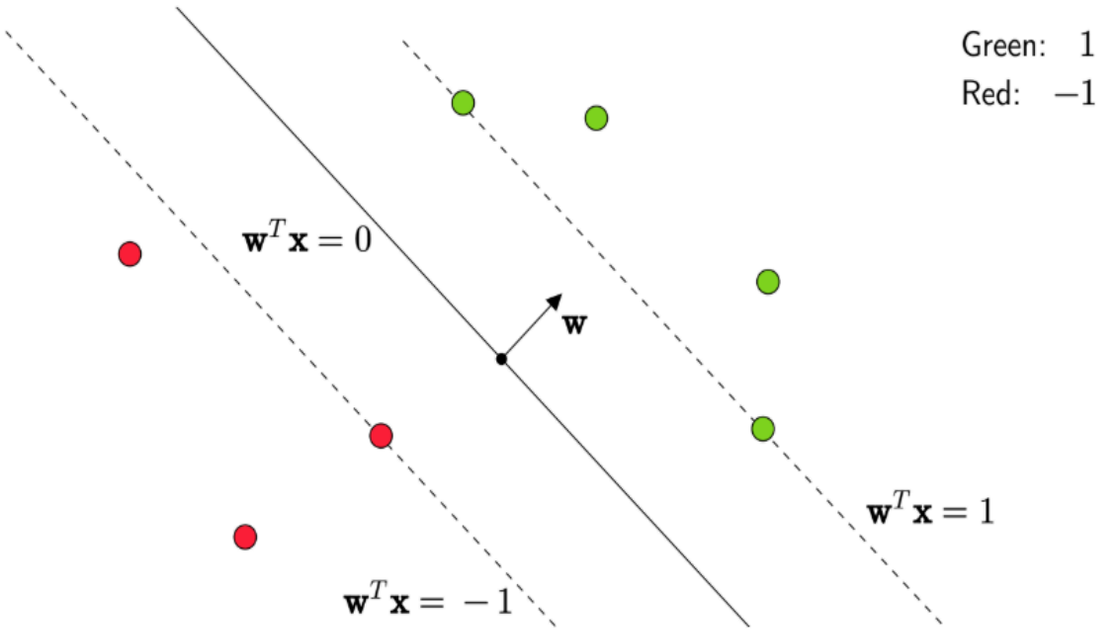


Since the closest points to the classifier lie on the margin, all other points have to lie beyond the margin, giving us the following n constraints:

Hard-Margin Support Vector Machines

We would like to find a classifier that has the maximum margin: $\max_{\mathbf{w}} \frac{1}{\|\mathbf{w}\|}$

But we also have constraints:



Since the closest points to the classifier lie on the margin, all other points have to lie beyond the margin, giving us the following n constraints:

$$\left(\mathbf{w}^T\mathbf{x}_i\right)y_i \geqslant 1,\ 1 \leqslant i \leqslant n$$

Optimization Problem

Optimization Problem

Putting everything together from previous observations, we have:

Optimization Problem

Putting everything together from previous observations, we have:

$$\max_{\mathbf{w}} \frac{1}{\|\mathbf{w}\|}$$

subject to

$$\left(\mathbf{w}^T \mathbf{x}_i\right) y_i \geqslant 1,\ 1 \leqslant i \leqslant n$$

Optimization Problem

Putting everything together from previous observations, we have:

$$\max_{\mathbf{w}} \frac{1}{\|\mathbf{w}\|}$$

subject to

$$\left(\mathbf{w}^T \mathbf{x}_i\right) y_i \geqslant 1,\ 1 \leqslant i \leqslant n$$

A slight modification to the objective function results in:

Optimization Problem

Putting everything together from previous observations, we have:

$$\max_{\mathbf{w}} \frac{1}{\|\mathbf{w}\|}$$

subject to

$$\left(\mathbf{w}^T \mathbf{x}_i\right) y_i \geqslant 1,\ 1 \leqslant i \leqslant n$$

A slight modification to the objective function results in:

$$\min_{\mathbf{w}} \frac{\|\mathbf{w}\|^2}{2}$$

subject to

$$\left(\mathbf{w}^T \mathbf{x}_i\right) y_i \geqslant 1,\ 1 \leqslant i \leqslant n$$

Primal and Dual

Primal and Dual

$$\min_{\mathbf{w}} \; f(\mathbf{w})$$

subject to

$$g(\mathbf{w}) \leqslant 0$$

Primal and Dual

Using Lagrangian multiplier

$$\min_{\mathbf{w}} \; f(\mathbf{w})$$

subject to

$$g(\mathbf{w}) \leqslant 0$$

Primal and Dual

Using Lagrangian multiplier

$$\max_{\alpha \geqslant 0} \; f(\mathbf{w}) \; + \; \alpha g(\mathbf{w}) = \begin{cases} f(\mathbf{w}) & g(\mathbf{w}) \leqslant 0 \\ \infty & g(\mathbf{w}) > 0 \end{cases}$$

$$\min_{\mathbf{w}} \; f(\mathbf{w})$$

subject to

$$g(\mathbf{w}) \leqslant 0$$

Primal and Dual

Using Lagrangian multiplier

$$\max_{\alpha \geqslant 0} \; f(\mathbf{w}) + \alpha g(\mathbf{w}) = \begin{cases} f(\mathbf{w}) & g(\mathbf{w}) \leqslant 0 \\ \infty & g(\mathbf{w}) > 0 \end{cases}$$

$$\min_{\mathbf{w}} \; f(\mathbf{w})$$

subject to

$$g(\mathbf{w}) \leqslant 0$$

$$\equiv \quad \min_{\mathbf{w}} \left[\max_{\alpha \geqslant 0} \; f(\mathbf{w}) + \alpha g(\mathbf{w}) \right]$$

The equivalent problem

Primal and Dual

Using Lagrangian multiplier

$$\max_{\alpha \geqslant 0} \; f(\mathbf{w}) + \alpha g(\mathbf{w}) = \begin{cases} f(\mathbf{w}) & g(\mathbf{w}) \leqslant 0 \\ \infty & g(\mathbf{w}) > 0 \end{cases}$$

$$\min_{\mathbf{w}} \; f(\mathbf{w})$$

subject to $\equiv$ $\min_{\mathbf{w}} \left[\max_{\alpha \geqslant 0} \; f(\mathbf{w}) + \alpha g(\mathbf{w}) \right]$

$$g(\mathbf{w}) \leqslant 0$$

The equivalent problem

which is called the **Primal Problem.**

Primal and Dual

Using Lagrangian multiplier

$$\max_{\alpha \geqslant 0} \; f(\mathbf{w}) + \alpha g(\mathbf{w}) = \begin{cases} f(\mathbf{w}) & g(\mathbf{w}) \leqslant 0 \\ \infty & g(\mathbf{w}) > 0 \end{cases}$$

$$\min_{\mathbf{w}} \; f(\mathbf{w})$$

subject to

$$g(\mathbf{w}) \leqslant 0$$

$$\equiv \quad \min_{\mathbf{w}} \left[\max_{\alpha \geqslant 0} \; f(\mathbf{w}) + \alpha g(\mathbf{w}) \right]$$

The equivalent problem

which is called the **Primal Problem.** Swapping min & max gives us **Dual Problem** as:

Primal and Dual

Using Lagrangian multiplier

$$\max_{\alpha \geqslant 0} \; f(\mathbf{w}) + \alpha g(\mathbf{w}) = \begin{cases} f(\mathbf{w}) & g(\mathbf{w}) \leqslant 0 \\ \infty & g(\mathbf{w}) > 0 \end{cases}$$

$$\min_{\mathbf{w}} \; f(\mathbf{w})$$

subject to

$$g(\mathbf{w}) \leqslant 0$$

$$\equiv \quad \min_{\mathbf{w}} \left[\max_{\alpha \geqslant 0} \; f(\mathbf{w}) + \alpha g(\mathbf{w}) \right]$$

The equivalent problem

which is called the **Primal Problem.** Swapping min & max gives us **Dual Problem** as:

$$\max_{\alpha \geqslant 0} \left[\min_{\mathbf{w}} \; f(\mathbf{w}) + \alpha g(\mathbf{w}) \right]$$

Primal and Dual

Using Lagrangian multiplier

$$\max_{\alpha \geqslant 0} \; f(\mathbf{w}) + \alpha g(\mathbf{w}) = \begin{cases} f(\mathbf{w}) & g(\mathbf{w}) \leqslant 0 \\ \infty & g(\mathbf{w}) > 0 \end{cases}$$

$$\min_{\mathbf{w}} \; f(\mathbf{w})$$

subject to

$$g(\mathbf{w}) \leqslant 0$$

$$\equiv \quad \min_{\mathbf{w}} \left[\max_{\alpha \geqslant 0} \; f(\mathbf{w}) + \alpha g(\mathbf{w}) \right]$$

The equivalent problem

which is called the **Primal Problem.** Swapping min & max gives us **Dual Problem** as:

$$\max_{\alpha \geqslant 0} \left[\min_{\mathbf{w}} \; f(\mathbf{w}) + \alpha g(\mathbf{w}) \right]$$

Strong duality refers to the condition where the optimal value of the primal problem equals the optimal value of the dual problem.

Solving the Optimization Problem

Solving the Optimization Problem

$$\min_{\mathbf{w}} \frac{\|\mathbf{w}\|^2}{2}$$

subject to

$$\left(\mathbf{w}^T \mathbf{x}_i\right) y_i \geqslant 1,\ 1 \leqslant i \leqslant n$$

Solving the Optimization Problem

$$\min_{\mathbf{w}} \frac{\|\mathbf{w}\|^2}{2}$$

subject to

$$\left(\mathbf{w}^T \mathbf{x}_i\right) y_i \geqslant 1,\ 1 \leqslant i \leqslant n$$

Using Lagrangian function with α_i being the ith Lagrange multiplier

Solving the Optimization Problem

$$\min_{\mathbf{w}} \frac{\|\mathbf{w}\|^2}{2}$$

subject to

$$\left(\mathbf{w}^T\mathbf{x}_i\right)y_i \geqslant 1,\ 1 \leqslant i \leqslant n$$

Using Lagrangian function with α_i being the ith Lagrange multiplier

$$\frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^{n} \alpha_i \left[1 - \left(\mathbf{w}^T\mathbf{x}_i\right)y_i\right]$$

Solving the Optimization Problem

$$\min_{\mathbf{w}} \frac{\|\mathbf{w}\|^2}{2}$$

subject to

$$\left(\mathbf{w}^T\mathbf{x}_i\right)y_i \geqslant 1,\ 1 \leqslant i \leqslant n$$

Using Lagrangian function with α_i being the ith Lagrange multiplier

$$\frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^{n} \alpha_i \left[1 - \left(\mathbf{w}^T\mathbf{x}_i\right)y_i\right]$$

Since the objective function is convex and since the inequality constraints involve convex functions, the entire problem is a convex optimization problem.

Solving the Optimization Problem

$$\min_{\mathbf{w}} \frac{\|\mathbf{w}\|^2}{2}$$

subject to

$$\left(\mathbf{w}^T\mathbf{x}_i\right)y_i \geqslant 1,\ 1 \leqslant i \leqslant n$$

Using Lagrangian function with α_i being the ith Lagrange multiplier

$$\frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^{n} \alpha_i \left[1 - \left(\mathbf{w}^T\mathbf{x}_i\right)y_i\right]$$

Since the objective function is convex and since the inequality constraints involve convex functions, the entire problem is a convex optimization problem.

Primal Problem

Solving the Optimization Problem

$$\min_{\mathbf{w}} \frac{\|\mathbf{w}\|^2}{2}$$

subject to

$$\left(\mathbf{w}^T\mathbf{x}_i\right)y_i \geqslant 1,\ 1 \leqslant i \leqslant n$$

Using Lagrangian function with α_i being the ith Lagrange multiplier

$$\frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^{n} \alpha_i \left[1 - \left(\mathbf{w}^T\mathbf{x}_i\right)y_i\right]$$

Since the objective function is convex and since the inequality constraints involve convex functions, the entire problem is a convex optimization problem.

Primal Problem

$$\min_{\mathbf{w}} \max_{\alpha \geqslant 0} \frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^{n} \alpha_i \left[1 - \left(\mathbf{w}^T\mathbf{x}_i\right)y_i\right]$$

Solving the Optimization Problem

$$\min_{\mathbf{w}} \frac{\|\mathbf{w}\|^2}{2}$$

subject to

$$\left(\mathbf{w}^T\mathbf{x}_i\right)y_i \geqslant 1,\ 1 \leqslant i \leqslant n$$

Using Lagrangian function with α_i being the ith Lagrange multiplier

$$\frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^{n} \alpha_i \left[1 - \left(\mathbf{w}^T\mathbf{x}_i\right)y_i\right]$$

Since the objective function is convex and since the inequality constraints involve convex functions, the entire problem is a convex optimization problem.

Primal Problem

$$\min_{\mathbf{w}} \max_{\alpha \geqslant 0} \frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^{n} \alpha_i \left[1 - \left(\mathbf{w}^T\mathbf{x}_i\right)y_i\right]$$

Optimizing variable is $\mathbf{w}$

Solving the Optimization Problem

$$\min_{\mathbf{w}} \frac{\|\mathbf{w}\|^2}{2}$$

subject to

$$\left(\mathbf{w}^T\mathbf{x}_i\right)y_i \geqslant 1,\ 1 \leqslant i \leqslant n$$

Using Lagrangian function with α_i being the ith Lagrange multiplier

$$\frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^{n} \alpha_i \left[1 - \left(\mathbf{w}^T\mathbf{x}_i\right)y_i\right]$$

Since the objective function is convex and since the inequality constraints involve convex functions, the entire problem is a convex optimization problem.

Primal Problem

$$\min_{\mathbf{w}} \max_{\alpha \geqslant 0} \frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^{n} \alpha_i \left[1 - \left(\mathbf{w}^T\mathbf{x}_i\right)y_i\right]$$

Optimizing variable is $\mathbf{w}$

Dual Problem

Solving the Optimization Problem

$$\min_{\mathbf{w}} \frac{\|\mathbf{w}\|^2}{2}$$

subject to

$$\left(\mathbf{w}^T\mathbf{x}_i\right)y_i \geqslant 1,\ 1 \leqslant i \leqslant n$$

Using Lagrangian function with α_i being the ith Lagrange multiplier

$$\frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^{n} \alpha_i \left[1 - \left(\mathbf{w}^T\mathbf{x}_i\right)y_i\right]$$

Since the objective function is convex and since the inequality constraints involve convex functions, the entire problem is a convex optimization problem.

Primal Problem

$$\min_{\mathbf{w}} \max_{\alpha \geqslant 0} \frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^{n} \alpha_i \left[1 - \left(\mathbf{w}^T\mathbf{x}_i\right)y_i\right]$$

Optimizing variable is $\mathbf{w}$

Dual Problem

$$\max_{\alpha \geqslant 0} \min_{\mathbf{w}} \frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^{n} \alpha_i \left[1 - \left(\mathbf{w}^T\mathbf{x}_i\right)y_i\right]$$

Solving the Optimization Problem

$$\min_{\mathbf{w}} \frac{\|\mathbf{w}\|^2}{2}$$

subject to

$$\left(\mathbf{w}^T\mathbf{x}_i\right)y_i \geqslant 1,\ 1 \leqslant i \leqslant n$$

Using Lagrangian function with α_i being the ith Lagrange multiplier

$$\frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^{n} \alpha_i\left[1 - \left(\mathbf{w}^T\mathbf{x}_i\right)y_i\right]$$

Since the objective function is convex and since the inequality constraints involve convex functions, the entire problem is a convex optimization problem.

Primal Problem

$$\min_{\mathbf{w}} \max_{\alpha \geqslant 0} \frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^{n} \alpha_i\left[1 - \left(\mathbf{w}^T\mathbf{x}_i\right)y_i\right]$$

Optimizing variable is $\mathbf{w}$

Dual Problem

$$\max_{\alpha \geqslant 0} \min_{\mathbf{w}} \frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^{n} \alpha_i\left[1 - \left(\mathbf{w}^T\mathbf{x}_i\right)y_i\right]$$

Optimizing variable is α

Formulating the Dual

Formulating the Dual

$$\max_{\alpha \geqslant 0} \min_{\mathbf{w}} \frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^{n} \alpha_i \Big[1 - \big(\mathbf{w}^T \mathbf{x}_i\big) y_i\Big]$$

Formulating the Dual

$$\max_{\alpha \geqslant 0} \boxed{\min_{\mathbf{w}} \frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^{n} \alpha_i \left[1 - \left(\mathbf{w}^T \mathbf{x}_i\right) y_i\right]}$$

$$\boxed{\mathbf{w} = \sum_{i=1}^{n} \alpha_i \mathbf{x}_i y_i}$$

Formulating the Dual

$$\max_{\alpha \geqslant 0} \boxed{\min_{\mathbf{w}} \frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^{n} \alpha_i \left[1 - \left(\mathbf{w}^T \mathbf{x}_i\right) y_i\right]} \qquad \boxed{\mathbf{w} = \sum_{i=1}^{n} \alpha_i \mathbf{x}_i y_i}$$

In matrix notation $\quad \mathbf{w} = \mathbf{X}\mathbf{Y}\boldsymbol{\alpha} \quad \mathbf{X} = \begin{bmatrix} | & & | \\ \mathbf{x}_1 & \cdots & \mathbf{x}_n \\ | & & | \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} y_1 & & \\ & \ddots & \\ & & y_n \end{bmatrix} \quad \boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}$

Formulating the Dual

$$\max_{\alpha \geqslant 0} \boxed{\min_{\mathbf{w}} \frac{\|\mathbf{w}\|^2}{2} + \sum_{i=1}^{n} \alpha_i \left[1 - \left(\mathbf{w}^T \mathbf{x}_i\right) y_i\right]} \qquad \boxed{\mathbf{w} = \sum_{i=1}^{n} \alpha_i \mathbf{x}_i y_i}$$

In matrix notation $\quad \mathbf{w} = \mathbf{X}\mathbf{Y}\boldsymbol{\alpha} \quad \mathbf{X} = \begin{bmatrix} | & & | \\ \mathbf{x}_1 & \cdots & \mathbf{x}_n \\ | & & | \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} y_1 & & \\ & \ddots & \\ & & y_n \end{bmatrix} \quad \boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}$

Plugging $\mathbf{w} = \mathbf{X}\mathbf{Y}\boldsymbol{\alpha}$ back into the dual objective, we get the following form:

$$\boxed{\max_{\alpha \geqslant 0} \boldsymbol{\alpha}^T \mathbf{1} - \frac{1}{2} \boldsymbol{\alpha}^T \left(\mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y}\right) \boldsymbol{\alpha}}$$

Advantages of the Dual

$$\max_{\alpha \geqslant 0} \ \boldsymbol{\alpha}^T \mathbf{1} - \frac{1}{2} \boldsymbol{\alpha}^T \left(\mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y} \right) \boldsymbol{\alpha}$$

(1) Simpler constraints, just bounds.

(2) The appearance of $\mathbf{X}^T\mathbf{X}$ points to kernels.

Support Vectors

$$\boldsymbol{\alpha}^* = \begin{bmatrix} \alpha_1^* \\ \vdots \\ \alpha_n^* \end{bmatrix} \qquad \mathbf{w}^* = \sum_{i=1}^{n} \alpha_i \mathbf{x}_i y_i$$

Support Vectors

$$\boldsymbol{\alpha}^* = \begin{bmatrix} \alpha_1^* \\ \vdots \\ \alpha_n^* \end{bmatrix} \qquad \mathbf{w}^* = \sum_{i=1}^{n} \alpha_i \mathbf{x}_i y_i$$

Definition: The points for which $\alpha^* > 0$ are called support vectors.

Support Vectors

$$\boldsymbol{\alpha}^* = \begin{bmatrix} \alpha_1^* \\ \vdots \\ \alpha_n^* \end{bmatrix} \qquad \mathbf{w}^* = \sum_{i=1}^{n} \alpha_i \mathbf{x}_i y_i$$

Definition: The points for which $\alpha^* > 0$ are called support vectors.

$$\alpha_i^* \left[1 - \left(\left(\mathbf{w}^* \right)^T \mathbf{x}_i \right) y_i \right] = 0$$

Complementary Slackness

Support Vectors

$$\boldsymbol{\alpha}^* = \begin{bmatrix} \alpha_1^* \\ \vdots \\ \alpha_n^* \end{bmatrix} \qquad \mathbf{w}^* = \sum_{i=1}^{n} \alpha_i \mathbf{x}_i y_i$$

Definition: The points for which $\alpha^* > 0$ are called support vectors.

$$\alpha_i^* \left[1 - \left(\left(\mathbf{w}^* \right)^T \mathbf{x}_i \right) y_i \right] = 0$$

Complementary Slackness

Every support vector lies on one of the two supporting hyperplanes $\left(\mathbf{w}^* \right)^T \mathbf{x}_i = \pm 1$

Every point that is **not** on one of the two supporting hyperplanes has $\alpha_i^* = 0$

Support Vectors

$$\boldsymbol{\alpha}^* = \begin{bmatrix} \alpha_1^* \\ \vdots \\ \alpha_n^* \end{bmatrix} \qquad \mathbf{w}^* = \sum_{i=1}^{n} \alpha_i \mathbf{x}_i y_i$$

Definition: The points for which $\alpha^* > 0$ are called support vectors.

$$\alpha_i^* \left[1 - \left(\left(\mathbf{w}^* \right)^T \mathbf{x}_i \right) y_i \right] = 0$$

Complementary Slackness

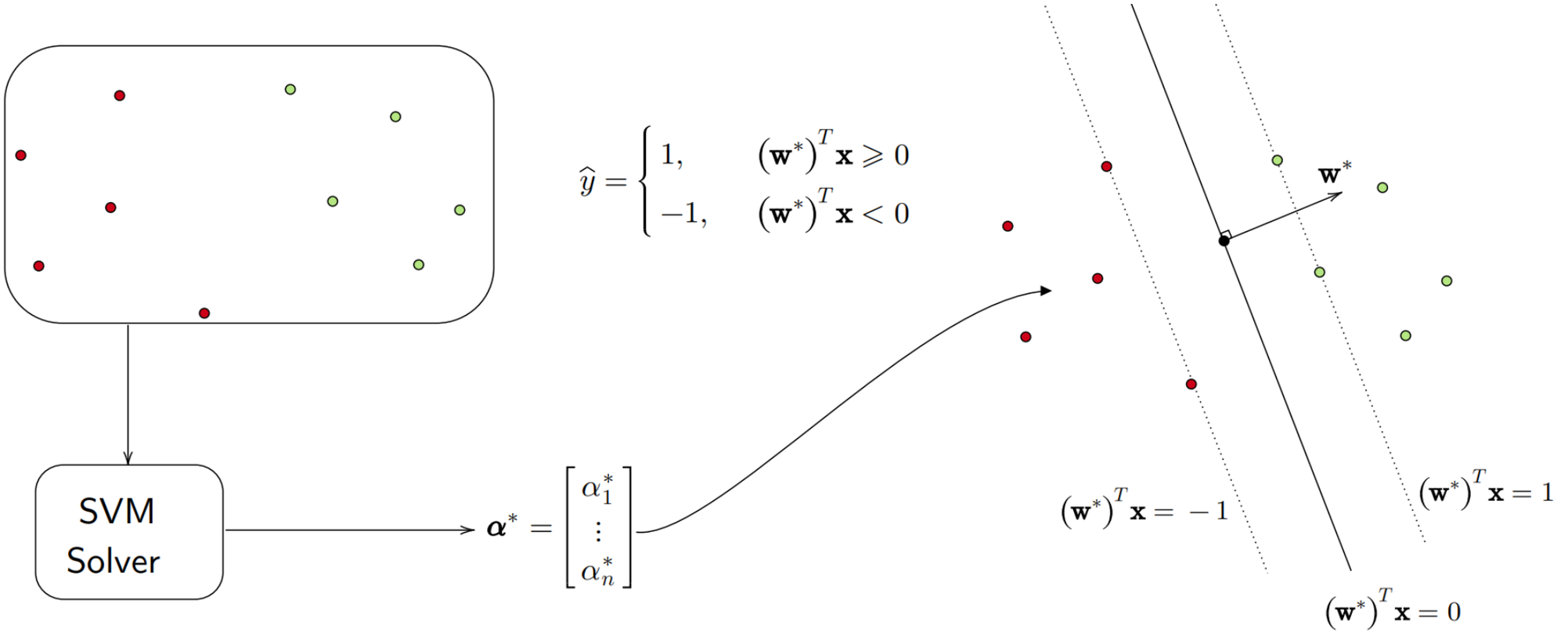
Every support vector lies on one of the two supporting hyperplanes $\left(\mathbf{w}^* \right)^T \mathbf{x}_i = \pm 1$

Is every point on one of the two supporting hyperplanes a support vector?

Every point that is **not** on one of the two supporting hyperplanes has $\alpha_i^* = 0$

The schematic diagram

The schematic diagram that summarizes the discussion so far:

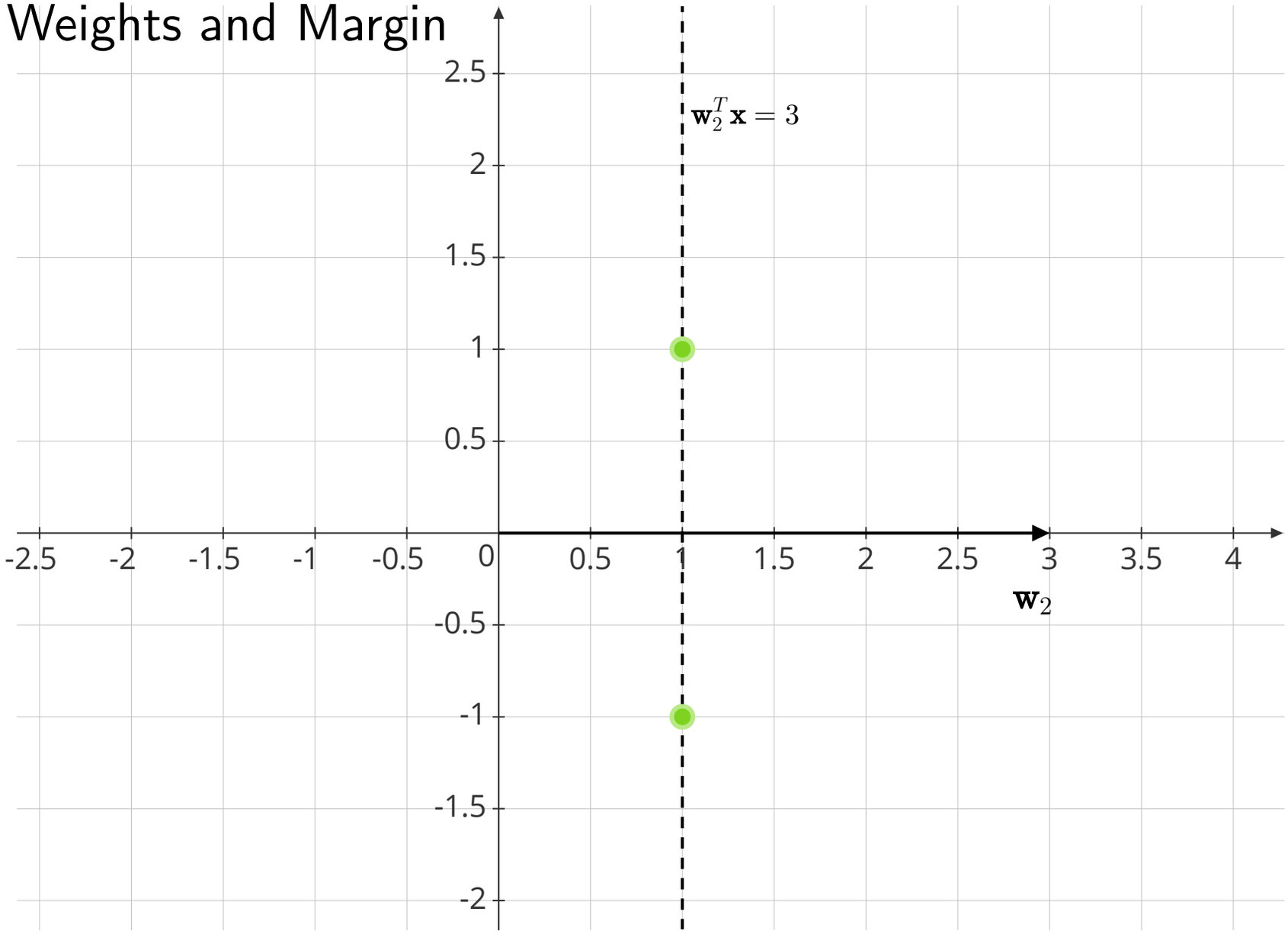


MLT: Week-10

Support Vector Machine
(practice)

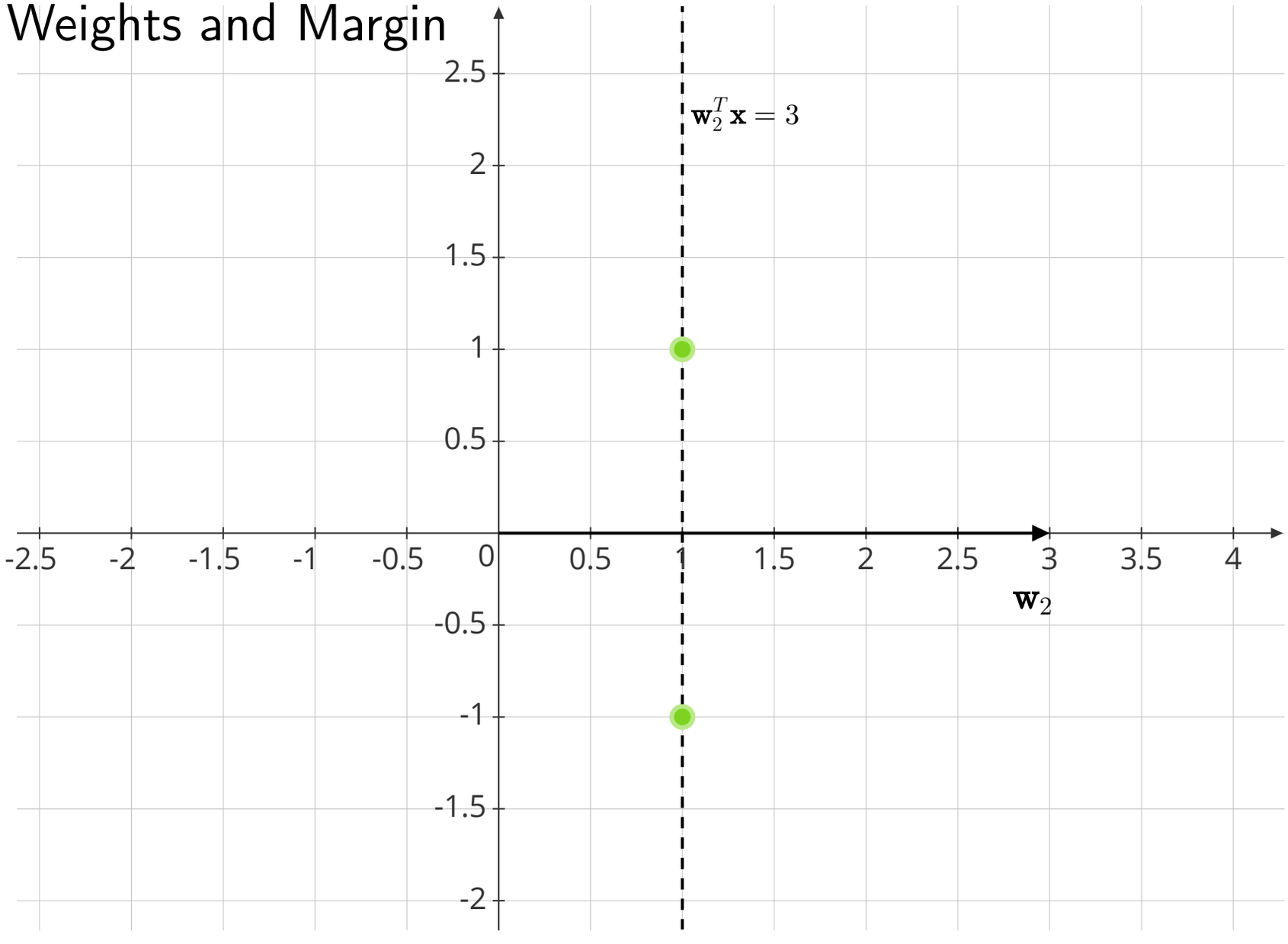
Pulkit Mangal

Weights and Margin



$$\mathbf{w}_2 = \begin{bmatrix} 3 & 0 \end{bmatrix}^T$$

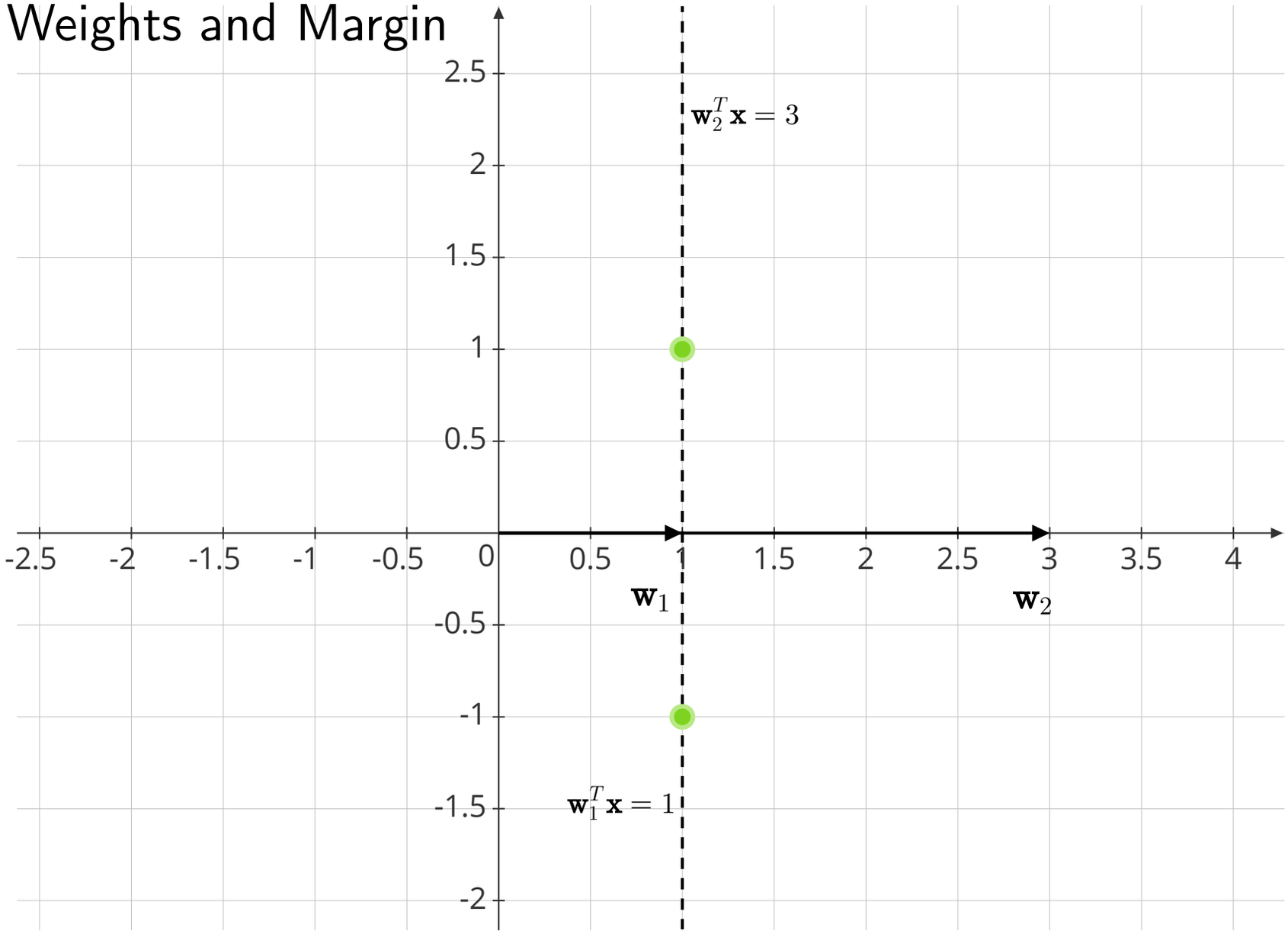
Weights and Margin



$$\mathbf{w}_2 = \begin{bmatrix} 3 & 0 \end{bmatrix}^T$$

$$\mathbf{w}_2^T\mathbf{x} = 3 \Longrightarrow \frac{1}{3}\left(\mathbf{w}_2^T\mathbf{x}\right) = \frac{1}{3} \cdot 3$$

Weights and Margin



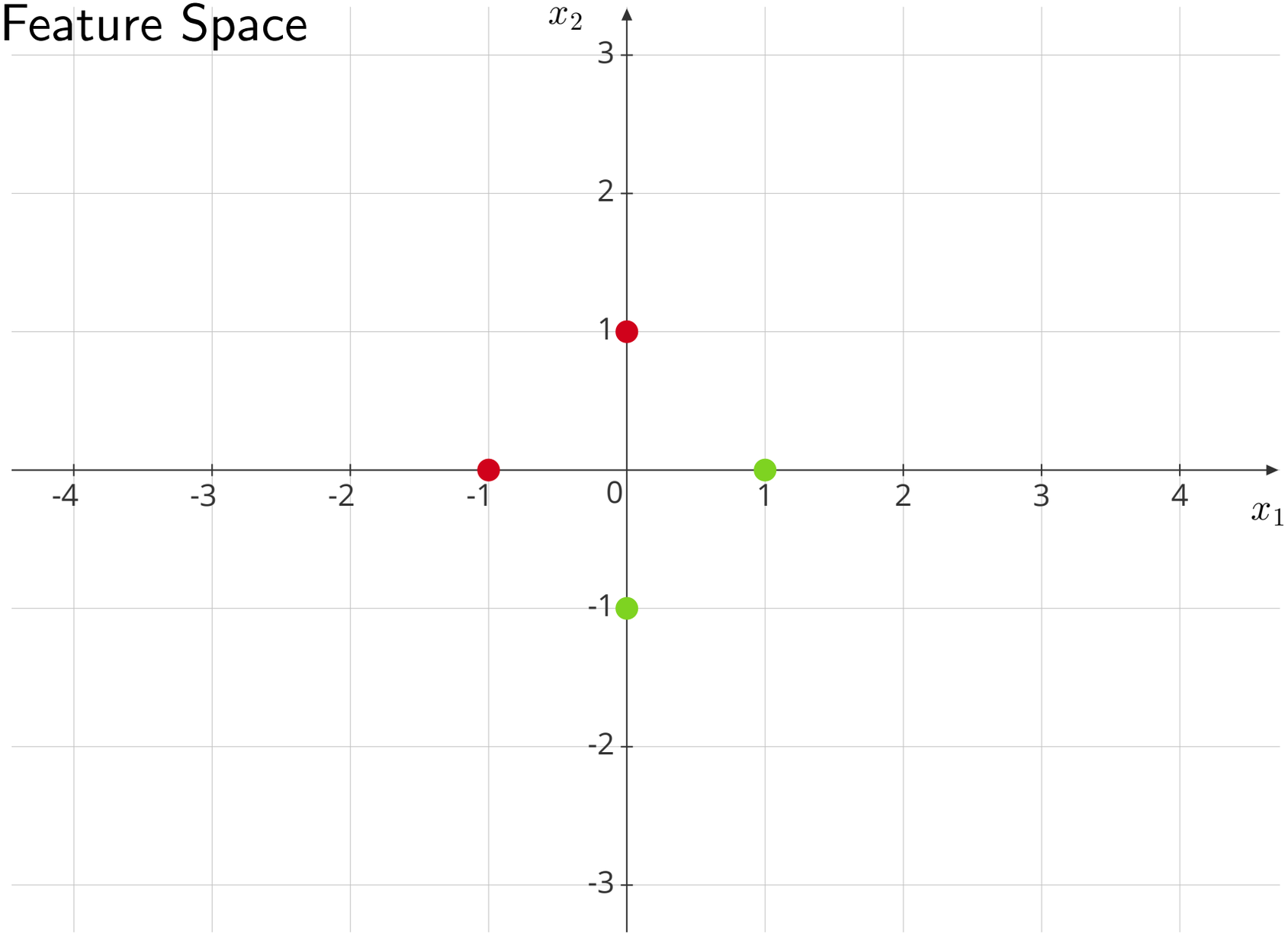
$$\mathbf{w}_2 = \begin{bmatrix} 3 & 0 \end{bmatrix}^T$$

$$\mathbf{w}_2^T\mathbf{x} = 3 \Longrightarrow \frac{1}{3}\left(\mathbf{w}_2^T\mathbf{x}\right) = \frac{1}{3}\cdot 3$$

$$\Longrightarrow \left(\frac{\mathbf{w}_2}{3}\right)^T \mathbf{x} = 1 \Longrightarrow \mathbf{w}_1^T\mathbf{x} = 1$$

$$\frac{\mathbf{w}_2}{3} = \begin{bmatrix} 1 & 0 \end{bmatrix}^T = \mathbf{w}_1$$

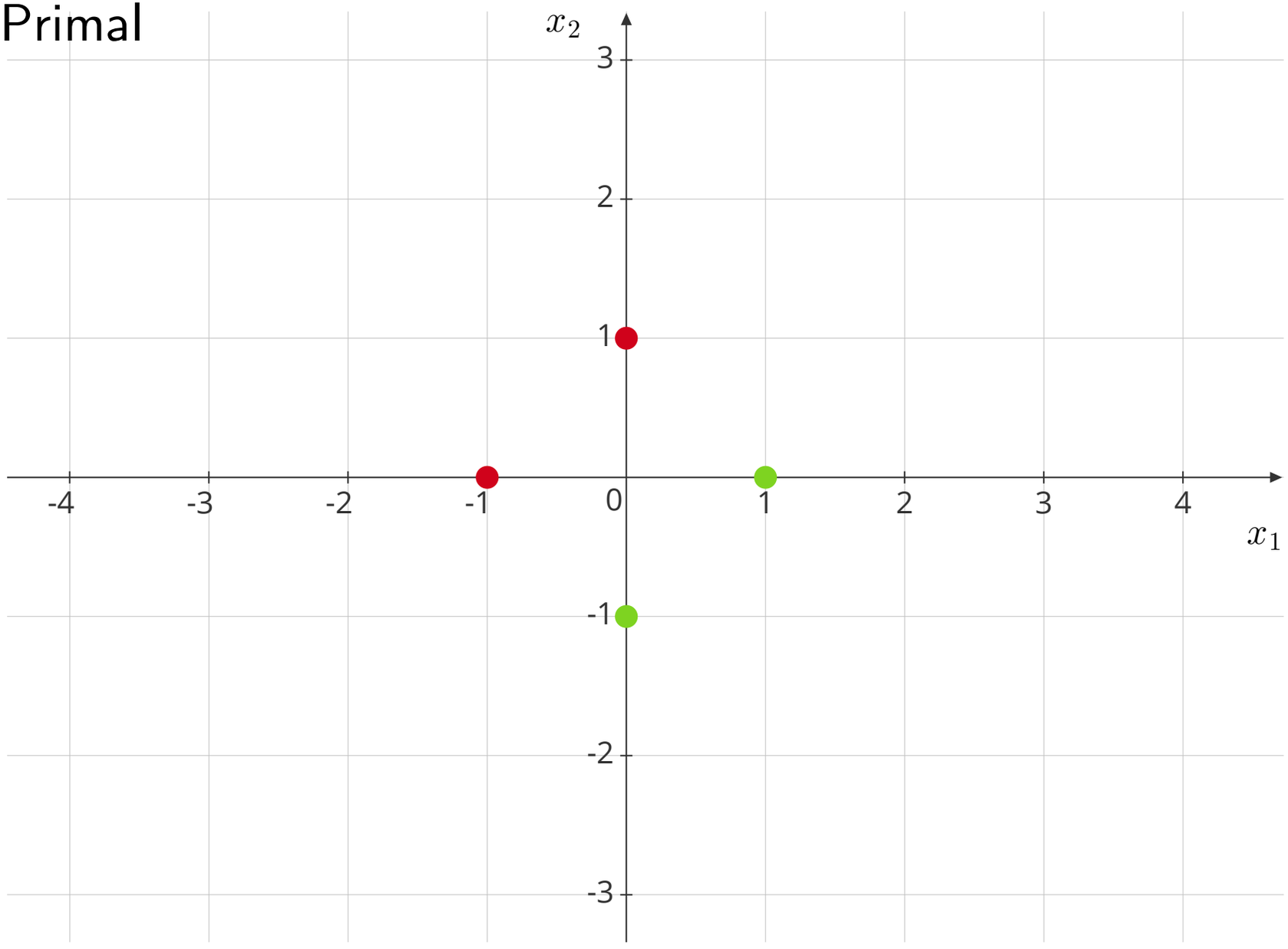
Feature Space



Example-1

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

Primal



Example-1

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\min_{\mathbf{w}} \quad \frac{||\mathbf{w}||^2}{2}$$

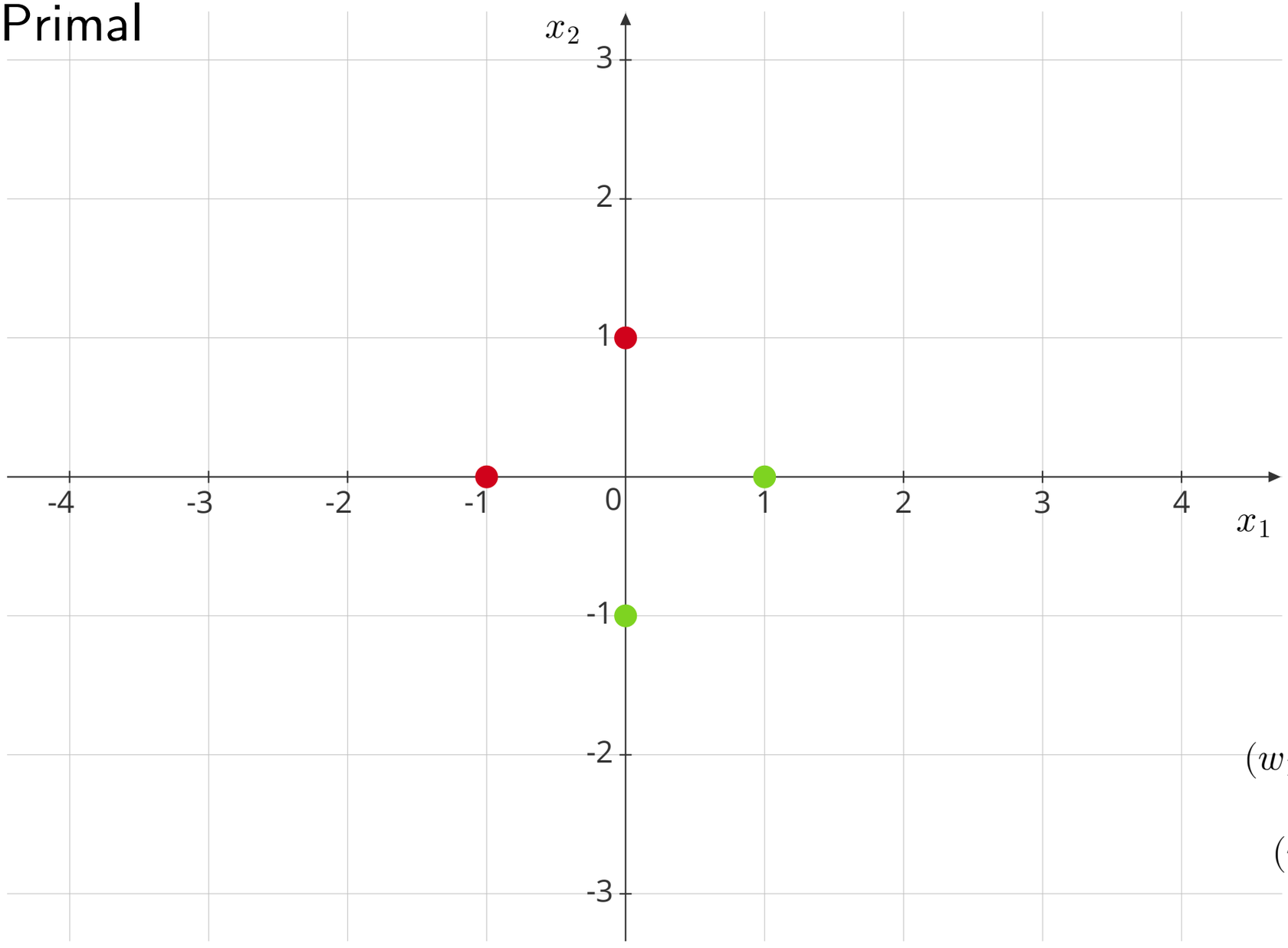
$$\left(\mathbf{w}^T \mathbf{x}_1\right) y_1 \geqslant 1 \quad (1)$$

$$\left(\mathbf{w}^T \mathbf{x}_2\right) y_2 \geqslant 1 \quad (2)$$

$$\left(\mathbf{w}^T \mathbf{x}_3\right) y_3 \geqslant 1 \quad (3)$$

$$\left(\mathbf{w}^T \mathbf{x}_4\right) y_4 \geqslant 1 \quad (4)$$

Primal



Example-1

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

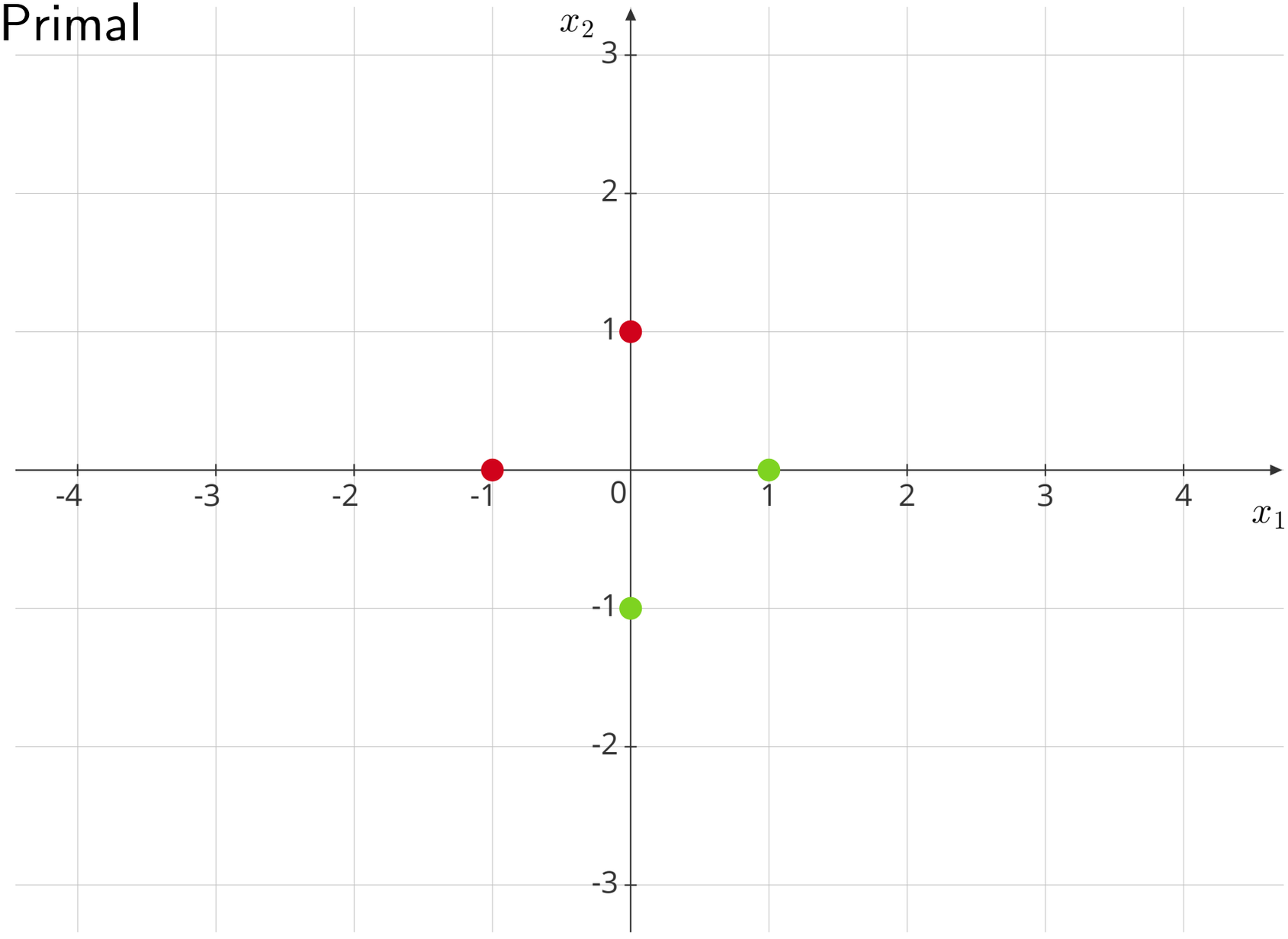
$$(w_1 \times 1 + w_2 \times 0) \cdot 1 \geqslant 1 \quad (1)$$

$$(w_1 \times 0 + w_2 \times -1) \cdot 1 \geqslant 1 \quad (2)$$

$$(w_1 \times (-1) + w_2 \times 0) \cdot (-1) \geqslant 1 \quad (3)$$

$$(w_1 \times (0) + w_2 \times 1) \cdot (-1) \geqslant 1 \quad (4)$$

Primal



Example-1

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

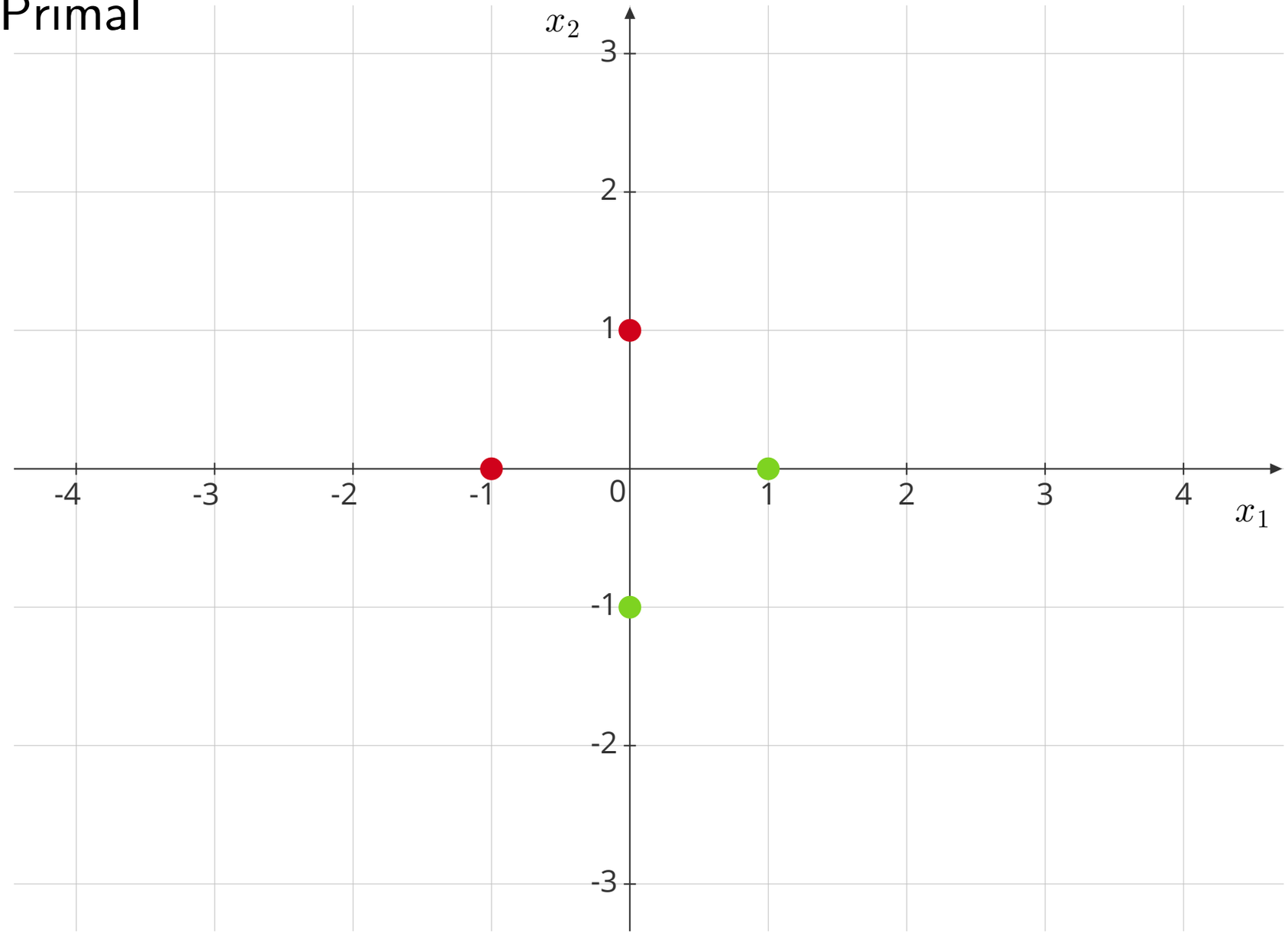
$$w_1 \geqslant 1 \qquad (1)$$

$$-w_2 \geqslant 1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$-w_2 \geqslant 1 \qquad (4)$$

Primal



Example-1

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

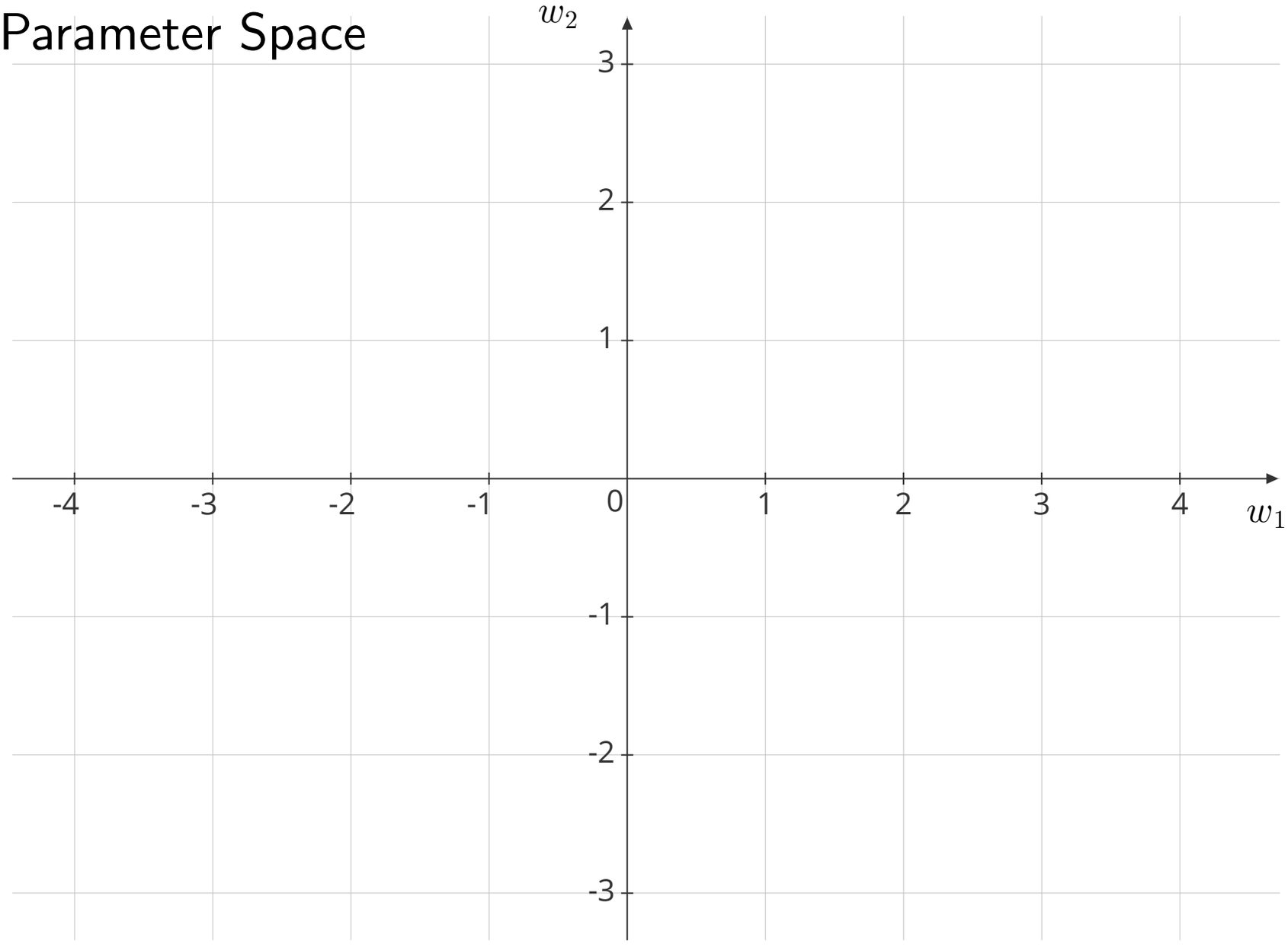
$$w_1 \geqslant 1 \qquad (1)$$

$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

Parameter Space



Example-1

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

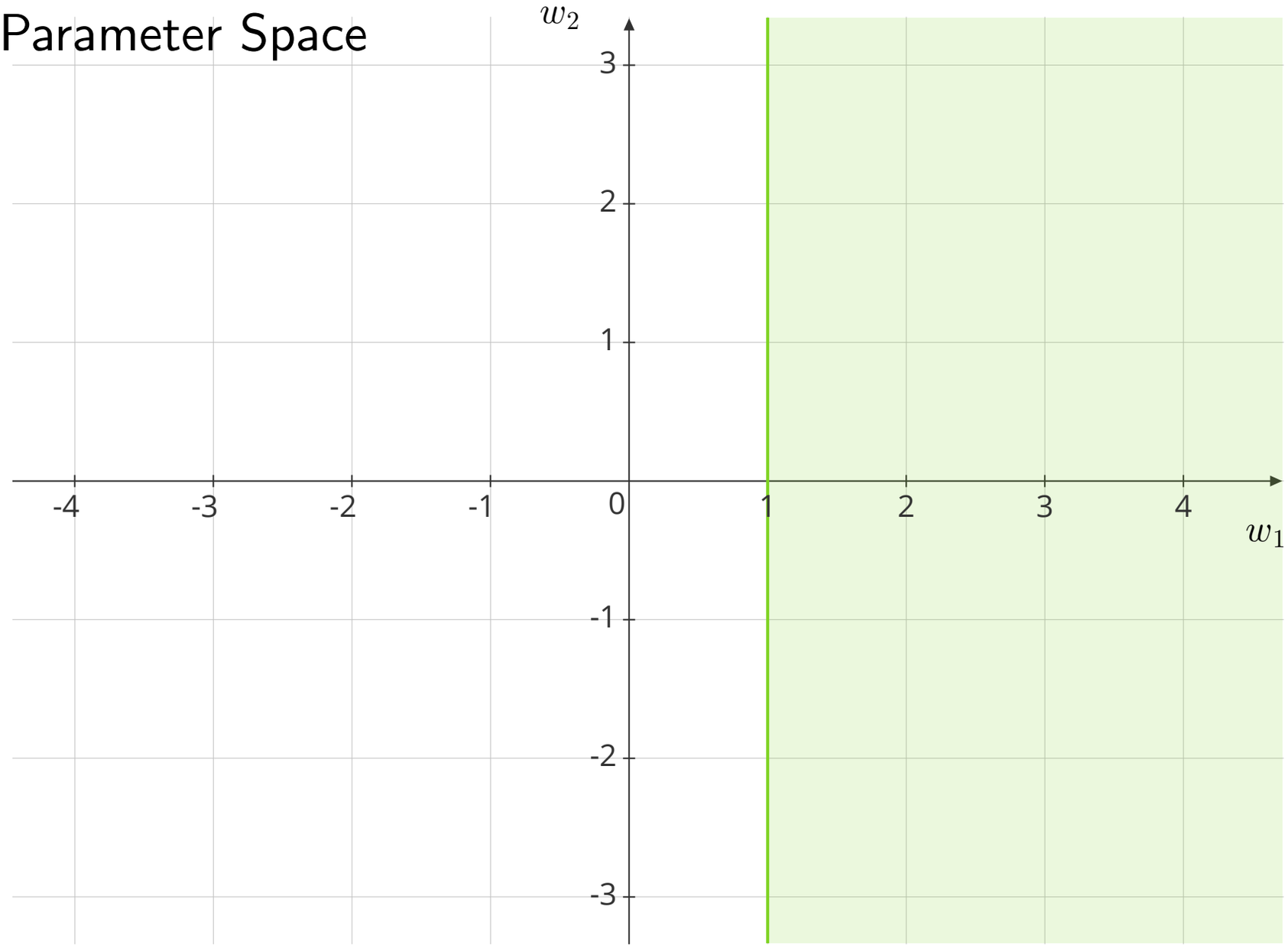
$$w_1 \geqslant 1 \qquad (1)$$

$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

Parameter Space



Example-1

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

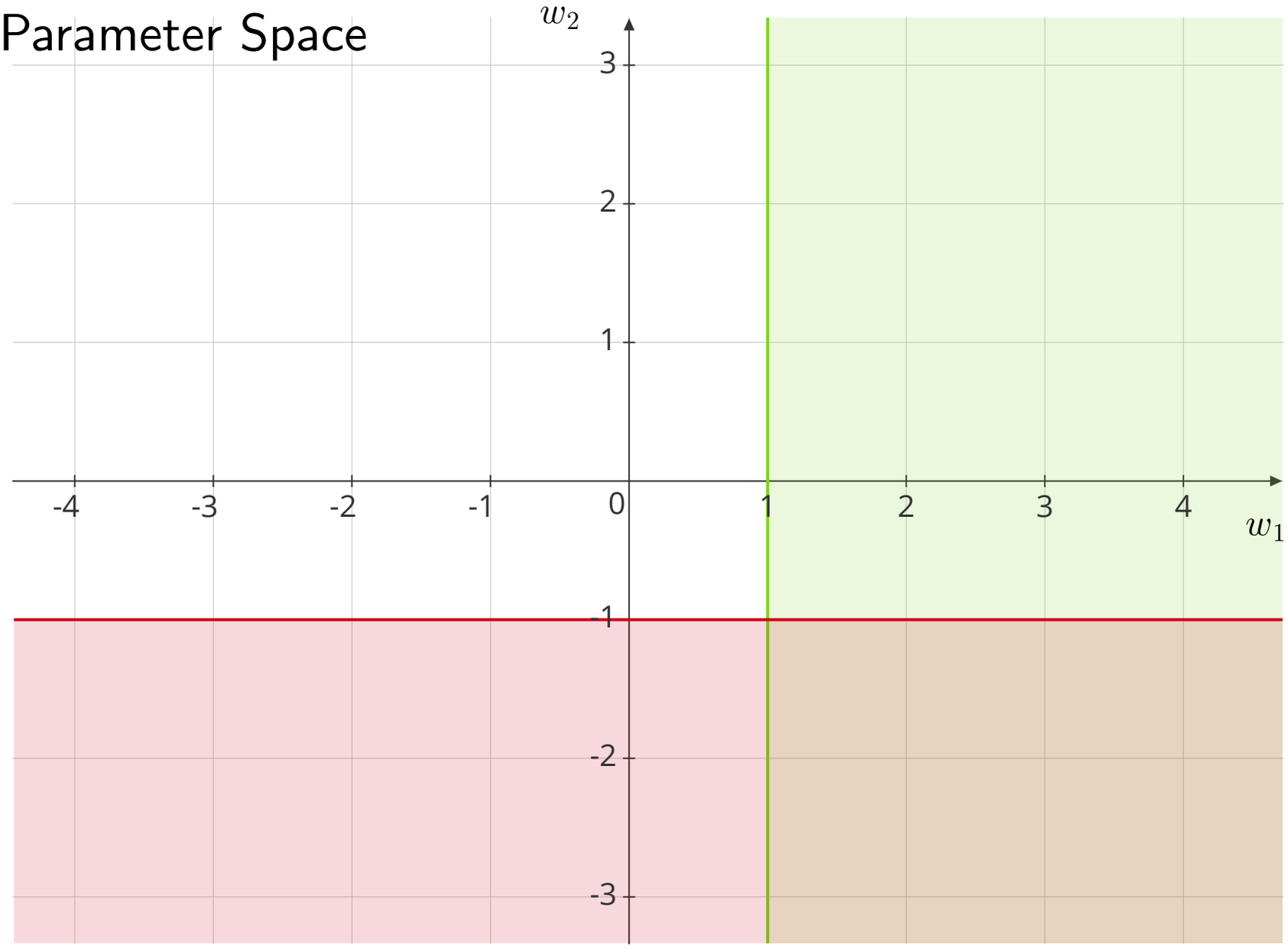
$$w_1 \geqslant 1 \qquad (1)$$

$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

Parameter Space



Example-1

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

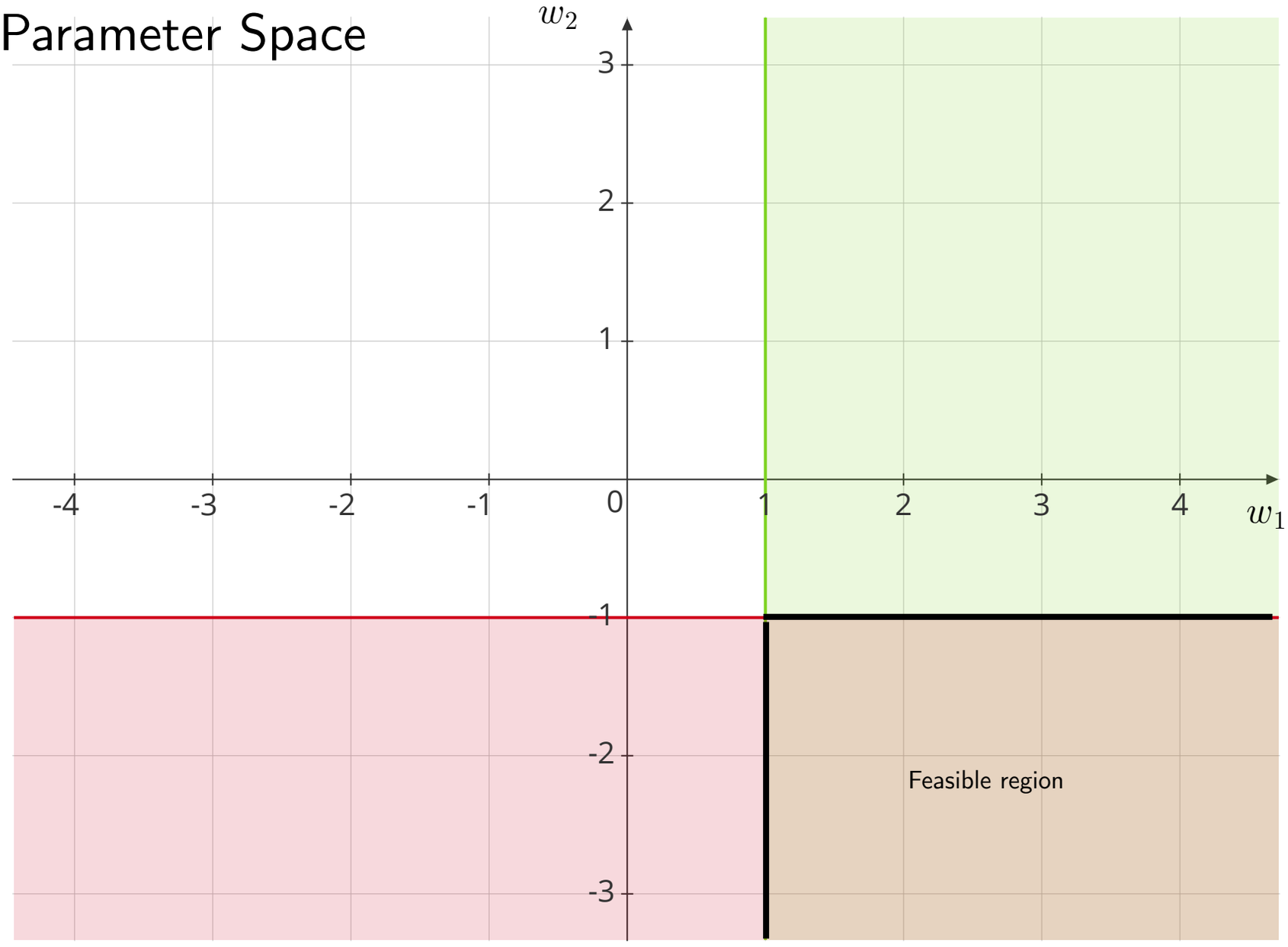
$$w_1 \geqslant 1 \qquad (1)$$

$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

Parameter Space



Example-1

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$w_1 \geqslant 1 \qquad (1)$$

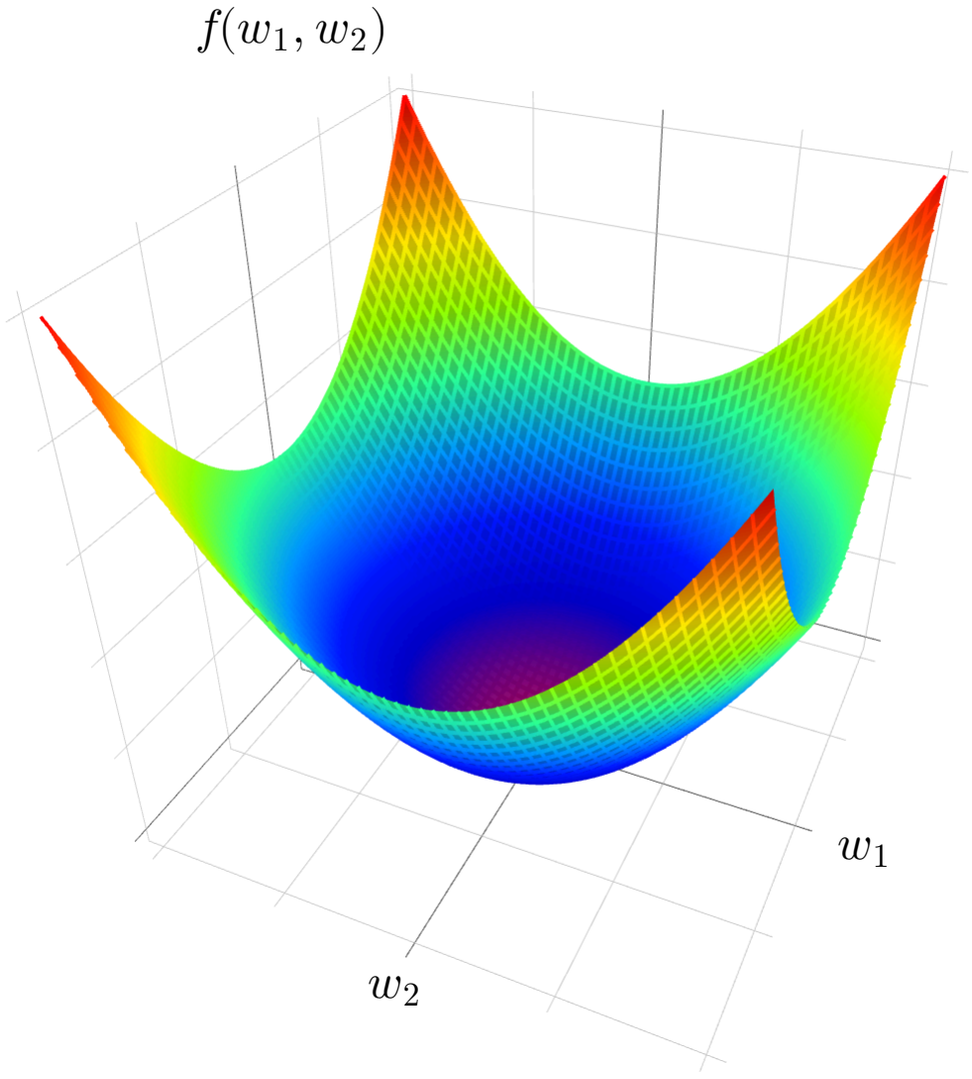
$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

Parameter Space - Objective Surface

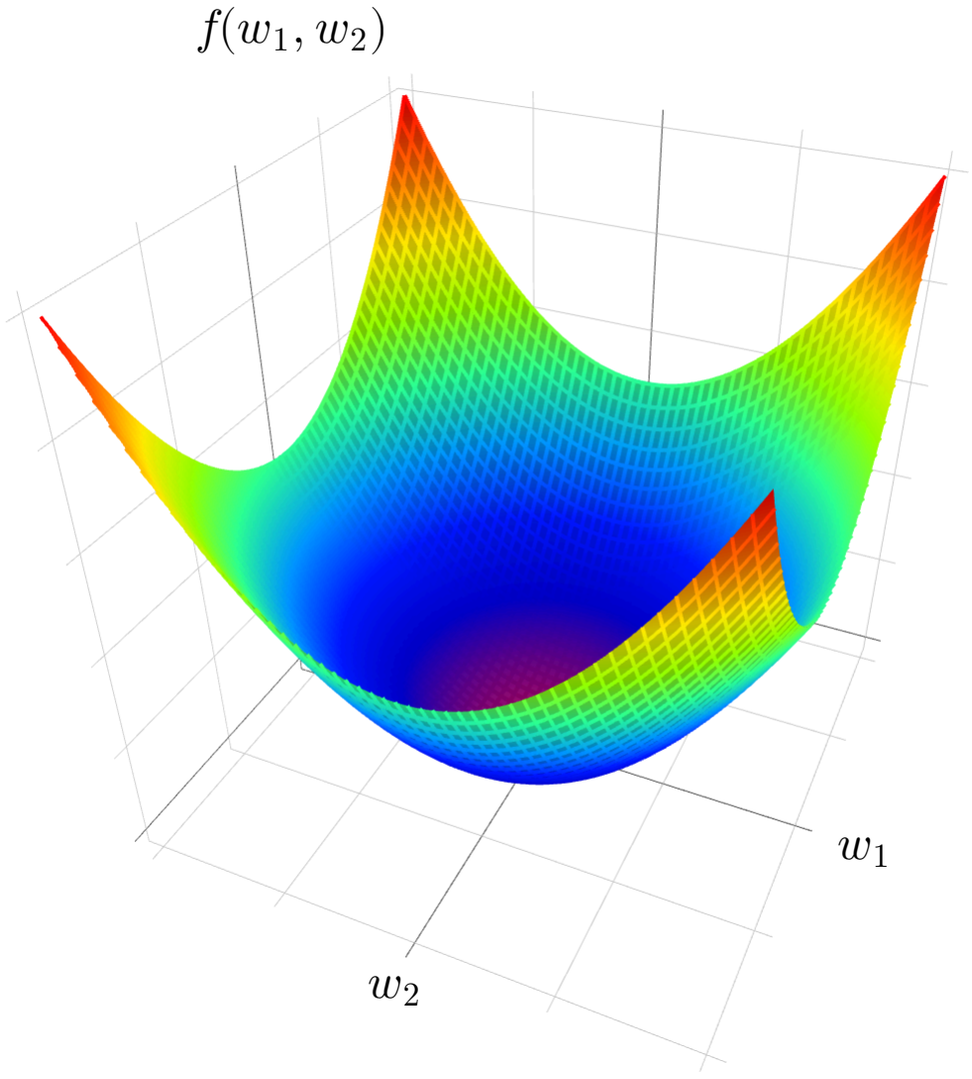
Example-1



$$f(w_1, w_2) = \frac{||\mathbf{w}||^2}{2} = \frac{w_1^2 + w_2^2}{2}$$

Parameter Space - Objective Surface

Example-1

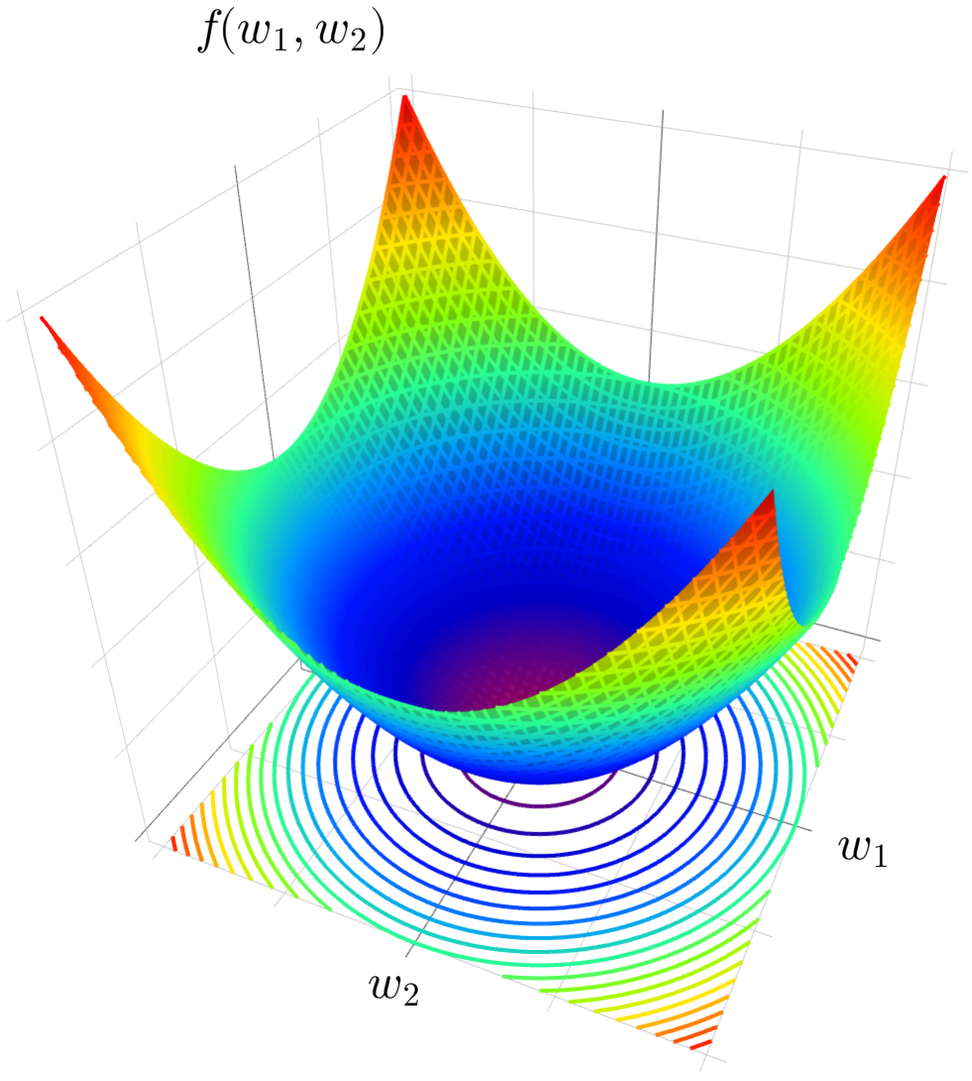


$$f(w_1, w_2) = \frac{||\mathbf{w}||^2}{2} = \frac{w_1^2 + w_2^2}{2}$$

$$f(w_1, w_2) = 0.5 \implies w_1^2 + w_2^2 = 1$$

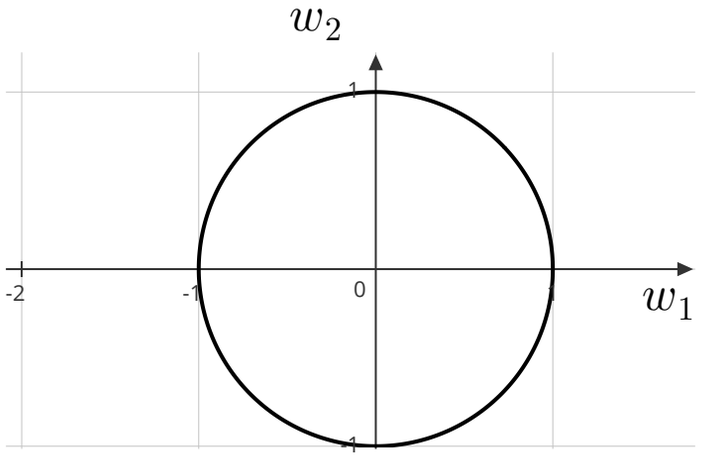
Parameter Space - Objective Surface

Example-1



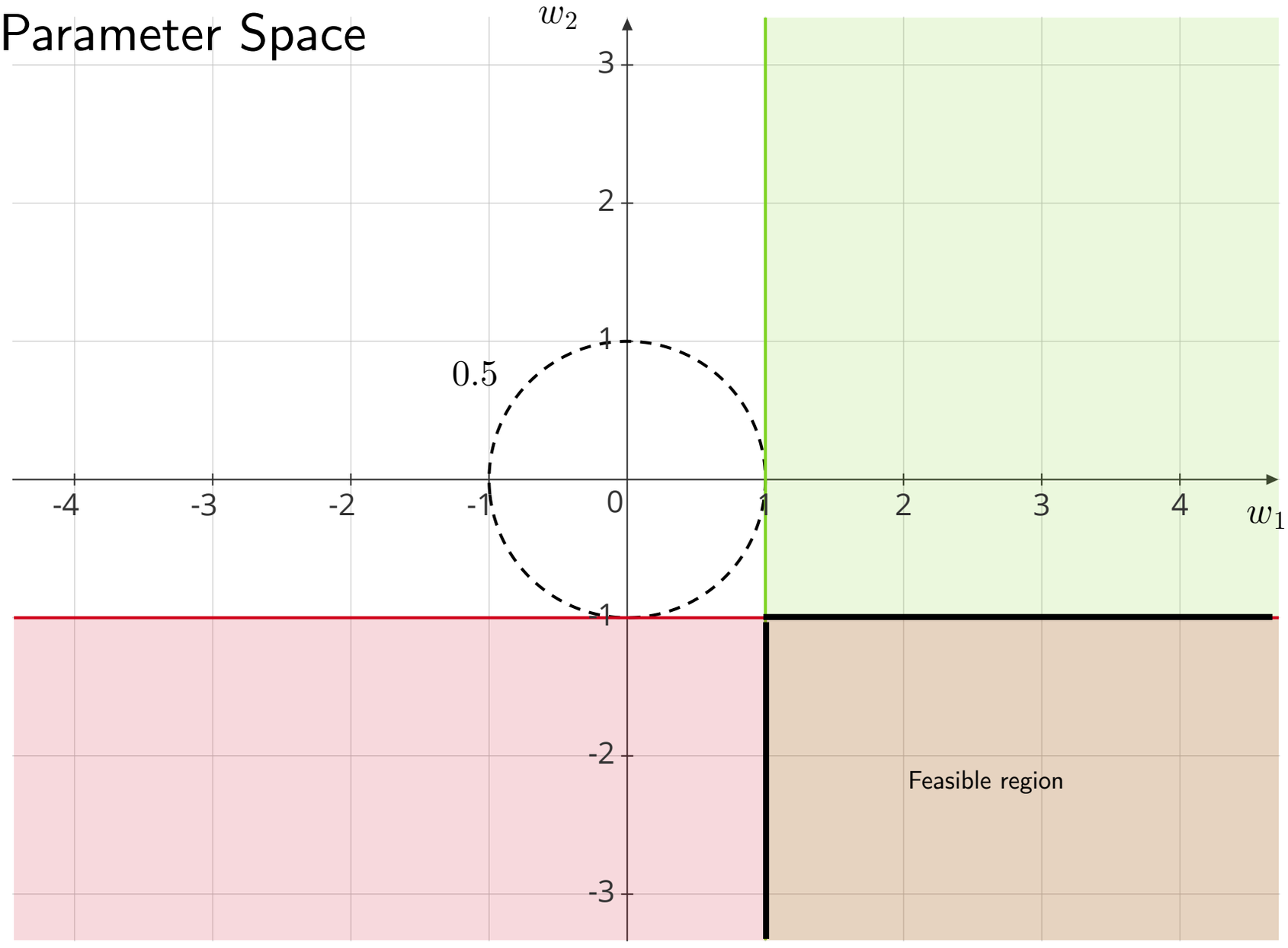
$$f(w_1, w_2) = \frac{||\mathbf{w}||^2}{2} = \frac{w_1^2 + w_2^2}{2}$$

$$f(w_1, w_2) = 0.5 \implies w_1^2 + w_2^2 = 1$$



$$w_1^2 + w_2^2 = 1$$

Parameter Space



Example-1

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

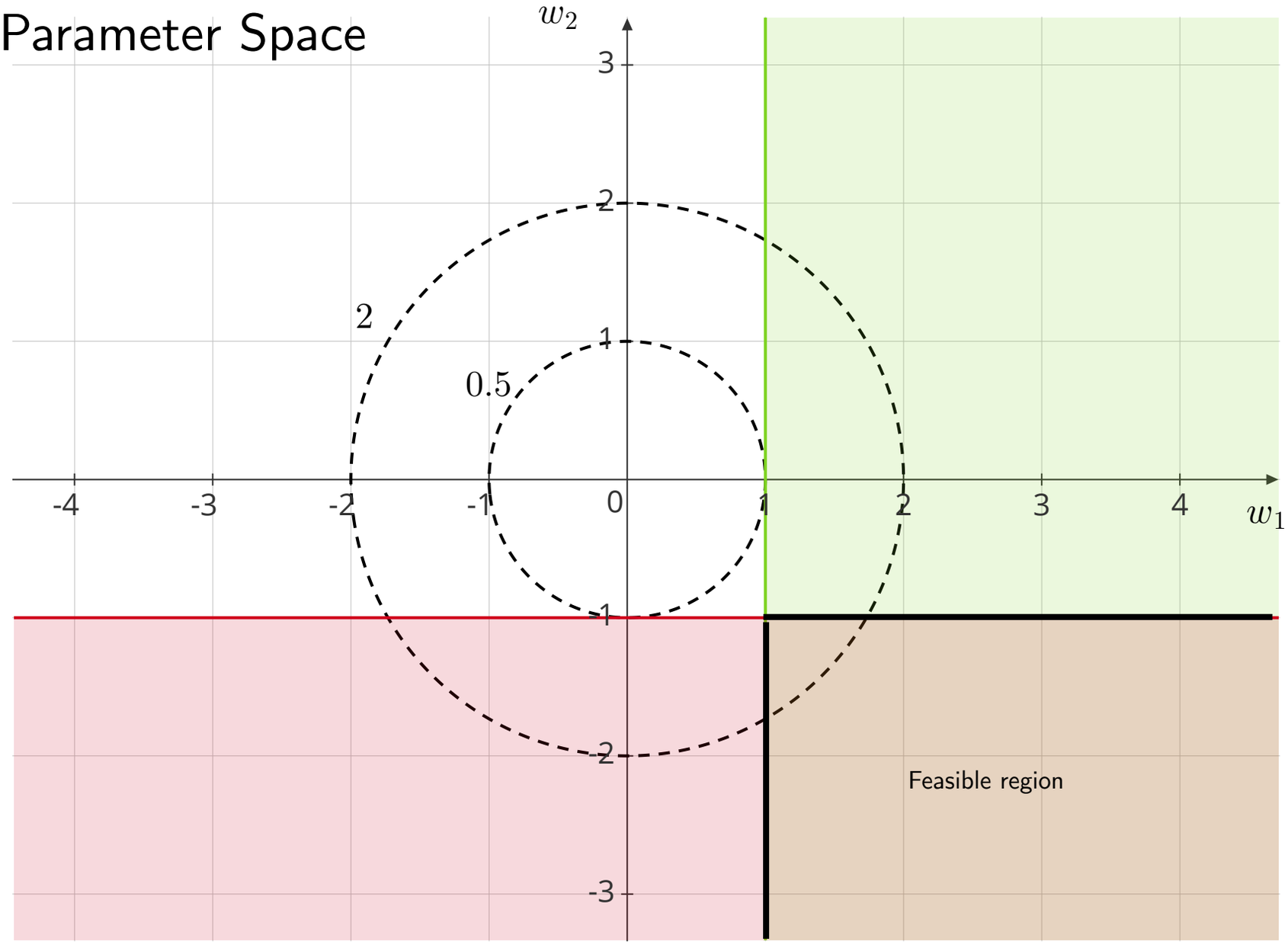
$$w_1 \geqslant 1 \qquad (1)$$

$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

Parameter Space



Example-1

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

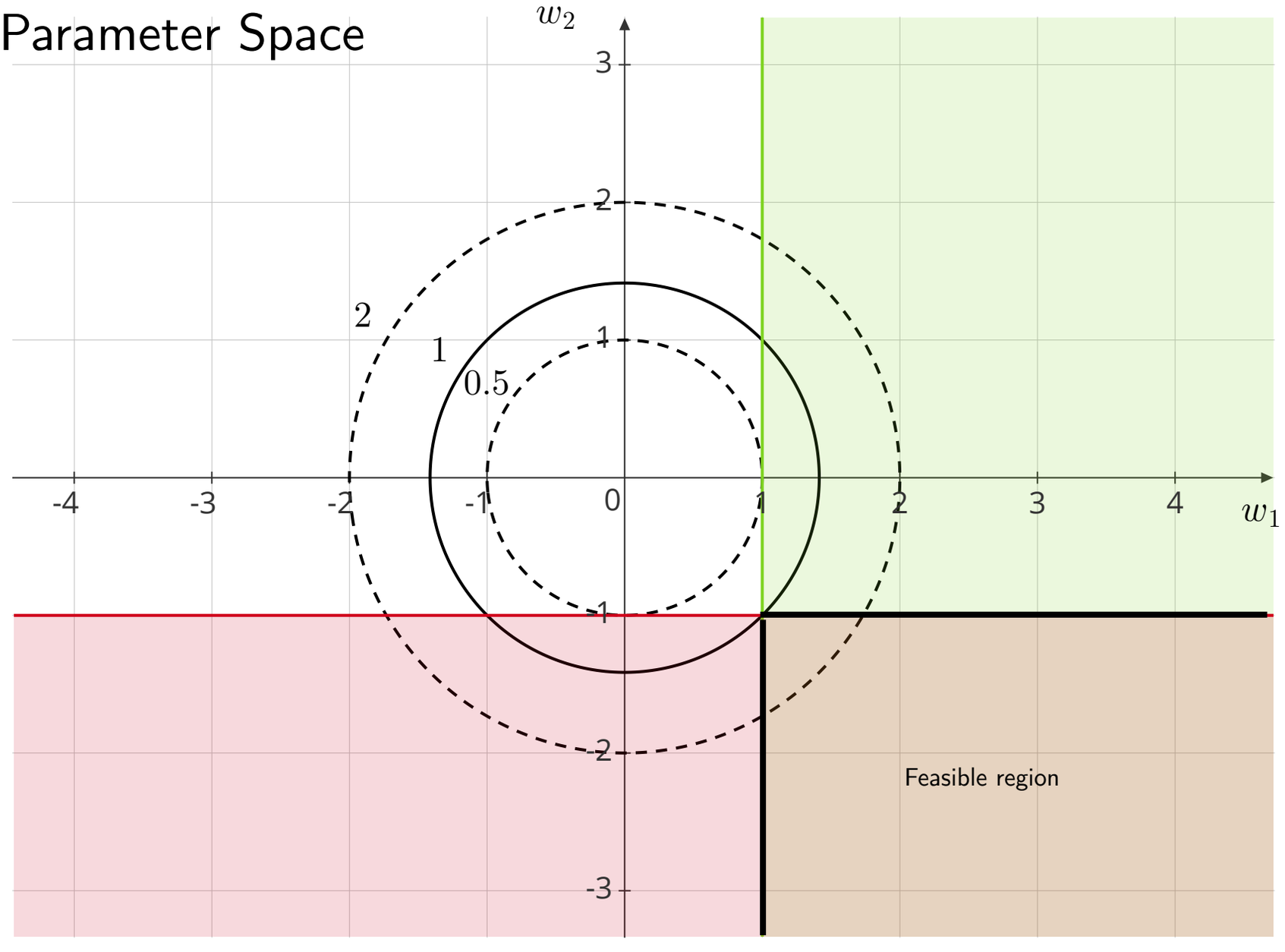
$$w_1 \geqslant 1 \qquad (1)$$

$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

Parameter Space



Example-1

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

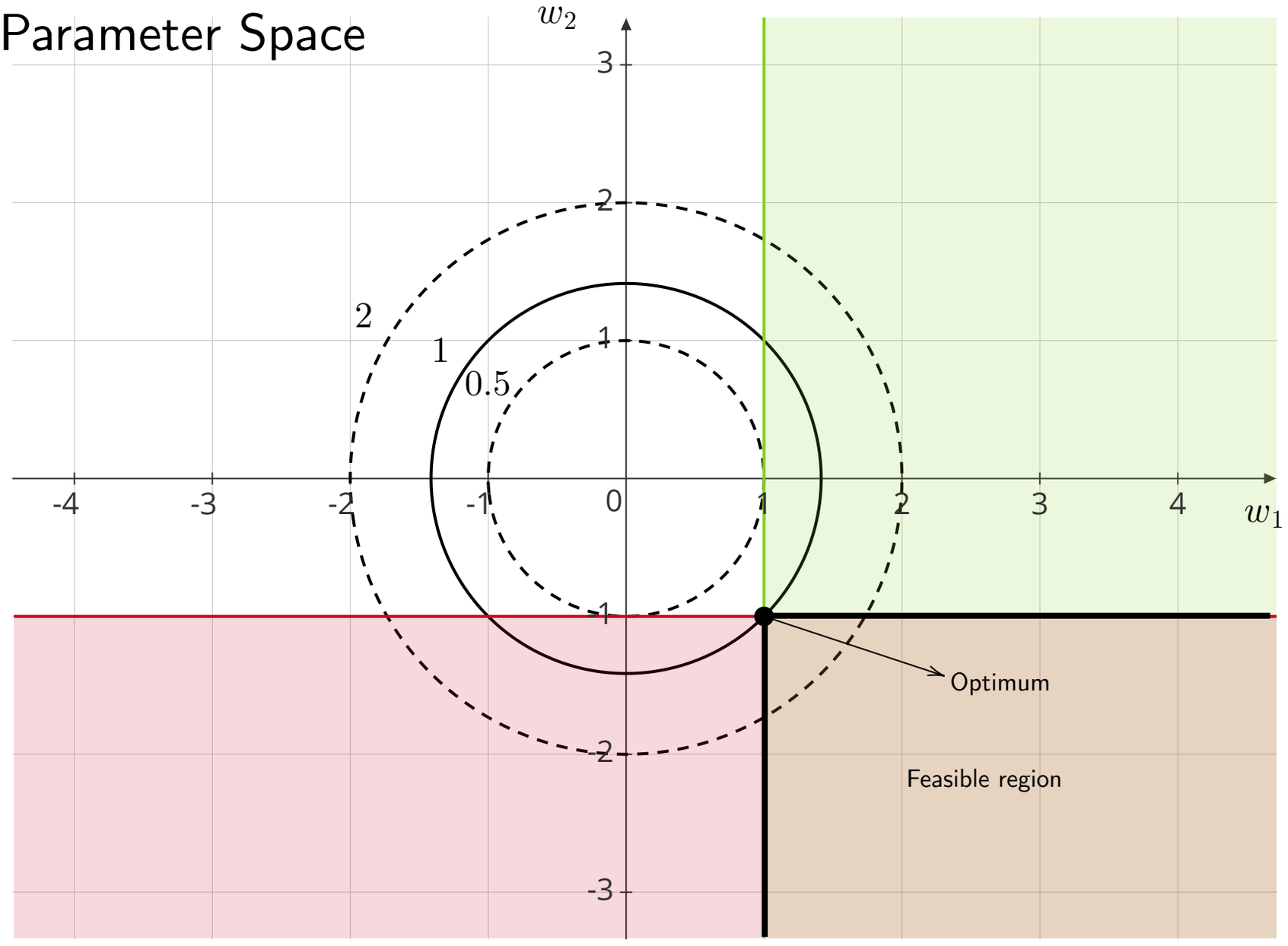
$$w_1 \geqslant 1 \qquad (1)$$

$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

Parameter Space



Example-1

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$w_1 \geqslant 1 \qquad (1)$$

$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

Primal $\rightarrow$ Dual

Example-1

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$w_1 \geqslant 1 \qquad (1)$$

$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

Primal $\rightarrow$ Dual

Example-1

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$w_1 \geqslant 1 \qquad (1)$$

$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

$$\max_{\boldsymbol{\alpha} \geqslant 0} \; \boldsymbol{\alpha}^T \mathbf{1} - \frac{1}{2} \cdot \boldsymbol{\alpha}^T \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y} \boldsymbol{\alpha}$$

$$\boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix}$$

Primal $\rightarrow$ Dual

Example-1

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$w_1 \geqslant 1 \qquad (1)$$

$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

$$\max_{\boldsymbol{\alpha} \geqslant 0} \; \boldsymbol{\alpha}^T \mathbf{1} - \frac{1}{2} \cdot \boldsymbol{\alpha}^T \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y} \boldsymbol{\alpha}$$

$$\boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix}$$

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\mathbf{Y} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

Primal $\rightarrow$ Dual

Example-1

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$w_1 \geqslant 1 \qquad (1)$$

$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

$$\max_{\boldsymbol{\alpha} \geqslant 0} \; \boldsymbol{\alpha}^T \mathbf{1} - \frac{1}{2} \cdot \boldsymbol{\alpha}^T \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y} \boldsymbol{\alpha}$$

$$\boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix}$$

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\mathbf{Y} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

$$\mathbf{Q} = \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y}$$

Primal $\rightarrow$ Dual

Example-1

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$w_1 \geqslant 1 \qquad (1)$$

$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

$$\max_{\boldsymbol{\alpha} \geqslant 0} \; \boldsymbol{\alpha}^T \mathbf{1} - \frac{1}{2} \cdot \boldsymbol{\alpha}^T \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y} \boldsymbol{\alpha}$$

$$\boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix}$$

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\mathbf{Y} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

$$\mathbf{Q} = \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y}$$

$$\mathbf{Q} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \\ -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

Primal $\rightarrow$ Dual

Example-1

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$w_1 \geqslant 1 \qquad (1)$$

$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

$$\max_{\boldsymbol{\alpha} \geqslant 0} \; \boldsymbol{\alpha}^T \mathbf{1} - \frac{1}{2} \cdot \boldsymbol{\alpha}^T \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y} \boldsymbol{\alpha}$$

$$\boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix}$$

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\mathbf{Y} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

$$\mathbf{Q} = \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y}$$

$$\mathbf{Q} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \\ -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Primal $\rightarrow$ Dual

Example-1

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$w_1 \geqslant 1 \qquad (1)$$

$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

$$\max_{\boldsymbol{\alpha} \geqslant 0} \; \boldsymbol{\alpha}^T \mathbf{1} - \frac{1}{2} \cdot \boldsymbol{\alpha}^T \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y} \boldsymbol{\alpha}$$

$$\boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix}$$

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\mathbf{Y} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

$$\mathbf{Q} = \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y}$$

$$\mathbf{Q} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \\ -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$\max_{\boldsymbol{\alpha} \geqslant 0} \; (\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4) - \frac{1}{2}\left(\alpha_1^2 + \alpha_2^2 + \alpha_3^2 + \alpha_4^2\right) - (\alpha_1 \alpha_3 + \alpha_2 \alpha_4)$$

Primal $\rightarrow$ Dual

Example-1

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$w_1 \geqslant 1 \qquad (1)$$

$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

$$\max_{\boldsymbol{\alpha} \geqslant 0} \; \boldsymbol{\alpha}^T \mathbf{1} - \frac{1}{2} \cdot \boldsymbol{\alpha}^T \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y} \boldsymbol{\alpha}$$

$$\boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix}$$

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\mathbf{Y} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

$$\mathbf{Q} = \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y}$$

$$\mathbf{Q} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \\ -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$\max_{\boldsymbol{\alpha} \geqslant 0} \; (\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4) - \frac{1}{2}\left(\alpha_1^2 + \alpha_2^2 + \alpha_3^2 + \alpha_4^2\right) - (\alpha_1 \alpha_3 + \alpha_2 \alpha_4)$$

$$\boldsymbol{\alpha}^* = \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \end{bmatrix}^T$$

Primal $\rightarrow$ Dual

Example-1

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$w_1 \geqslant 1 \qquad (1)$$

$$w_2 \leqslant -1 \qquad (2)$$

$$w_1 \geqslant 1 \qquad (3)$$

$$w_2 \leqslant -1 \qquad (4)$$

$$\max_{\boldsymbol{\alpha} \geqslant 0} \; \boldsymbol{\alpha}^T \mathbf{1} - \frac{1}{2} \cdot \boldsymbol{\alpha}^T \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y} \boldsymbol{\alpha}$$

$$\boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix}$$

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\mathbf{Y} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

$$\mathbf{Q} = \mathbf{Y}^T \mathbf{X}^T \mathbf{X} \mathbf{Y}$$

$$\mathbf{Q} = \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \\ -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}$$

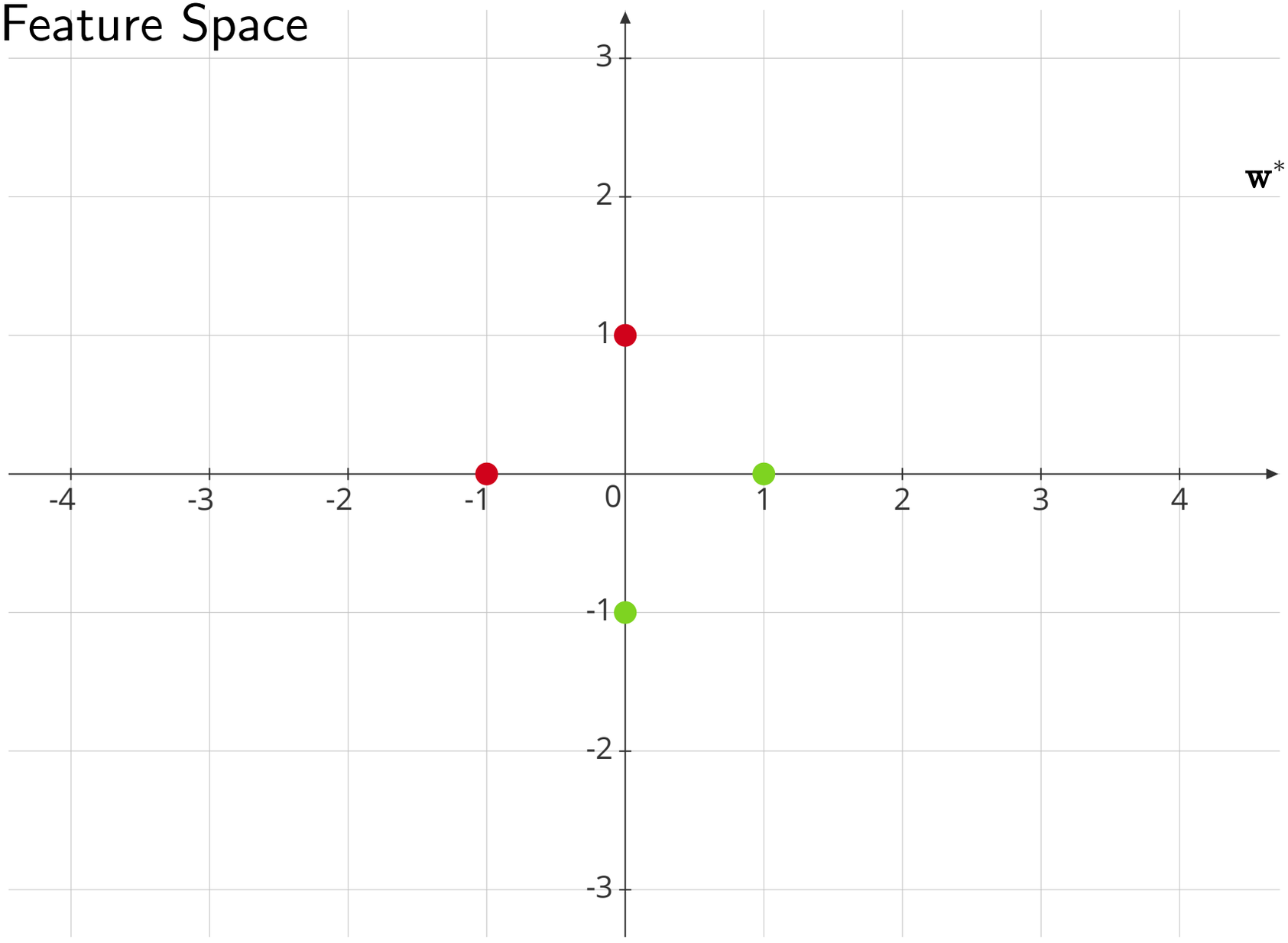
$$\mathbf{Q} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$\max_{\boldsymbol{\alpha} \geqslant 0} \; (\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4) - \frac{1}{2}\left(\alpha_1^2 + \alpha_2^2 + \alpha_3^2 + \alpha_4^2\right) - (\alpha_1 \alpha_3 + \alpha_2 \alpha_4)$$

$$\boldsymbol{\alpha}^* = \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \end{bmatrix}^T \quad \text{OR} \quad \boldsymbol{\alpha}^* = \begin{bmatrix} 1 & 0 & 0 & 1 \end{bmatrix}^T$$

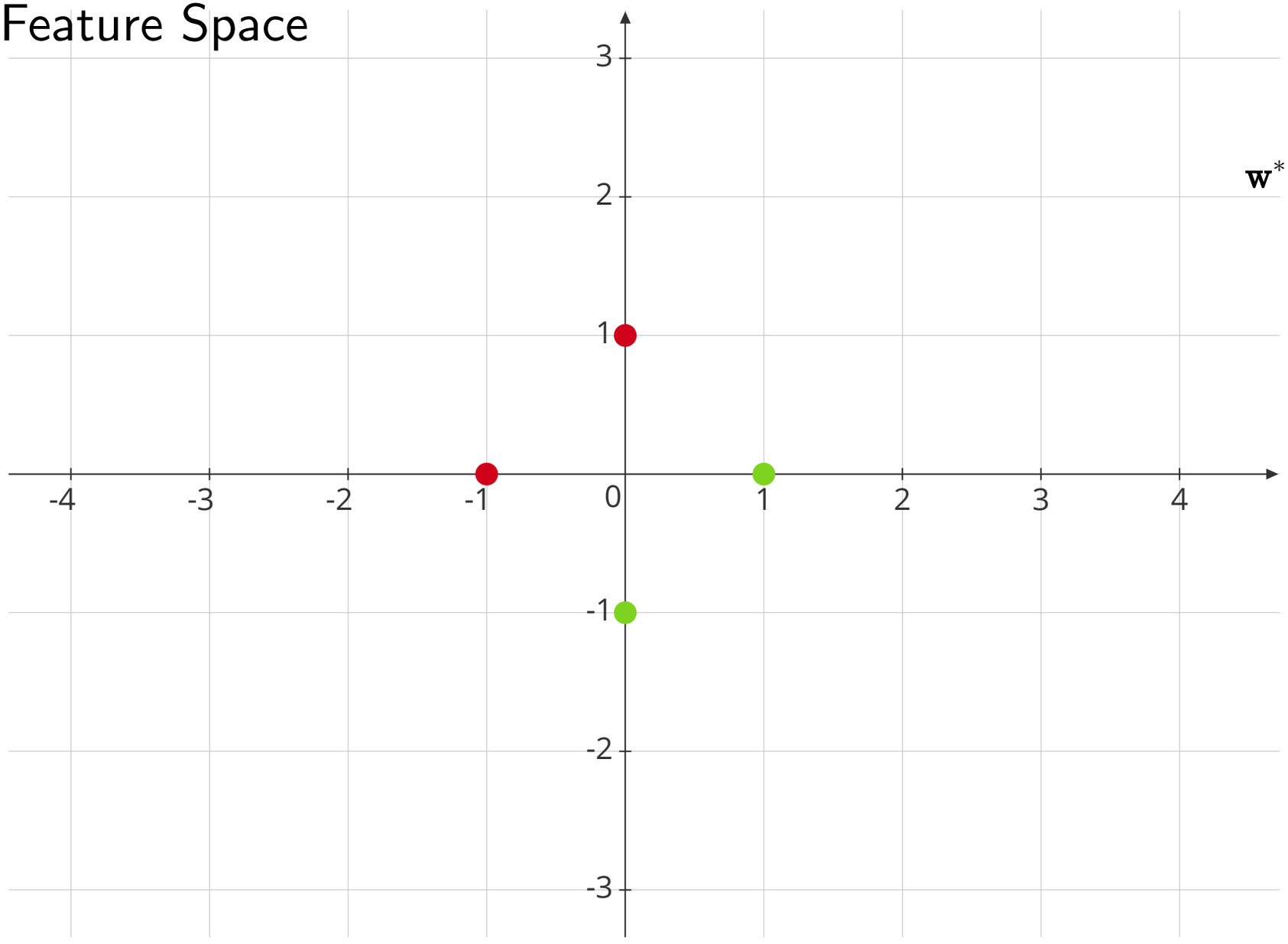
Feature Space

Example-1



$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

Feature Space

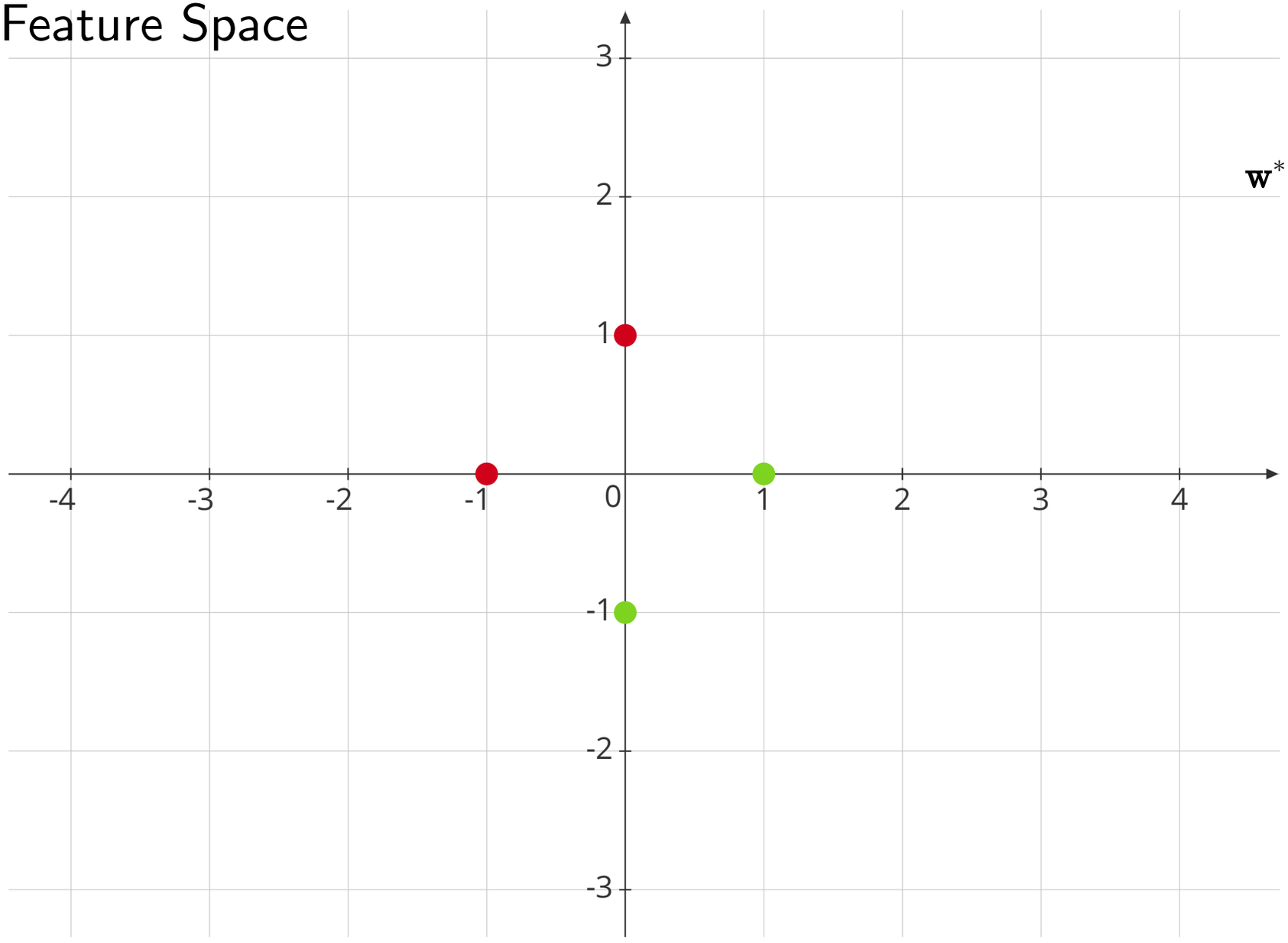


Example-1

$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

$$= 0.5 \cdot \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

Feature Space



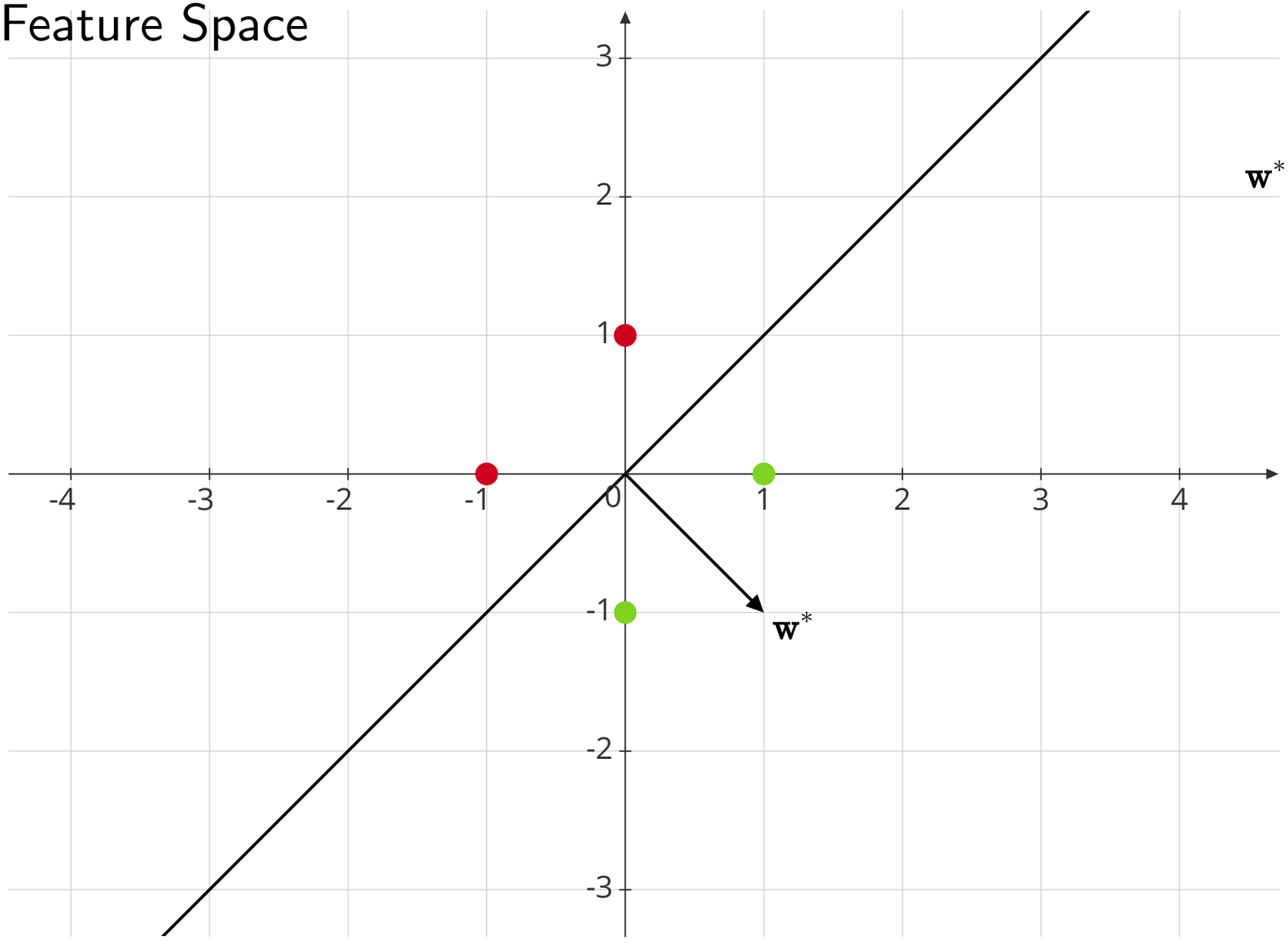
Example-1

$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

$$= 0.5 \cdot \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Feature Space



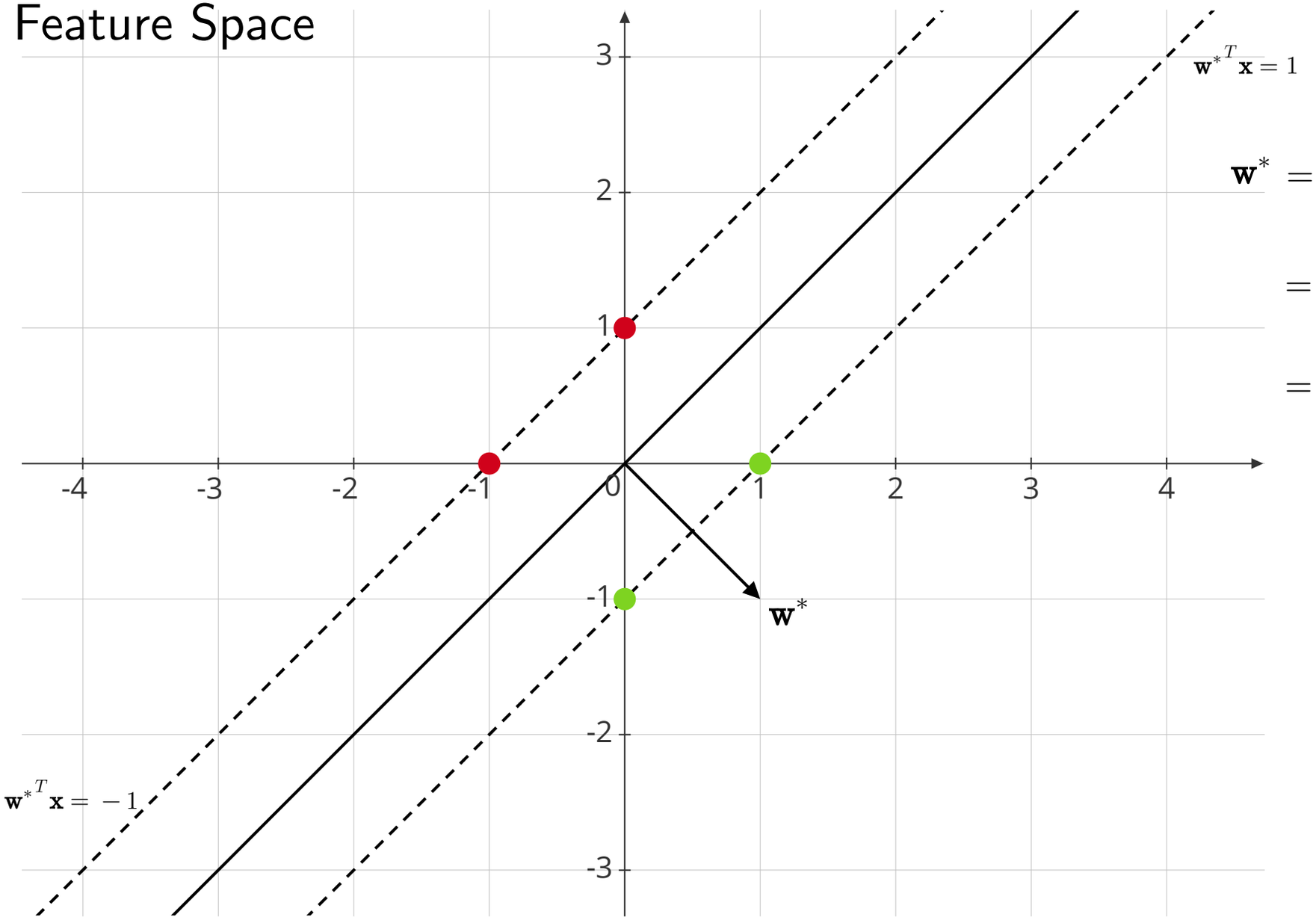
Example-1

$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

$$= 0.5 \cdot \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Feature Space



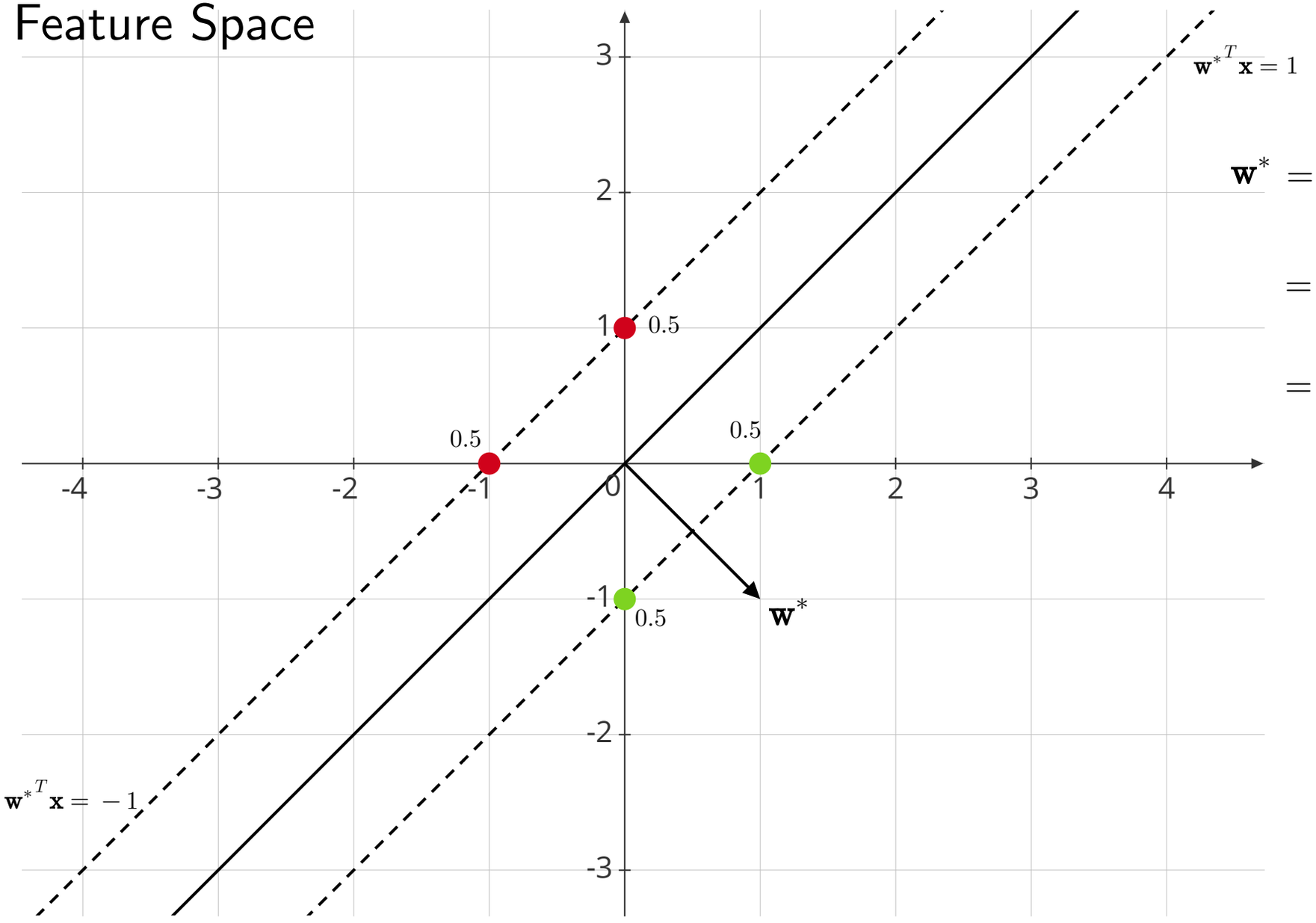
Example-1

$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

$$= 0.5 \cdot \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Feature Space



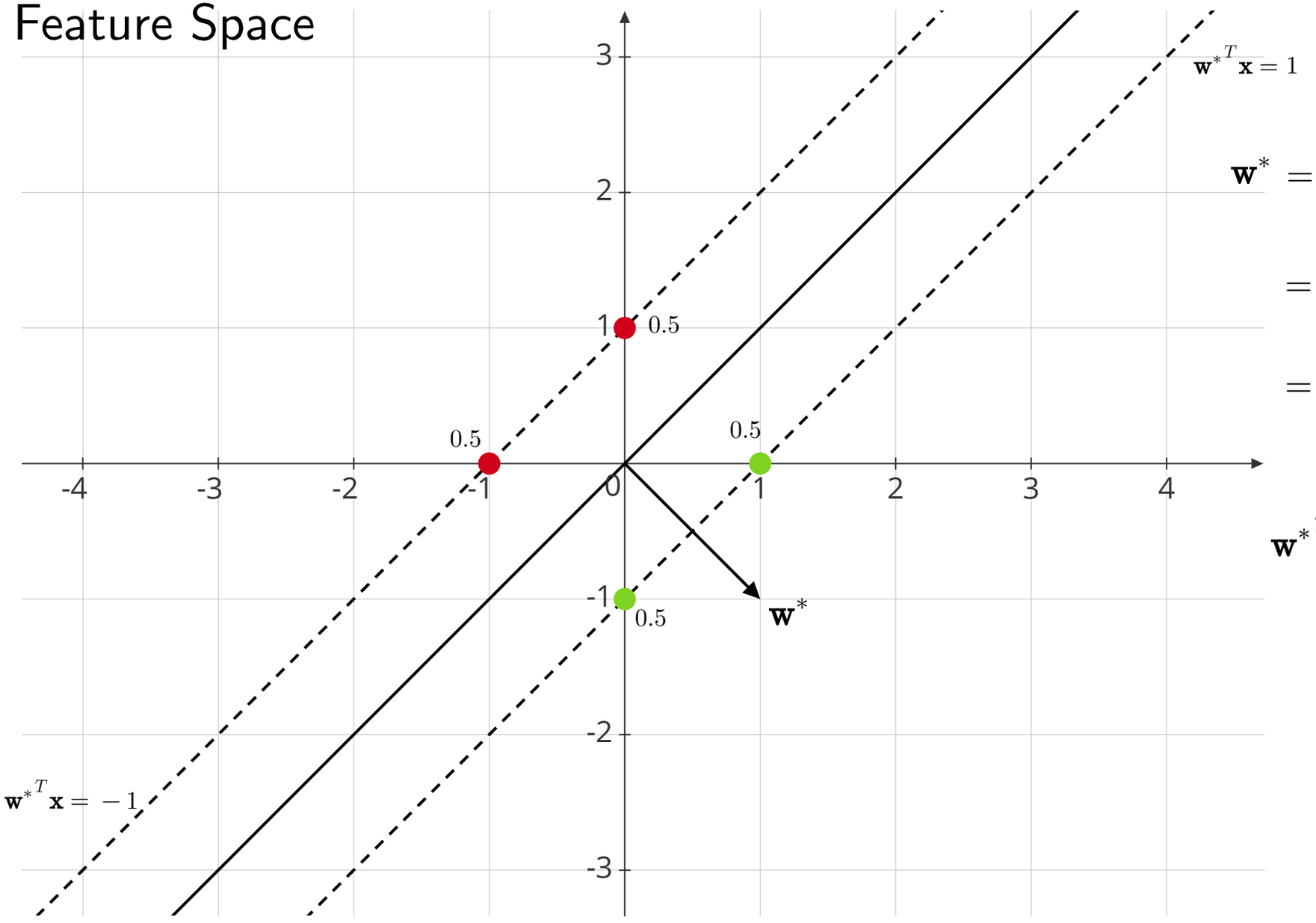
Example-1

$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

$$= 0.5 \cdot \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Feature Space



Example-1

$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

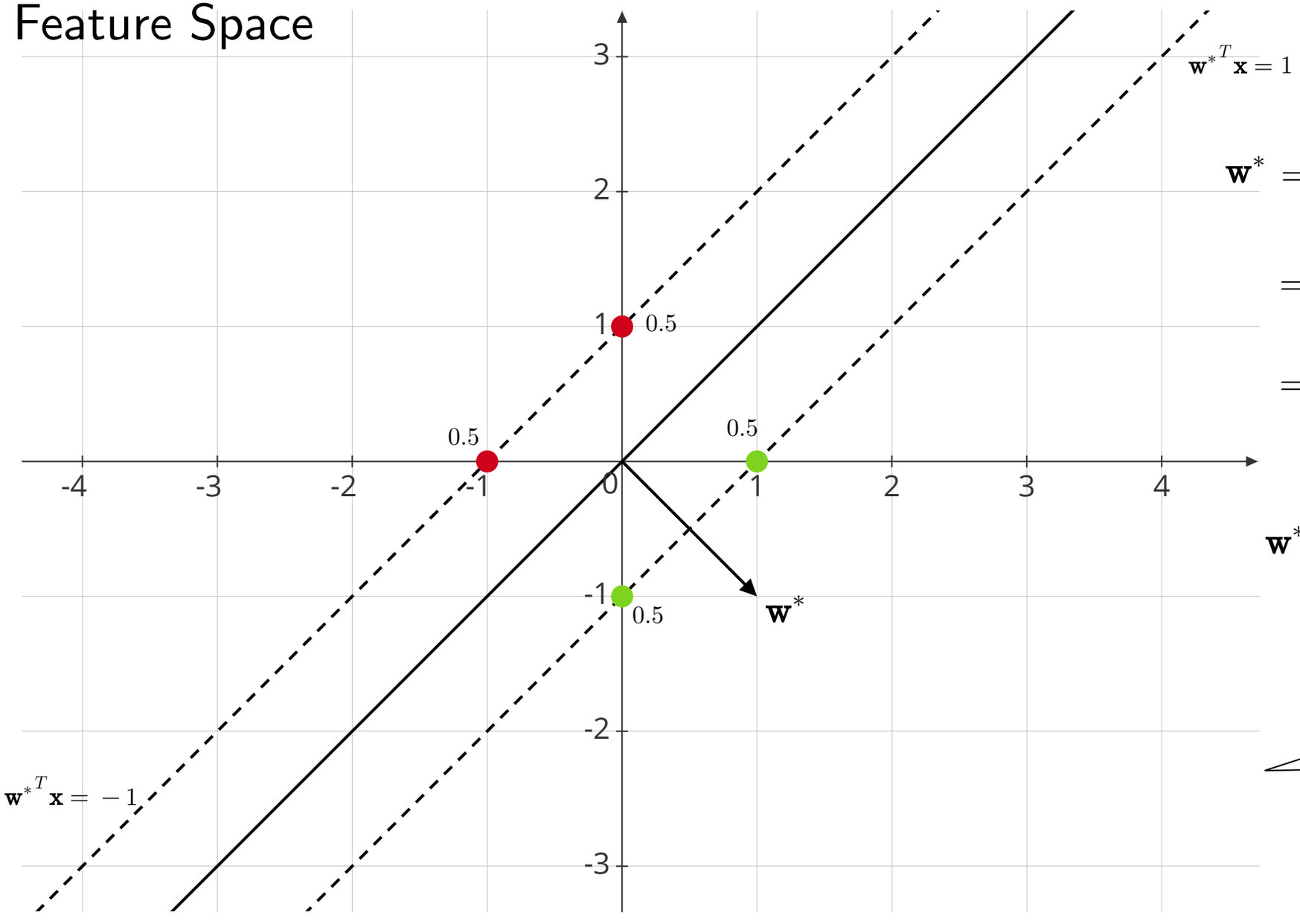
$$= 0.5 \cdot \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\mathbf{w}^{*T}\mathbf{x}_{\text{test}} = \sum_{i=1}^{n} \alpha_i \cdot \underbrace{\left[(y_i \mathbf{x}_i)^T \mathbf{x}_{\text{test}} \right]}_{\text{similarity}}$$

Feature Space

Example-1

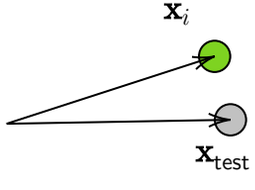


$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

$$= 0.5 \cdot \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

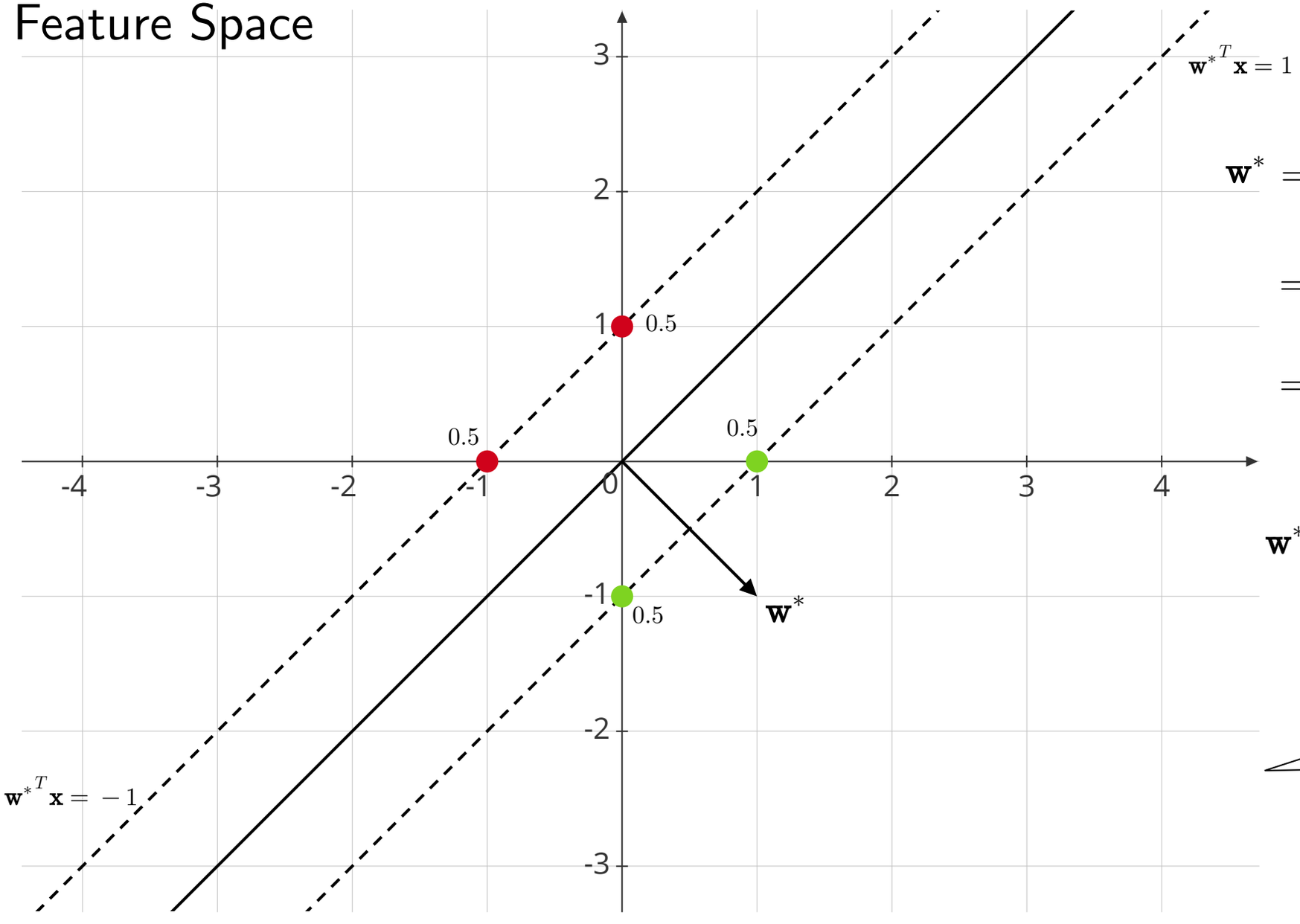
$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\mathbf{w}^{*T}\mathbf{x}_{\text{test}} = \sum_{i=1}^{n} \alpha_i \cdot \underbrace{\left[(y_i \mathbf{x}_i)^T \mathbf{x}_{\text{test}} \right]}_{\text{similarity}}$$



Feature Space

Example-1

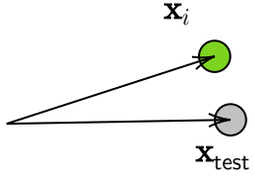


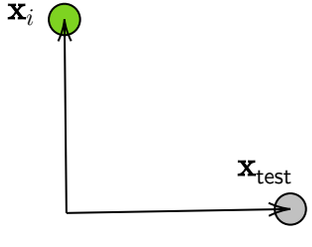
$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

$$= 0.5 \cdot \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

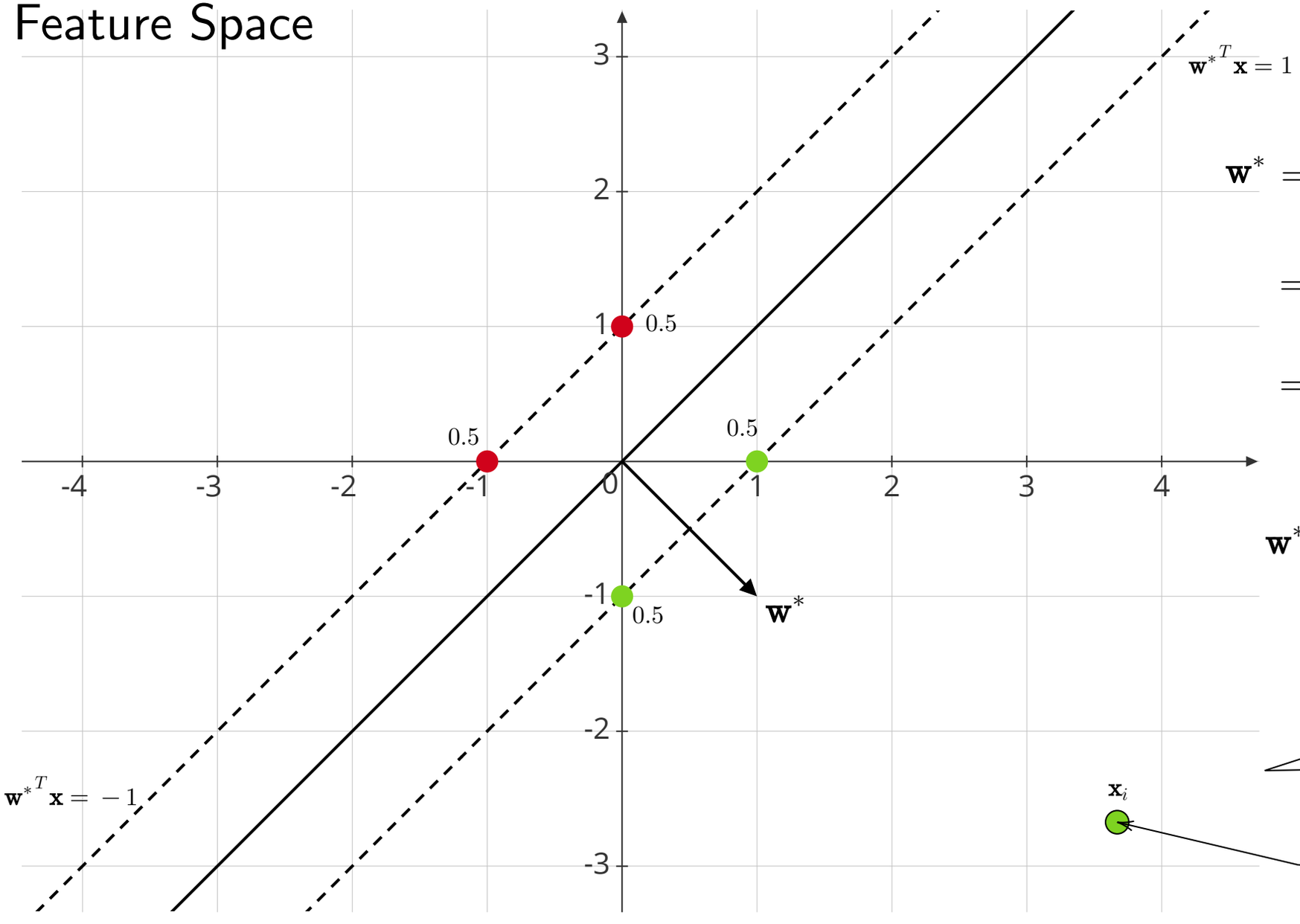
$$\mathbf{w}^{*T}\mathbf{x}_{\text{test}} = \sum_{i=1}^{n} \alpha_i \cdot \underbrace{\left[(y_i \mathbf{x}_i)^T \mathbf{x}_{\text{test}} \right]}_{\text{similarity}}$$





Feature Space

Example-1

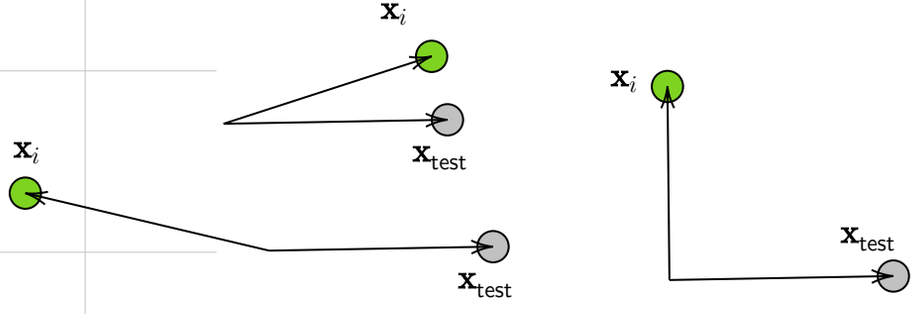


$$\mathbf{w}^* = \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i$$

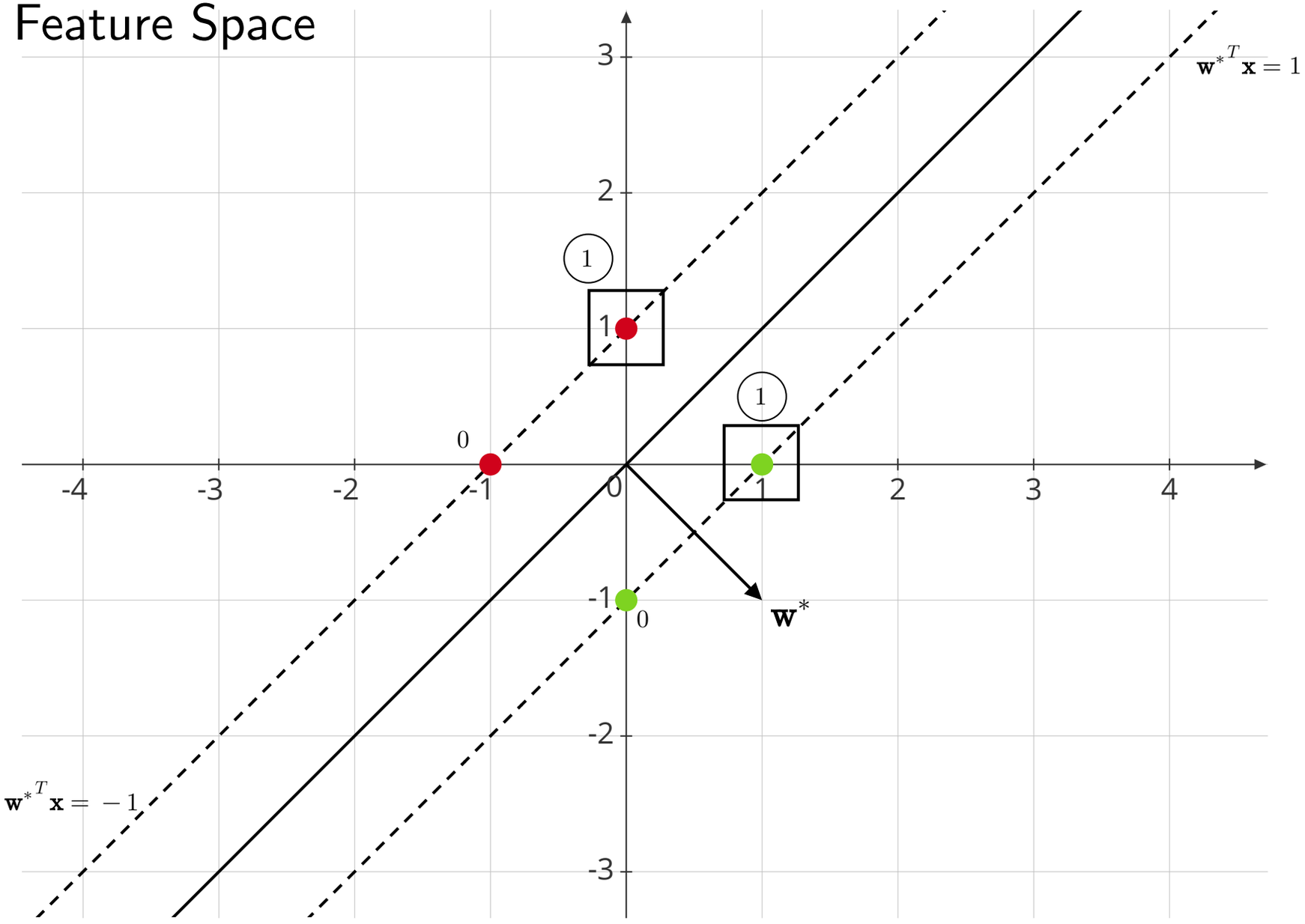
$$= 0.5 \cdot \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\mathbf{w}^{*T}\mathbf{x}_{\text{test}} = \sum_{i=1}^{n} \alpha_i \cdot \underbrace{\left[(y_i \mathbf{x}_i)^T \mathbf{x}_{\text{test}} \right]}_{\text{similarity}}$$



Feature Space



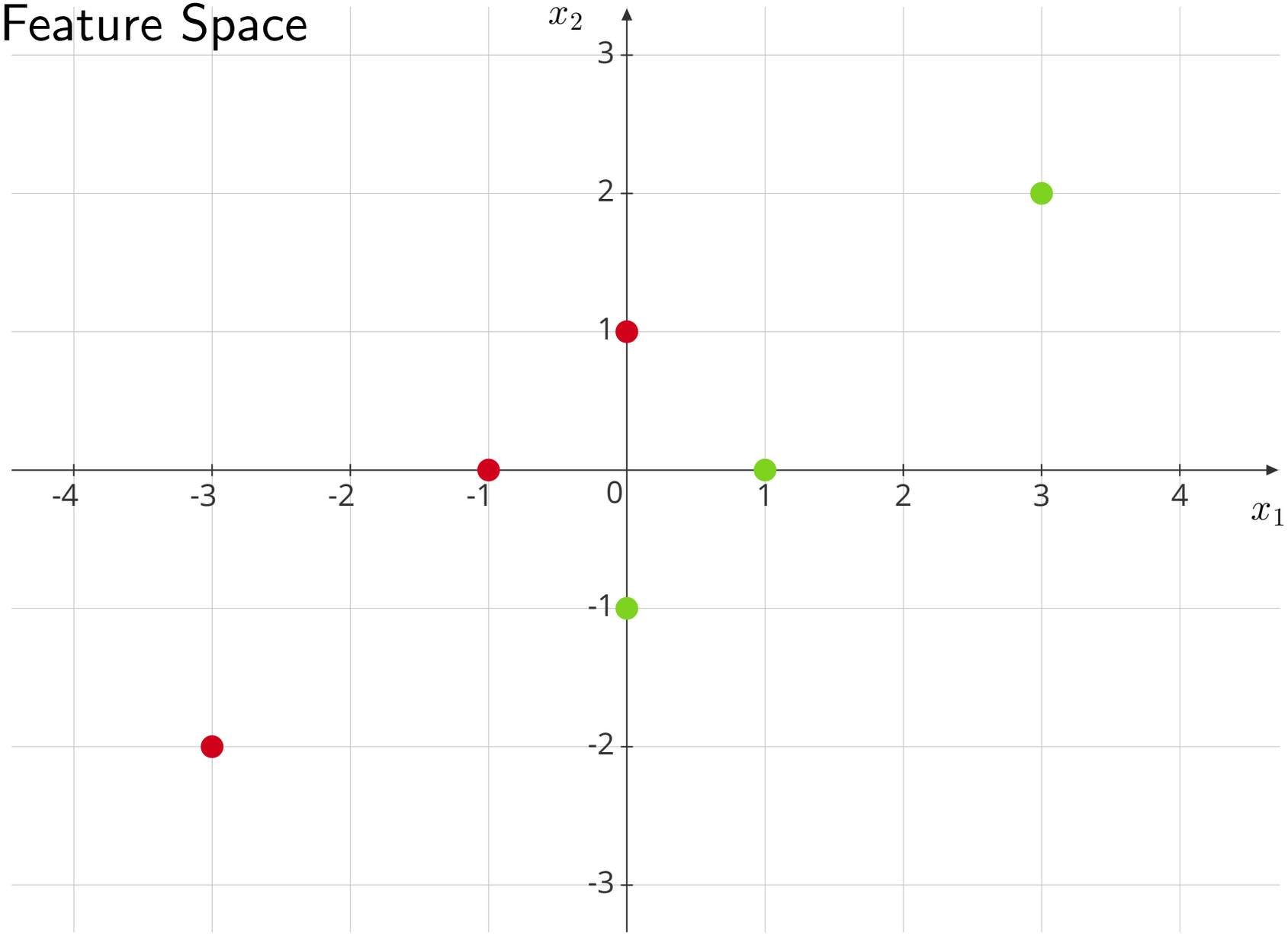
Example-1

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\boldsymbol{\alpha}^* = \begin{bmatrix} 1 & 0 & 0 & 1 \end{bmatrix}^T$$

$$\begin{aligned} \mathbf{w}^* &= \sum_{i=1}^{4} \alpha_i^* \mathbf{x}_i y_i \\ &= 1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} - 1 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ -1 \end{bmatrix} \end{aligned}$$

Feature Space

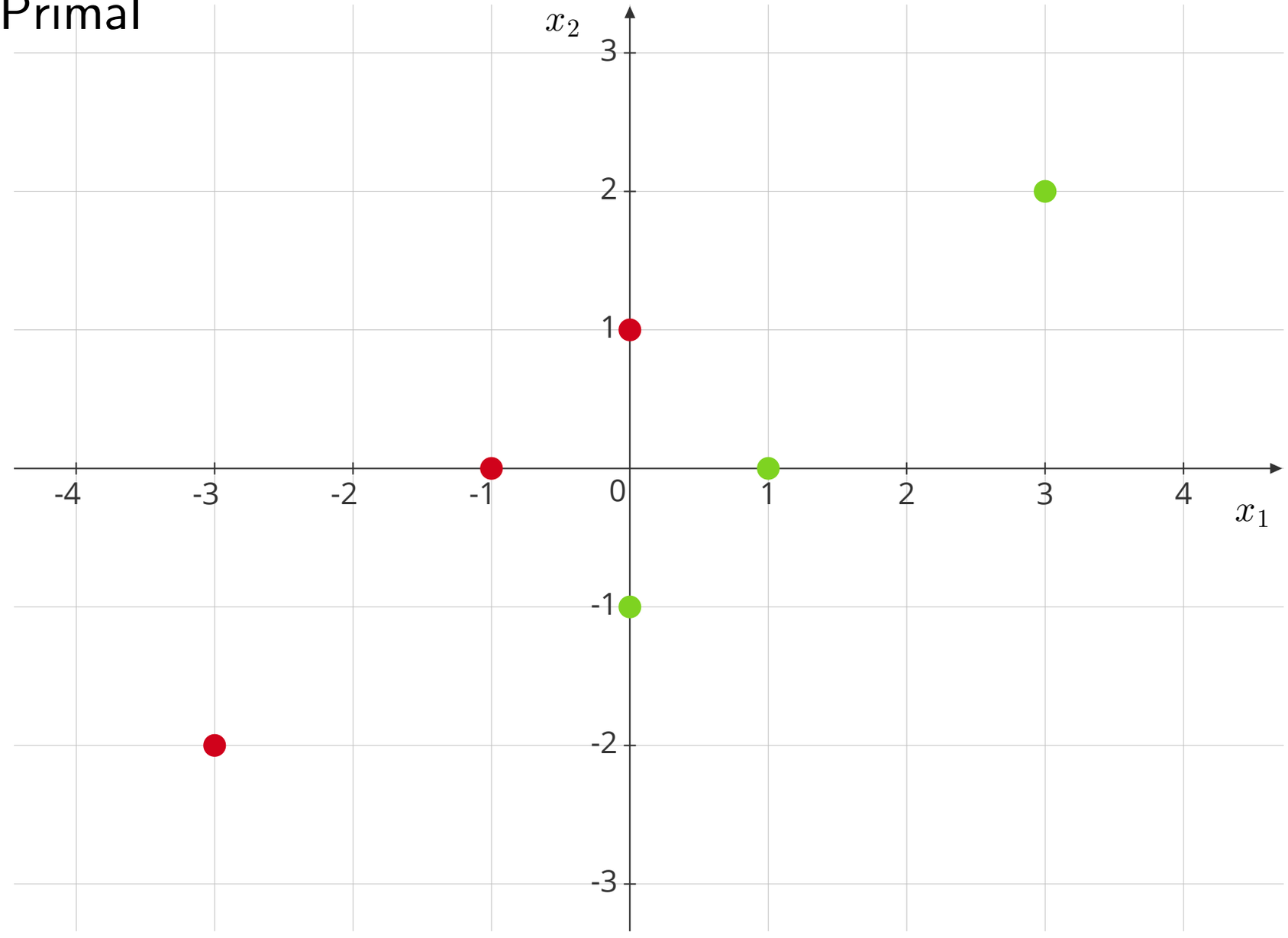


Example-2

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & -1 & 0 & -3 \\ 0 & -1 & 2 & 0 & 1 & -2 \end{bmatrix},$$

$$\mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \\ -1 \\ -1 \end{bmatrix}$$

Primal



Example-2

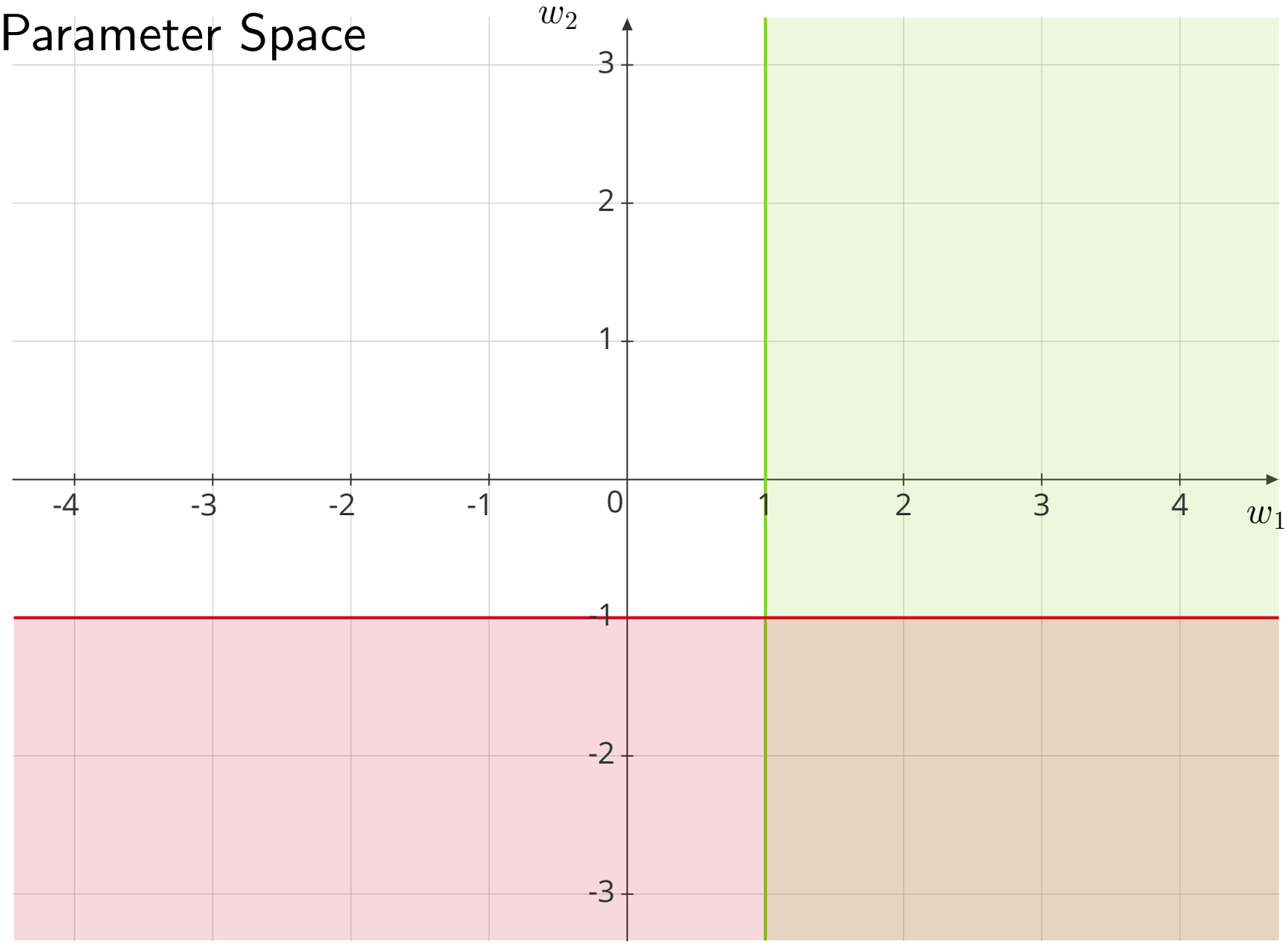
$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & -1 & 0 & -3 \\ 0 & -1 & 2 & 0 & 1 & -2 \end{bmatrix},$$

$$\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 & -1 & -1 & -1 \end{bmatrix}^T$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$\begin{aligned} w_1 &\geqslant 1 && (1) \\ w_2 &\leqslant -1 && (2) \\ 3w_1 + 2w_2 &\geqslant 1 && (3) \\ w_1 &\geqslant 1 && (4) \\ w_2 &\leqslant -1 && (5) \\ 3w_1 + 2w_2 &\geqslant 1 && (6) \end{aligned}$$

Parameter Space



Example-2

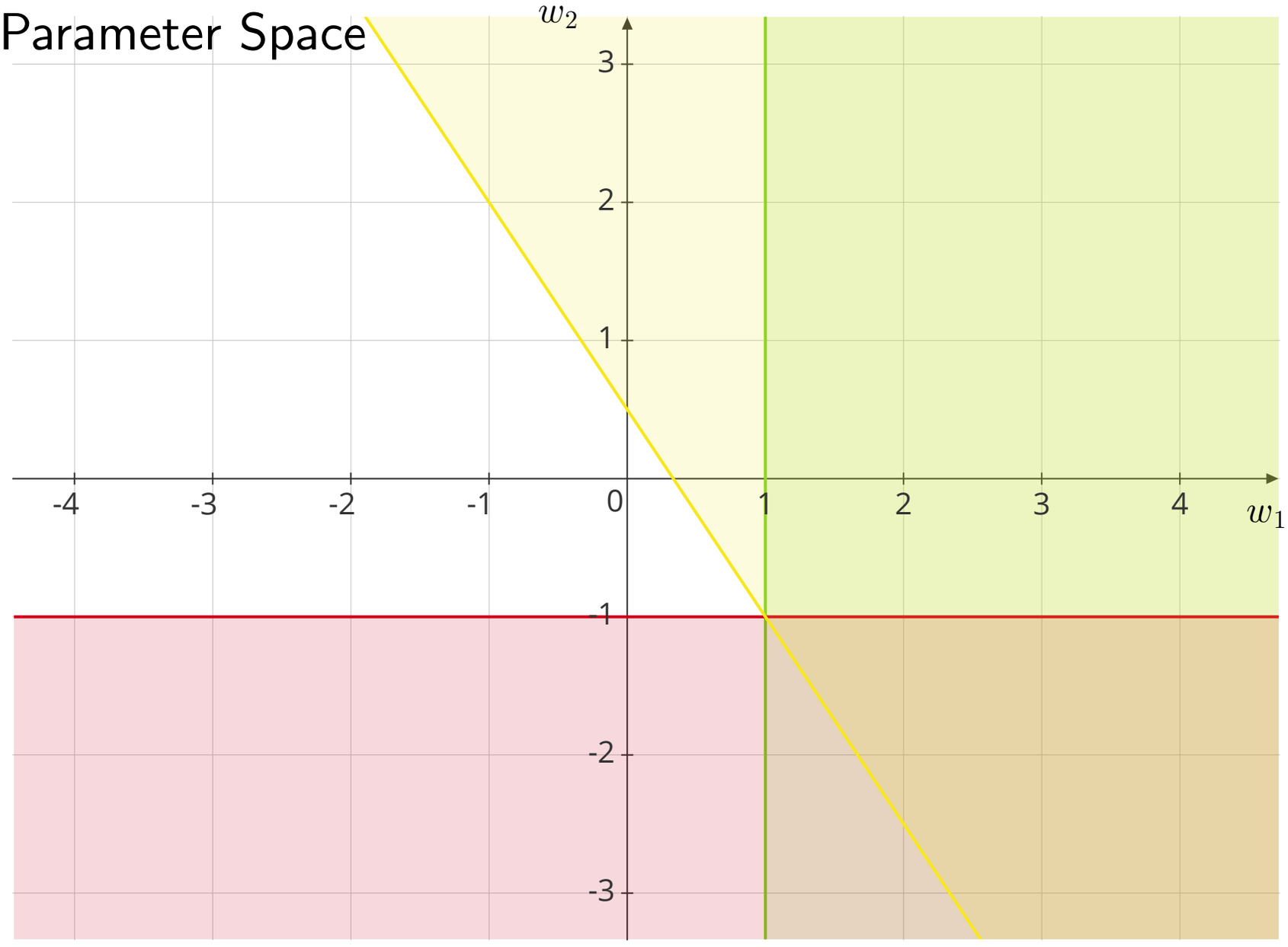
$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & -1 & 0 & -3 \\ 0 & -1 & 2 & 0 & 1 & -2 \end{bmatrix},$$

$$\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 & -1 & -1 & -1 \end{bmatrix}^T$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$\begin{aligned} w_1 &\geqslant 1 && (1) \\ w_2 &\leqslant -1 && (2) \\ 3w_1 + 2w_2 &\geqslant 1 && (3) \\ w_1 &\geqslant 1 && (4) \\ w_2 &\leqslant -1 && (5) \\ 3w_1 + 2w_2 &\geqslant 1 && (6) \end{aligned}$$

Parameter Space



Example-2

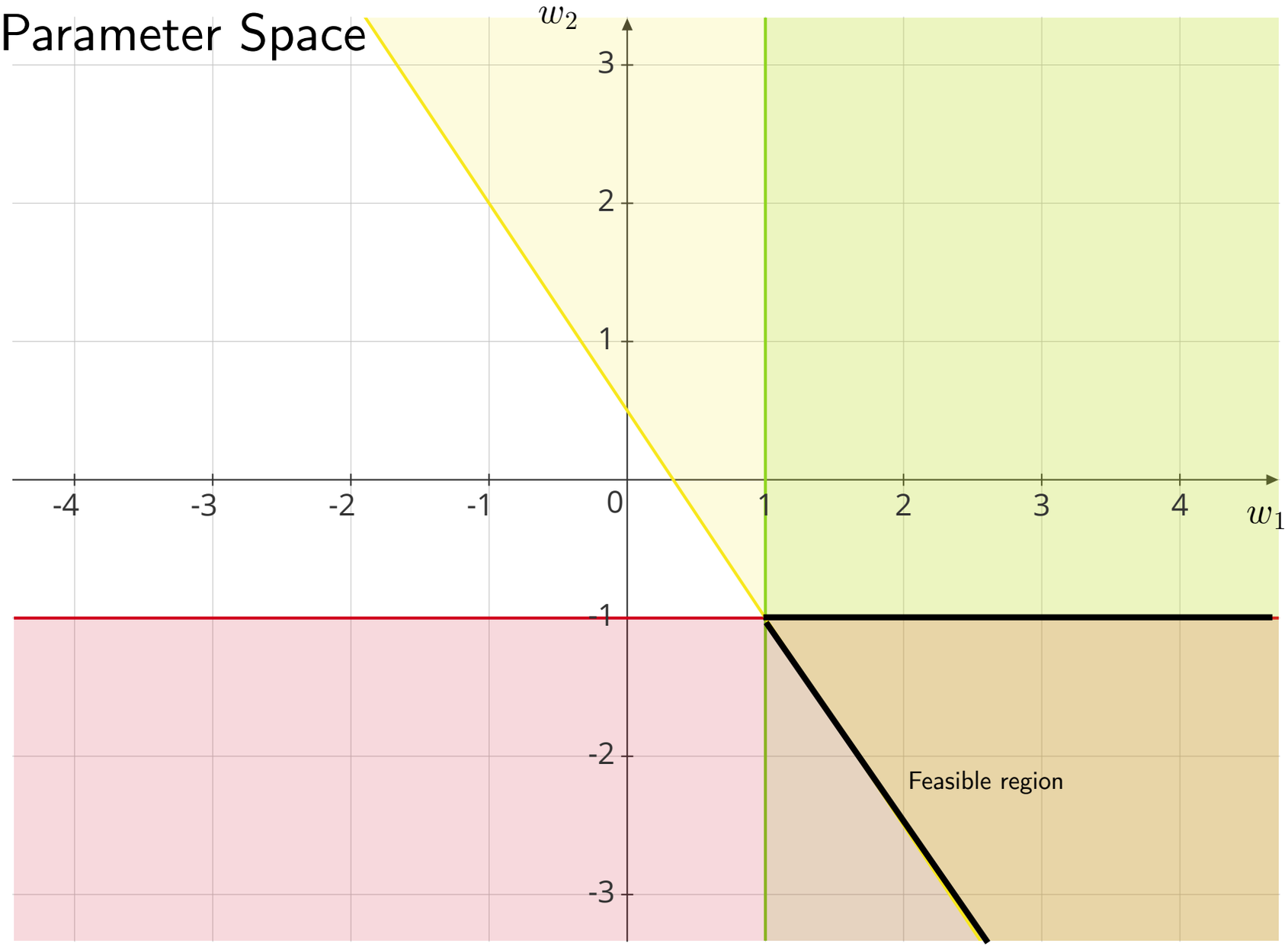
$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & -1 & 0 & -3 \\ 0 & -1 & 2 & 0 & 1 & -2 \end{bmatrix},$$

$$\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 & -1 & -1 & -1 \end{bmatrix}^T$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$\begin{aligned} w_1 &\geqslant 1 && (1) \\ w_2 &\leqslant -1 && (2) \\ 3w_1 + 2w_2 &\geqslant 1 && (3) \\ w_1 &\geqslant 1 && (4) \\ w_2 &\leqslant -1 && (5) \\ 3w_1 + 2w_2 &\geqslant 1 && (6) \end{aligned}$$

Parameter Space



Example-2

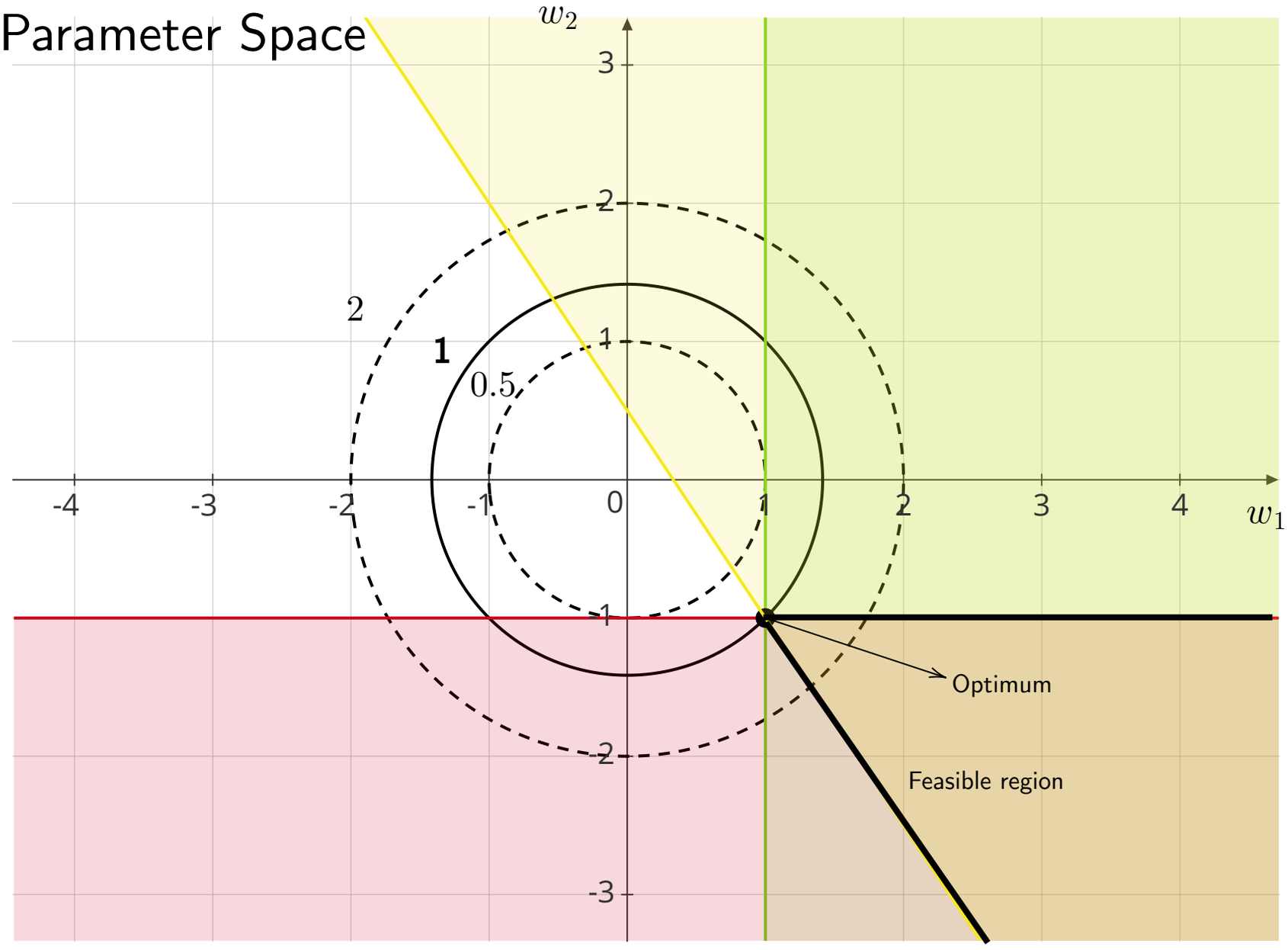
$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & -1 & 0 & -3 \\ 0 & -1 & 2 & 0 & 1 & -2 \end{bmatrix},$$

$$\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 & -1 & -1 & -1 \end{bmatrix}^T$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$\begin{aligned} w_1 &\geqslant 1 && (1) \\ w_2 &\leqslant -1 && (2) \\ 3w_1 + 2w_2 &\geqslant 1 && (3) \\ w_1 &\geqslant 1 && (4) \\ w_2 &\leqslant -1 && (5) \\ 3w_1 + 2w_2 &\geqslant 1 && (6) \end{aligned}$$

Parameter Space



Example-2

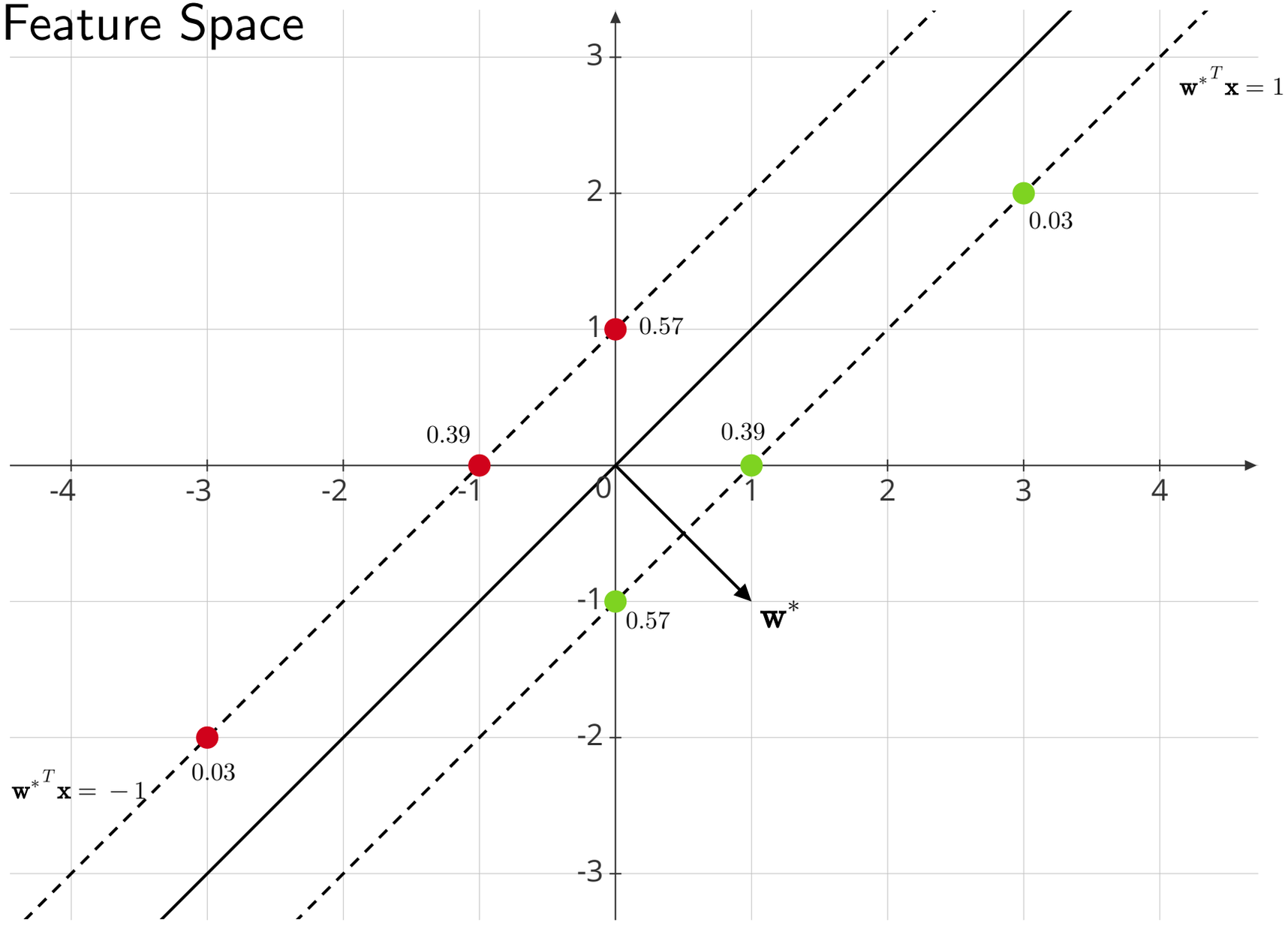
$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & -1 & 0 & -3 \\ 0 & -1 & 2 & 0 & 1 & -2 \end{bmatrix},$$

$$\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 & -1 & -1 & -1 \end{bmatrix}^T$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$\begin{aligned} w_1 &\geqslant 1 && (1) \\ w_2 &\leqslant -1 && (2) \\ 3w_1 + 2w_2 &\geqslant 1 && (3) \\ w_1 &\geqslant 1 && (4) \\ w_2 &\leqslant -1 && (5) \\ 3w_1 + 2w_2 &\geqslant 1 && (6) \end{aligned}$$

Feature Space



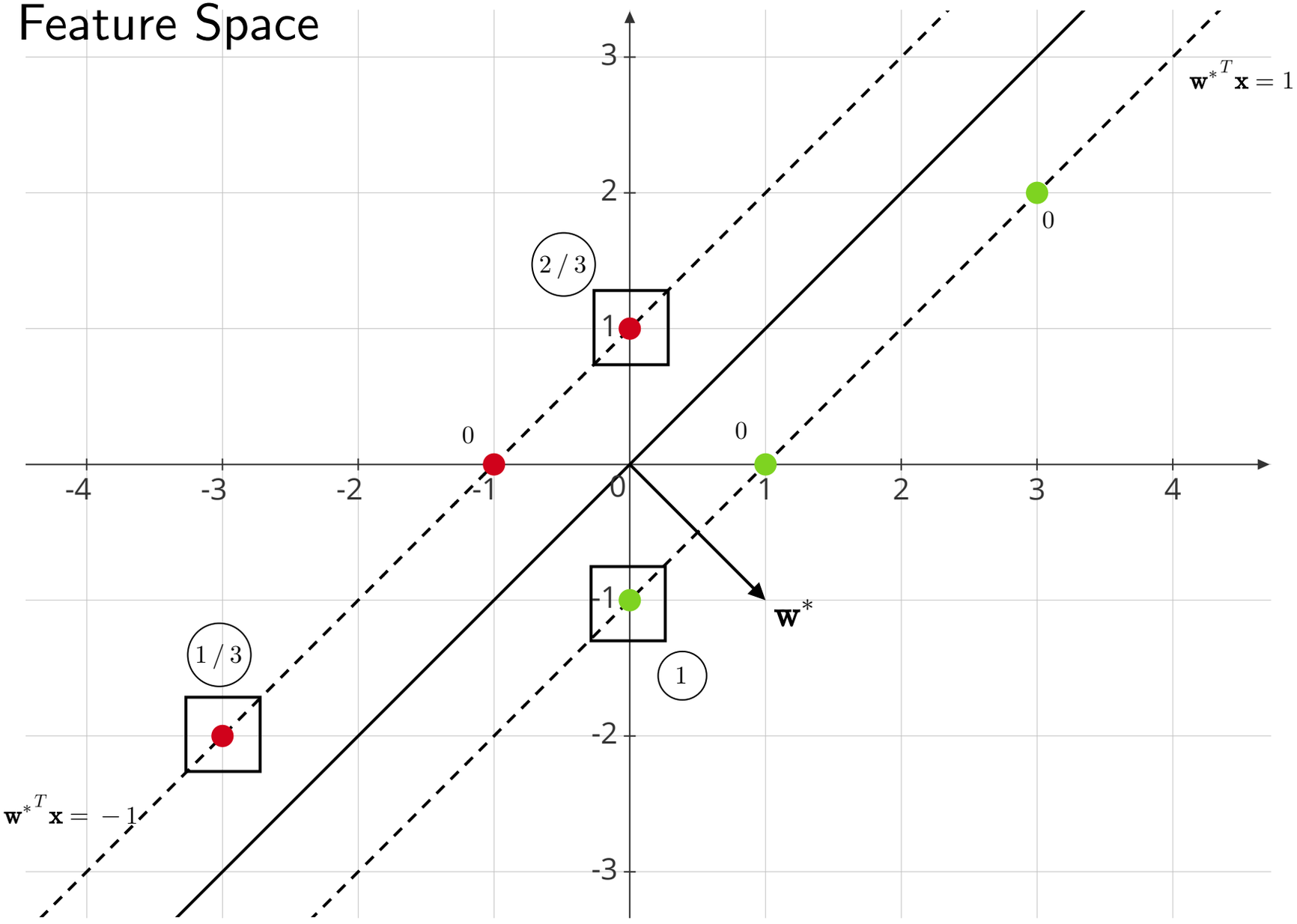
Example-2

$$\boldsymbol{\alpha}^* = \begin{bmatrix} 0.39 \\ 0.57 \\ 0.03 \\ 0.39 \\ 0.57 \\ 0.03 \end{bmatrix}$$

$$\mathbf{w}^* = \sum_{i=1}^{6} \alpha_i^* \mathbf{x}_i y_i$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Feature Space



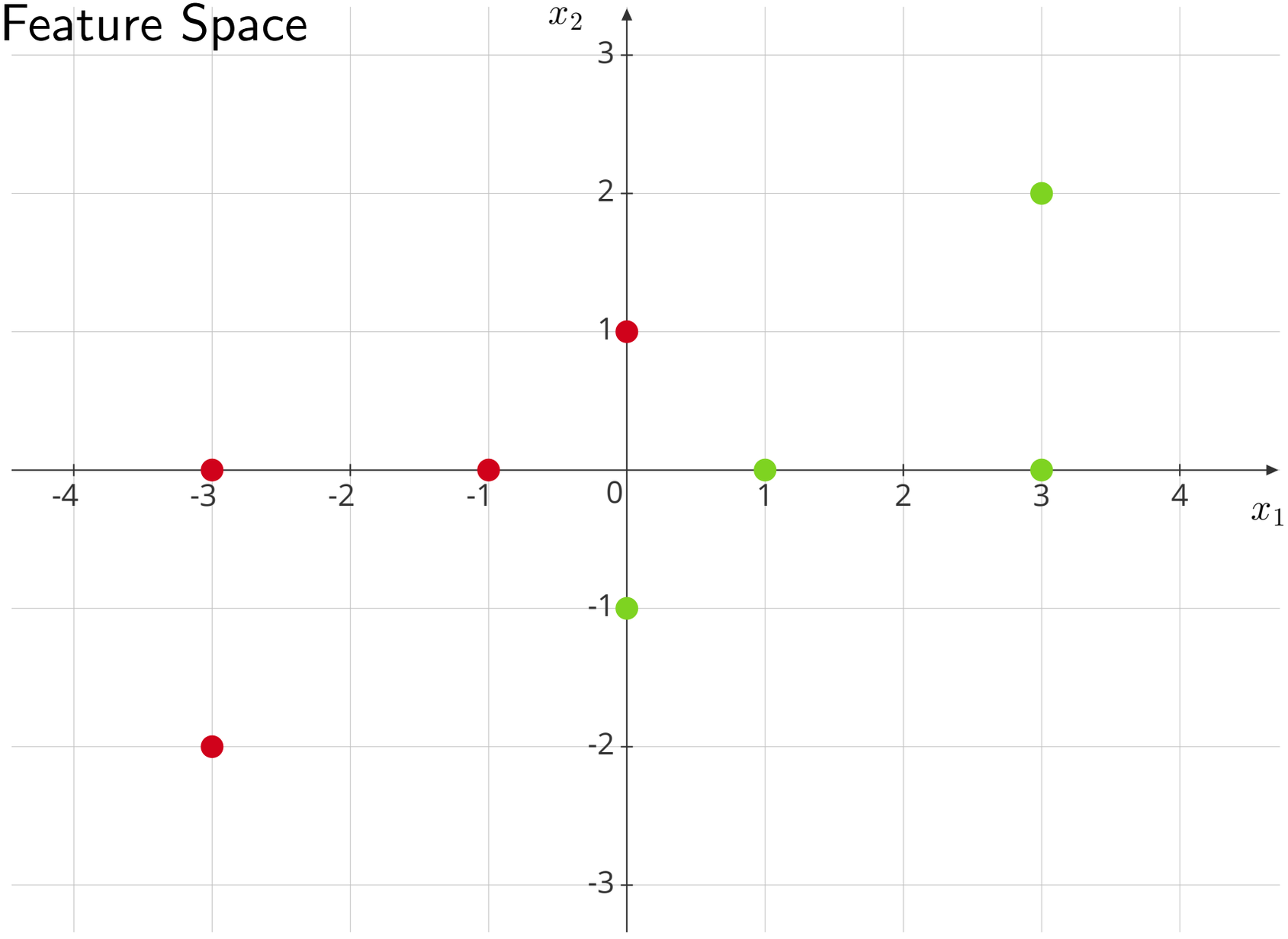
Example-2

$$\boldsymbol{\alpha}^* = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 2/3 \\ 1/3 \end{bmatrix}$$

$$\mathbf{w}^* = \sum_{i=1}^{6} \alpha_i^* \mathbf{x}_i y_i$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Feature Space

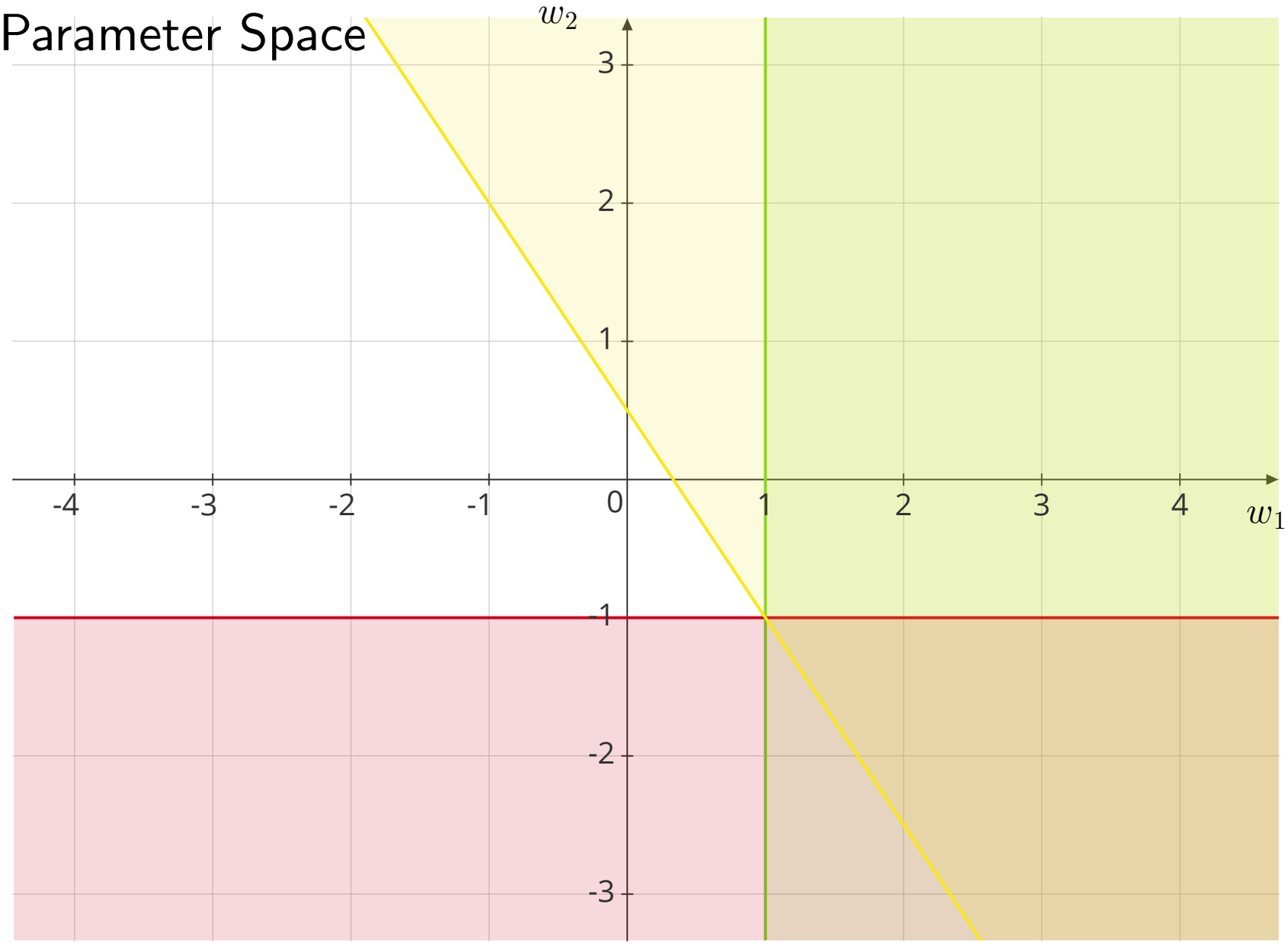


Example-3

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & 3 & -1 & 0 & -3 & -3 \\ 0 & -1 & 2 & 0 & 0 & 1 & -2 & 0 \end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ -1 \\ -1 \\ -1 \\ -1 \end{bmatrix}$$

Parameter Space



Example-3

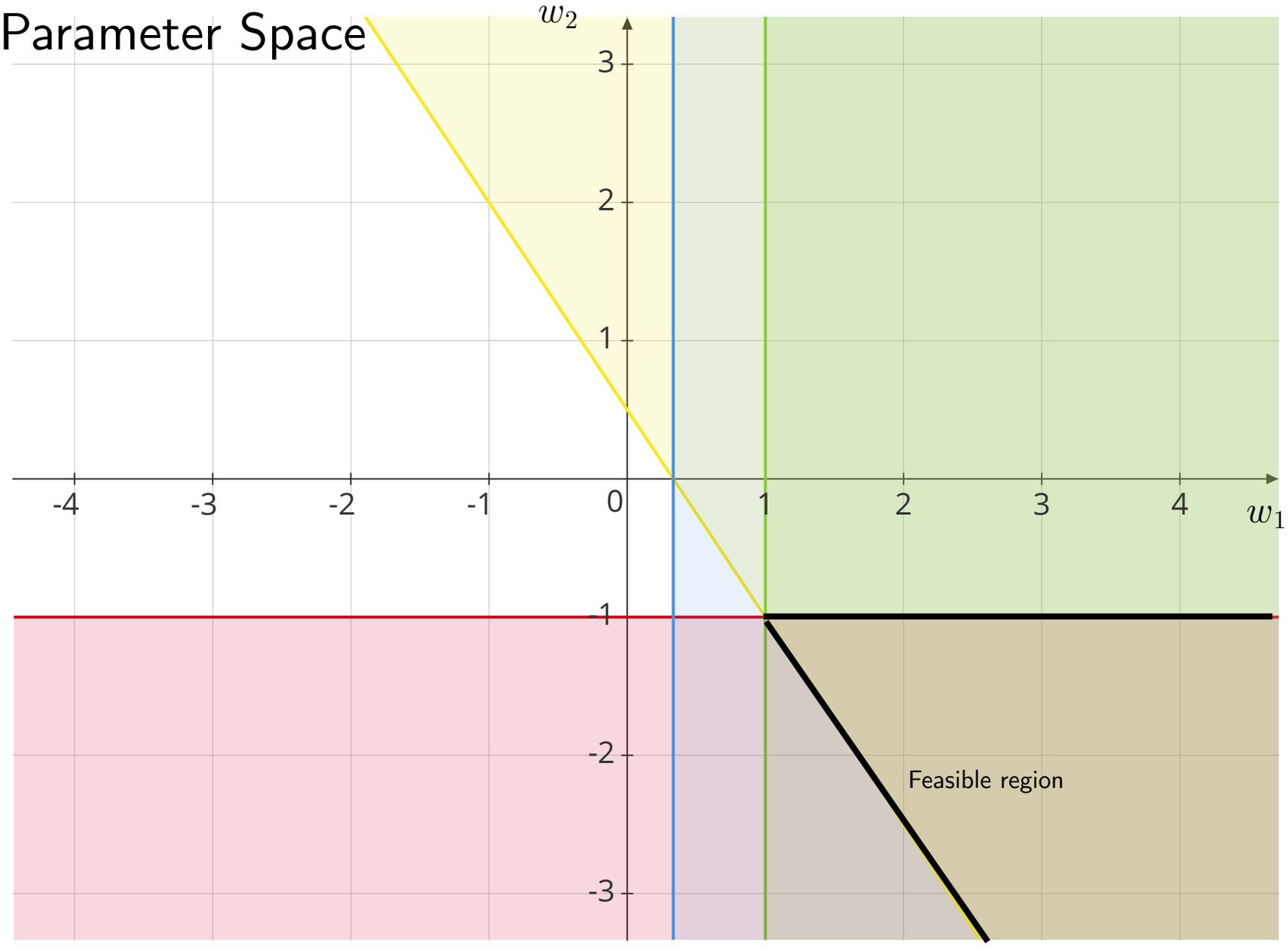
$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & 3 & -1 & 0 & -3 & -3 \\ 0 & -1 & 2 & 0 & 0 & 1 & -2 & 0 \end{bmatrix},$$

$$\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \end{bmatrix}^T$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$\begin{aligned} w_1 &\geqslant 1 && (1) \\ w_2 &\leqslant -1 && (2) \\ 3w_1 + 2w_2 &\geqslant 1 && (3) \\ w_1 &\geqslant 1/3 && (4) \\ w_1 &\geqslant 1 && (5) \\ w_2 &\leqslant -1 && (6) \\ 3w_1 + 2w_2 &\geqslant 1 && (7) \\ w_1 &\geqslant 1/3 && (8) \end{aligned}$$

Parameter Space



Example-3

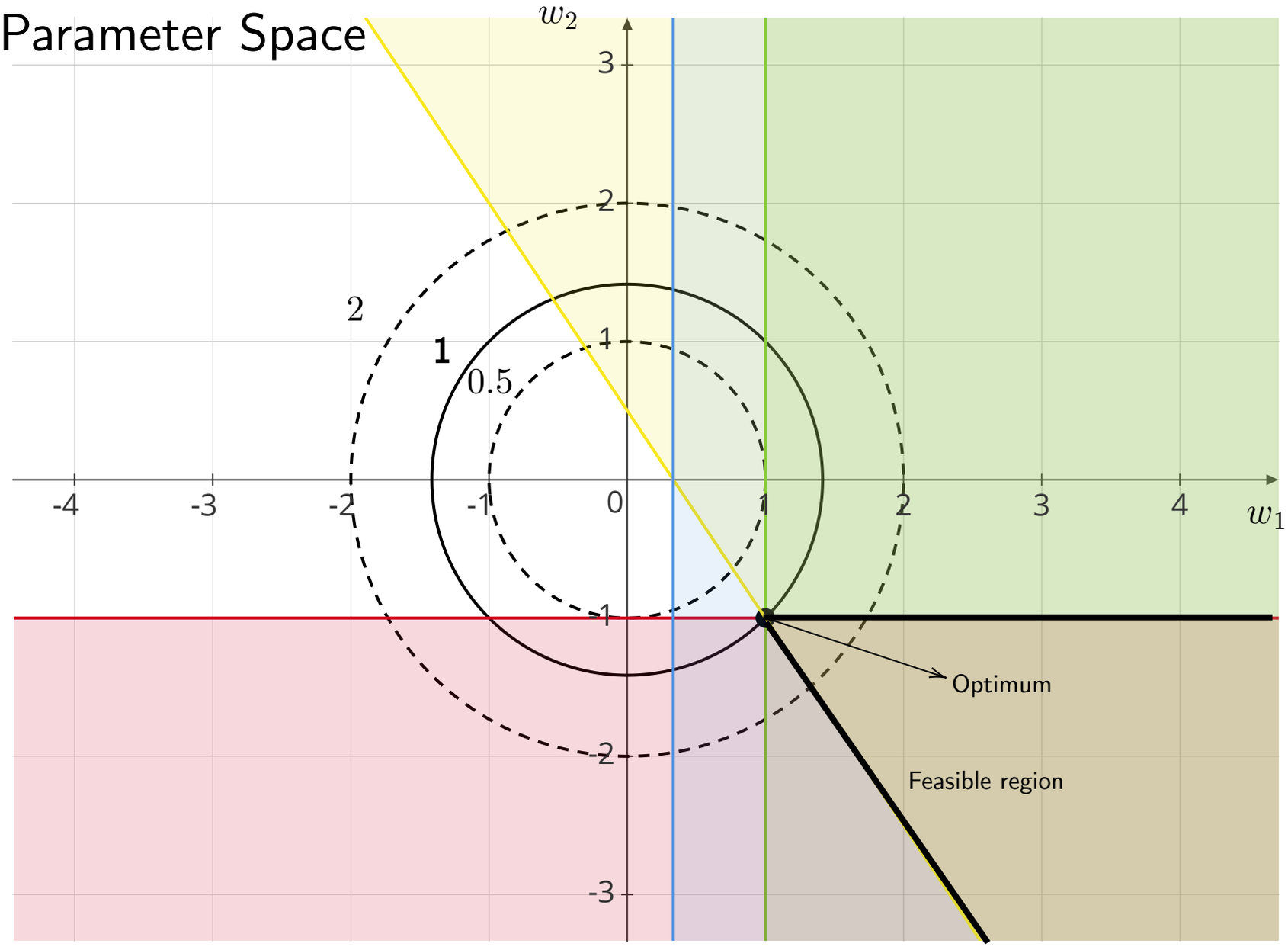
$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & 3 & -1 & 0 & -3 & -3 \\ 0 & -1 & 2 & 0 & 0 & 1 & -2 & 0 \end{bmatrix},$$

$$\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \end{bmatrix}^T$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$\begin{aligned} w_1 &\geqslant 1 && (1) \\ w_2 &\leqslant -1 && (2) \\ 3w_1 + 2w_2 &\geqslant 1 && (3) \\ w_1 &\geqslant 1/3 && (4) \\ w_1 &\geqslant 1 && (5) \\ w_2 &\leqslant -1 && (6) \\ 3w_1 + 2w_2 &\geqslant 1 && (7) \\ w_1 &\geqslant 1/3 && (8) \end{aligned}$$

Parameter Space



Example-3

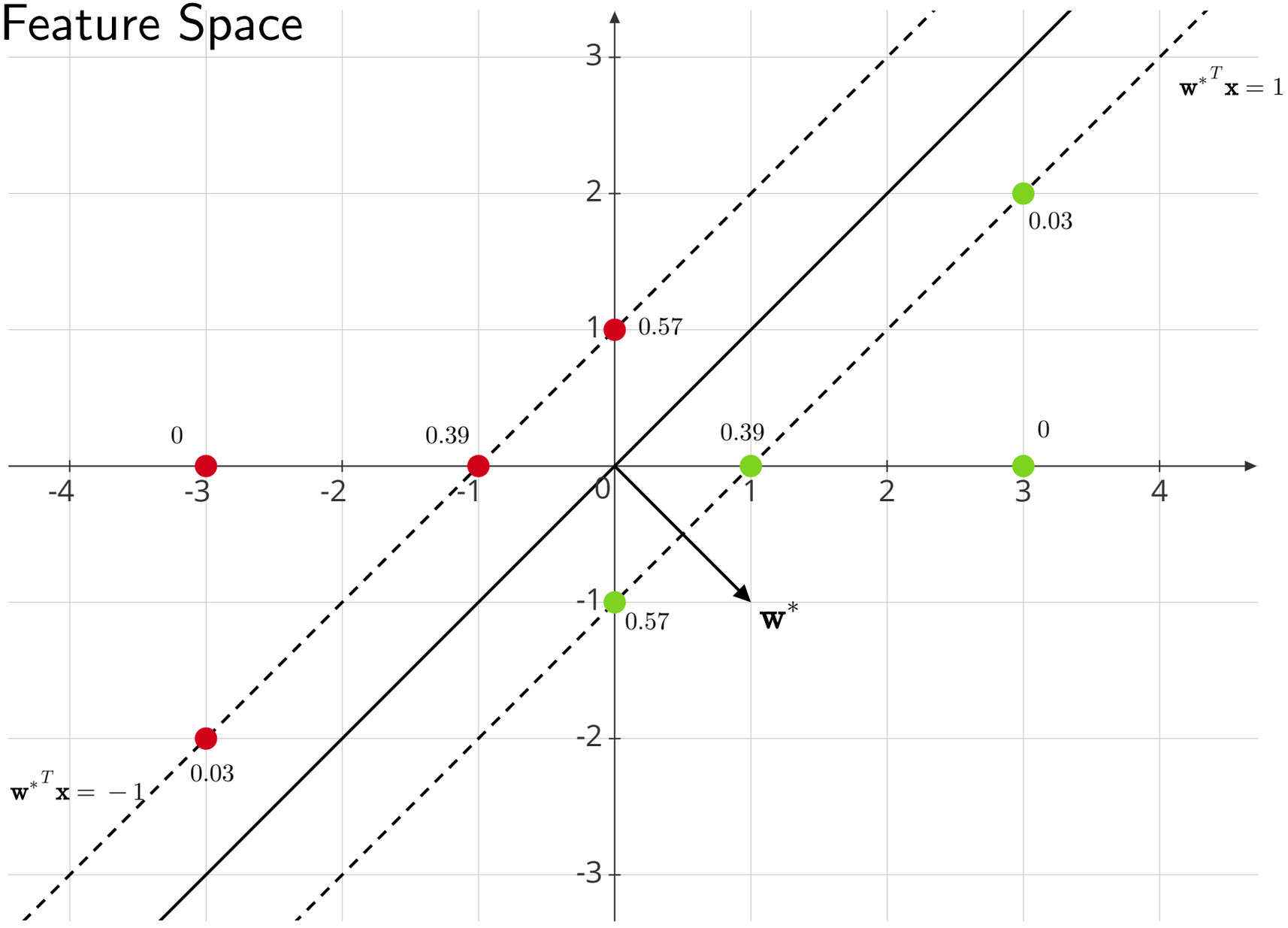
$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 3 & 3 & -1 & 0 & -3 & -3 \\ 0 & -1 & 2 & 0 & 0 & 1 & -2 & 0 \end{bmatrix},$$

$$\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \end{bmatrix}^T$$

$$\min_{\mathbf{w}} \quad \frac{w_1^2 + w_2^2}{2}$$

$$\begin{aligned} w_1 &\geqslant 1 && (1) \\ w_2 &\leqslant -1 && (2) \\ 3w_1 + 2w_2 &\geqslant 1 && (3) \\ w_1 &\geqslant 1/3 && (4) \\ w_1 &\geqslant 1 && (5) \\ w_2 &\leqslant -1 && (6) \\ 3w_1 + 2w_2 &\geqslant 1 && (7) \\ w_1 &\geqslant 1/3 && (8) \end{aligned}$$

Feature Space



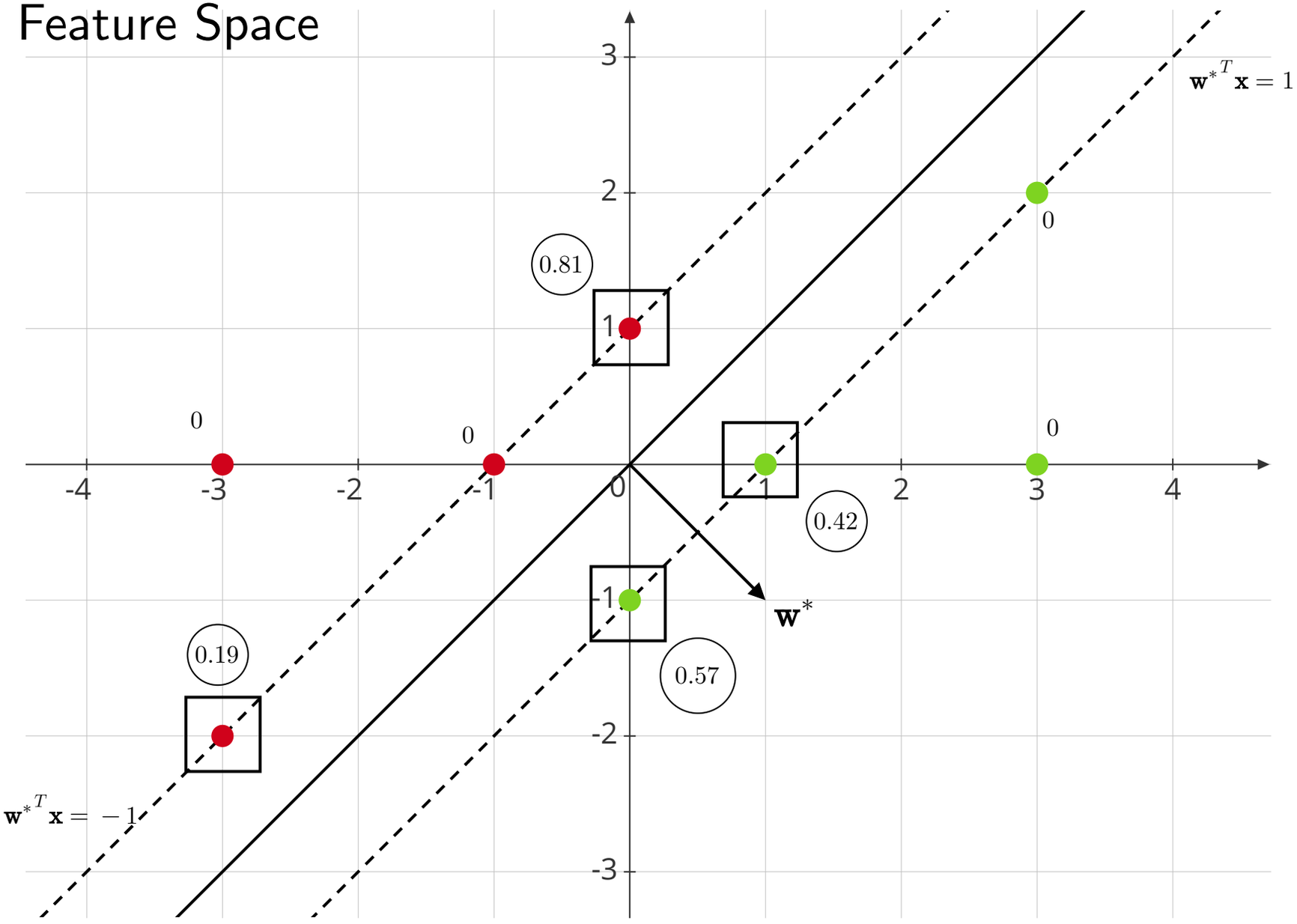
Example-2

$$\boldsymbol{\alpha}^* = \begin{bmatrix} 0.39 \\ 0.57 \\ 0.03 \\ 0 \\ 0.39 \\ 0.57 \\ 0.03 \\ 0 \end{bmatrix}$$

$$\mathbf{w}^* = \sum_{i=1}^{6} \alpha_i^* \mathbf{x}_i y_i$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Feature Space



Example-3

$$\boldsymbol{\alpha}^* = \begin{bmatrix} 0.42 \\ 0.57 \\ 0 \\ 0 \\ 0 \\ 0.81 \\ 0.19 \\ 0 \end{bmatrix}$$

$$\mathbf{w}^* = \sum_{i=1}^{6} \alpha_i^* \mathbf{x}_i y_i$$

$$= \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

References

- The content included in these slides has been referenced from the lectures of Professor Arun Rajkumar.
- Both the theoretical content and the examples in this presentation have been drawn, in part, from the session slides of Instructor Karthik Thiagarajan.

Thank You